32. Which of the following is the equation of the line tangent to the graph of $f(x) = \begin{cases} x - \pi, & x \le 0 \\ 1 - 2\sin x, & x > 0 \end{cases}$ at $(\pi, f(\pi))$?

A. $y = 2x - 2\pi + 1$
B. $y = 2x - 2\pi - 1$
C. $y = -2x + 2\pi + 1$
D. $y = -2x + 2\pi - 1$

Solution: It is found f '(x)

Since $f'(x) = (1 - 2\sin x)'$ for $x > 0$

$f'(x) = -2\cos x$

$f'(\pi) = -2\cos\pi$

$= 2$

and $f'(x) = (x - \pi)'$ for $x \le 0$

$f'(x) = 1$

Equation of tangent line at $(\pi, f(\pi)) = (\pi, 1)$ will be

$\frac{y - y_1}{x - x_1} = m$

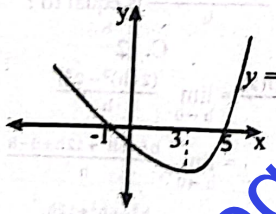
$\frac{y-1}{x-\pi} = 2$

$y - 1 = 2x - 2\pi$

$y = 2x - 2\pi + 1$

∴ Answer A

33. Suppose f(x) is differentiable at all $x \in (-\infty, \infty)$ and the graph of $y = f'(x)$ is as shown in the figure below. Which of the following is true?



A. f(x) increases on $[3, \infty)$.
B. f(x) attains its minimum value at $x = 3$.
C. f(x) decreases on $[-1, 5]$.
D. f(x) attains its maximum value at $x = 5$.

Solution:

✓ f(x) increases in $[5, \infty)$ and $(-\infty, -1]$ where $f'(x) > 0$.

✓ f(x) attains its maximum at $x = -1$ since f '(x) shifts from positive to negative at $x = -1$

✓ f(x) attains its minimum value at $x = 5$ since shifts from negative to positive at $x = 5$.

∴ Answer C

34. Let $h(x) = x + 3^x$ and $F(x) = (foh)(x)$ such that $f'(4) = -1$ Then F'(1) is equal to:

A. $1 + 3\ln 3$ B. $-1 - 3\ln 3$ C. $4 + 12\ln 3$ D. $-4 - 12\ln 3$

Solution: $h(x) = x + 3^x$

$F(x) = f(h(x))$

$f'(4) = -1$

$f'(1) = ?$

$F'(x) = f'(h(x)) \times h'(x)$

But $h'(x) = (x + 3^x) = 1 + 3^x \ln 3$

So $F'(x) = f'(h(x)) \times (1 + 3^x \ln 3)$

$f'(1) = f'(h(1) \times (1 + 3^1 \ln 3)$

$f'(1) = f'(4) \times (1 + 3\ln 3)$

$f'(1) = -1(1 + 3\ln 3)$

$f'(1) = -1 - 3\ln 3$

∴ Answer B

35. If $f(x) = \ln(2 + \sec x)$, then f '(x) equal to:

A. $\frac{\tan x}{2+\sec x}$ B. $\frac{\tan x}{2+\cos x}$ C. $\frac{\cos x}{1+2\cos x}$ D. $\frac{\tan x}{1+2\cos x}$

Solution: $f'(x) = \left(\frac{1}{2+\sec x}\right)(2 + \sec x)'$

$f'(x) = \left(\frac{1}{2+\sec x}\right)(\sec x \tan x)$

$f'(x) = \left(\frac{1}{2+\sec x}\right) \times \left(\frac{1}{\cos x} \times \frac{\sin x}{\cos x}\right)$

$f'(x) = \left(\frac{1}{2+\sec x}\right) \times \left(\frac{\sin x}{\cos^2 x}\right)$

$f'(x) = \frac{\sin x}{\cos x}\left(\frac{1}{\cos x}\left(\frac{1}{2+\sec x}\right)\right)$

$f'(x) = \frac{\tan x}{2\cos x + \cos x \sec x} = \frac{\tan x}{2\cos x + 1}$

∴ Answer D

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36. Which of the following is the set of all critical numbers of $f(x) = \begin{cases} x-1, & x<1 \\ x^2-x, & x \geq 1 \end{cases}$?

A. {1} B. $\{\frac{1}{2}\}$ C. $\{\frac{1}{2}, 1\}$ D. { }

Solution: $\lim_{x\to 1^+} \frac{f(x)-f(1)}{x-1} = \lim_{x\to 1^+} \frac{x^2-x-0}{x-1} = \lim_{x\to 1^+} \frac{x(x-1)}{x-1}$

$= \lim_{x\to 1^+} x = 1$

and $\lim_{x\to 1^-} \frac{f(x)-f(1)}{x-1} = \lim_{x\to 1^-} \frac{x-1}{x-1} = \lim_{x\to 1^-} 1 = 1$

$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^-} f(x) = 1$

Thus f(x) has no critical numbers.

∴ ***Answer D***

37. The rate of certain chemical reaction is given by $r(x) = kx(a-x)$, for $0 \leq x \leq a$, $a > 0$ and $k > 0$.

For what values of x does the rate of reaction increase?

A. $0 \leq x \leq \frac{a}{2}$ C. $-\infty \leq x \leq \frac{a}{2}$

B. $\frac{a}{2} \leq x \leq a$ D. $\frac{a}{2} \leq x \leq \frac{3a}{4}$

Solution: Reaction rate increase on the intervals such that $r'(x) \geq 0$

$r(x) = kax - kx^2$

$r'(x) = ka - 2kx$

$r'(x) \geq 0$

$\Rightarrow ka - 2kx \geq 0$

$\Rightarrow -2kx \geq -ka$

$\Rightarrow x \leq \frac{-ka}{-2k}$

$\Rightarrow x \leq \frac{a}{2}$

But $x \geq 0$ is given by $0 \leq x \leq \frac{a}{2}$

∴ ***Answer A***

38. Let $f(x) = x^3 - 12x + 1$ be defined on [0, 3]. What is the range of f?

A. $(-\infty, \infty)$ B. $(-\infty, 1]$ C. [-15, 1] D. [-15, 17]

Solution: $f'(x) = 0 \Rightarrow 3x^2 - 12 = 0$

$\Rightarrow 3x^2 = 12 \Rightarrow x = \pm 2$

But, only $2 \in [0,3]$ is the critical number.

So $f(0) = 0^3 - 12(0) + 1 = 1$

$f(2) = 2^3 - 12(2) + 1 = -15$

$f(3) = 3^3 - 12(3) + 1 = -8$

∴ 1 is the local maximum while -15 is the local minimum.

So, the range of f(x) is [-15, 1]

∴ ***Answer C***

39. Let F(x) and G(x) be anti-derivatives of the function f(x) in the interval [a, b]. Which of the following is necessarily true?

A. $F(x) = G(x)$ for all $x \in [a, b]$.

B. $F(x) - G(x) = C$, where C is a constant, for all $x \in [a, b]$.

C. $\int f(x)dx = F(x) + G(x) + C$, where C is a constant,

D. $\int_a^b f(x)dx = G(b) - F(a)$.

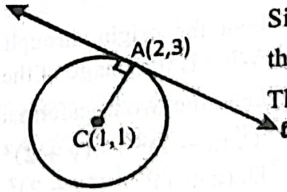
Solution: If F(x) and G(x) are anti-derivatives of f(x) in the interval [a, b] i.e $F(x) - G(x) = C$, where C is a constant for all $x \in [a, b]$.

∴ ***Answer B***

40. Consider a circle that passes through A = (2, 3) and centered at C(1,1). The equation of the line tangent to the circle at A is

A. $3x + y = 9$ B. $x + 2y = 8$ C. $3x - 2y = 0$ D. $x + y = 5$

Solution: Consider the following Figure



Thus slope of BO $= \frac{3-1}{2-1} = 2$

Since AC prpndicular to ℓ, the slope of the line ℓ is $-\frac{1}{2}$.

Thus equation of a line ℓ is given by:

$\frac{y-3}{x-2} = \frac{-1}{2}$

$\Rightarrow 2y - 6 = -x + 2$

$\Rightarrow 2y + x = 8$

∴ The equation of the line tangent to the circle at A is $2y + x = 8$

∴ ***Answer B***

41. If the equation of a line ℓ is given by the dot product: $(x-1, y+3).(2,-1) = 0$, which of the following is the equation of the circle with center at (-2,1) and ℓ is its tangent line?

A. $(x+2)^2 + (y-1)^2 = 2\sqrt{5}$ C. $(x+2)^2 + (y-1)^2 = 5\sqrt{2}$

B. $(x+2)^2 + (y-1)^2 = 10$ D. $(x+2)^2 + (y-1)^2 = 20$

Solution: Equation of a line given by the dot product

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$$(x-1, y+3)(2,-1) = 0$$
$$2(x-1) - 1(y+3) = 0$$
$$2x - 2 - y - 3 = 0$$
$$2x - y - 5 = 0$$

The perpendicular distance between the centre c and the line is called radius and is given by : $r = \frac{|AX_0 + BY_0 + C|}{\sqrt{A^2 + B^2}}$

Where (x_0, y_0) is the centre and A and B are coefficients of x_0 and y_0 and c is a constant in the equation format $Ax_0 + BY_0 + C = 0$

$$r = \frac{|AX_0 + BY_0 + C|}{\sqrt{A^2 + B^2}}$$
$$r = \frac{|2(-2) + (-1)(1) + (-5)|}{\sqrt{2^2 + (-1)^2}}$$
$$r = \frac{|-4-1-5|}{\sqrt{5}}$$
$$r = \frac{|-10|}{\sqrt{5}} = \frac{10}{\sqrt{5}} = 2\sqrt{5}$$

Therefore, the equation of the circle is $(x - x_0)^2 + (y - y_0)^2 = r^2$

$$\Rightarrow (x - (-2))^2 + (y - 1)^2 = (2\sqrt{5})^2$$
$$\Rightarrow (x + 2)^2 + (y - 1)^2 = 20$$

∴ Answer D

42. Suppose anti-clockwise rotation about the origin through 90° is followed by reflection in the x-axis. What is the image of the ellipse given by $\frac{1}{4}(x-1)^2 + (y+2)^2 = 1$ under the two transformations?

A. $4(x-2)^2 + (y+1)^2 = 4$ C. $(x-1)^2 + 4(y+2)^2 = 4$

B. $4(x-2)^2 + (y-1)^2 = 4$ D. $(x+1)^2 + 4(y-2)^2 = 4$

Solution: Image of the ellipse about the origin through 90° is followed by reflection in the x-axis.

Center of ellipse is C(1,-2) and a = 2 and b = 1. After 90° anticlockwise rotation C'(2,1) and a = 1, b = 2

Reflection through the x-axis C"(2,-1) and a = 1, b = 1

Therefore, equation of image ellipse is $\frac{(x-2)^2}{1^2} + \frac{(y+1)^2}{2^2} = 1$

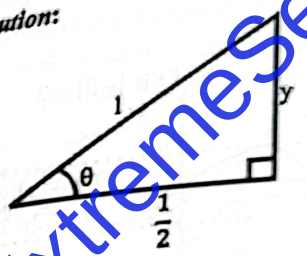
$$\Rightarrow 4(x-2)^2 + (y+1)^2 = 4$$

∴ Answer A

43. If $\theta = \arccos\left(\frac{1}{2}\right)$, what is the value of $\tan\theta$?

A. $\frac{1}{\sqrt{3}}$ B. $\sqrt{3}$ C. $\frac{\sqrt{3}}{4}$ D. 1

Solution:



$$\theta = \arccos\left(\frac{1}{2}\right) \Rightarrow \cos\theta = \frac{1}{2}$$
$$\left(\frac{1}{2}\right)^2 + y^2 = 1^2$$
$$y^2 = 1 - \frac{1}{4}$$
$$y^2 = \frac{3}{4}$$
$$y = \frac{\sqrt{3}}{2}$$
$$\therefore \tan\theta = \frac{y}{x} = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3}$$

∴ Answer B

44. If $x = \arcsin\left(\frac{3}{4}\right)$, the value of cos2x is

A. $\frac{3}{4}$ B. $\frac{-3}{4}$ C. $\frac{-1}{8}$ D. $\frac{1}{8}$

Solution: $x = \arcsin\left(\frac{3}{4}\right)$

$$\Rightarrow \sin x = \frac{3}{4}$$

Thus $\cos 2x = \cos^2 x - \sin^2 x$ ..... Equation(1)

But $\sin^2 x + \cos^2 x = 1$

$$\Rightarrow \cos^2 x = 1 - \sin^2 x$$ ..... Equation(2)

Substituting equation (2) in equation (1) we will get

$$\cos 2x = \cos^2 x - \sin^2 x$$
$$= 1 - \sin^2 x - \sin^2 x$$
$$= 1 - 2\sin^2 x$$
$$= 1 - 2\left(\frac{3}{4}\right)^2$$
$$= 1 - \frac{18}{16}$$
$$= \frac{-1}{8}$$

∴ Answer C

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45. Let $f(x) = -3\sin\left(\frac{1}{2}x\right)$. Which of the following is true?

A. The amplitude of the graph of f is -3.

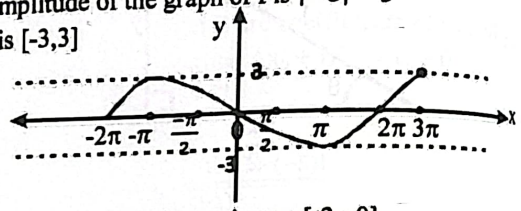
B. The range of f is $[-1,1]$

C. f is decreasing on $[-2\pi, 0]$

D. f is increasing on $[\pi, 3\pi]$

Solution: The amplitude of the graph of f is $|-3| = 3$

✓ The range of f is $[-3,3]$



f is neither increasing nor decreasing on $[-2\pi,0]$

From the graph above f is increasing on $[\pi,3\pi]$.

$\therefore$ ***Answer D***

46. Which of the following is equal to $\tan\left(x - \frac{\pi}{4}\right)$ if $\cos x \neq 0$?

A. $\frac{\sin x - \cos x}{\sin x + \cos x}$ B. $\frac{\cos x - \sin x}{\cos x + \sin x}$ C. $\frac{\sin x + \cos x}{\sin x - \cos x}$ D. $\frac{\cos x + \sin x}{\cos x - \sin x}$

Solution: Note: $\tan(a-b) = \frac{\tan a - \tan b}{1 + \tan a \tan b}$

Thus $\tan\left(x - \frac{\pi}{4}\right) = \frac{\tan x - \tan\frac{\pi}{4}}{1 + \tan x \tan\frac{\pi}{4}}$

$= \frac{\tan x - 1}{1 + \tan x}$

$= \frac{\frac{\sin x}{\cos x} - 1}{1 + \frac{\sin x}{\cos x}}$

$= \frac{\frac{\sin x - \cos x}{\cos x}}{\frac{\cos x + \sin x}{\cos x}}$

$= \frac{\sin x - \cos x}{\sin x + \cos x}$

$\therefore$ ***Answer A***

47. In the interval $[0, 2\pi)$, what is the solution set of the equation $\cos^2 x + \sin x \cos x = 1$?

A. $\left\{0, \frac{\pi}{4}, \pi, \frac{5\pi}{4}\right\}$ B. $\left\{0, \frac{\pi}{4}, \frac{3\pi}{4}, \pi\right\}$ C. $\{0, \pi\}$ D. $\left\{0, \frac{\pi}{4}, \pi\right\}$

Solution: $\cos^2 x + \sin x \cos x = 1$

$\Rightarrow \sin x \cos x = 1 - \cos^2 x \Rightarrow \sin x \cos x = \sin^2 x$

For $\sin x \neq 0$, $\cos x = \sin x$

$\Rightarrow x = \frac{\pi}{4}$ or $x = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$ on $[0, 2\pi)$

and for $\sin x = 0$, $x = 0$ or $x = \pi$ on $[0, 2\pi)$

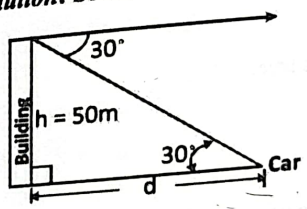
Therefore, the solution set is $\left\{0, \frac{\pi}{4}, \frac{5\pi}{4}, \pi\right\}$

$\therefore$ ***Answer A***

48. From the top of a building 50m high, the angle of depression of a parked car is 30°. How far is the car from the foot of the building measured horizontally?

A. $\frac{50}{\sqrt{3}}$m B. $55\sqrt{3}$m C. $50 + \sqrt{3}$m D. $50\sqrt{3}$m

Solution: See the following figure.



$\tan 30° = \frac{h}{d}$

$\frac{\sqrt{3}}{3} = \frac{50m}{d}$

$\sqrt{3}d = 50m \times 3$

$d = \frac{50m \times 3}{\sqrt{3}}$

$\Rightarrow d = 50\sqrt{3}$m

$\therefore$ The car far from the foot of the building is $50\sqrt{3}$m.

$\therefore$ ***Answer D***

49. The expression $\frac{\cos A}{1+\sin A} + \frac{\cos A}{1-\sin A}$ is equal to

A. 2cscA B. 2secA C. 2cosA D. 2sinA

Solution: $\frac{\cos A}{1+\sin A} + \frac{\cos A}{1-\sin A} = \frac{\cos A(1-\sin A) + \cos A(1+\sin A)}{(1+\sin A)(1-\sin A)}$

$= \frac{\cos A - \sin A \cos A + \cos A + \sin A \cos A}{1 - \sin^2 A}$

$= \frac{\cos A + \cos A}{\cos^2 A}$

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$$= \frac{2\cos A}{\cos^2 A}$$

$$= \frac{2}{\cos A}$$

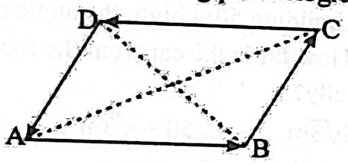
$$= 2\sec A$$

∴ ***Answer B***

50. Let ABCD be a parallelogram and $\overrightarrow{AC}$ and $\overrightarrow{BD}$ be its diagonals. Which of the following is **NOT** true?

A. $\overrightarrow{AB} = \overrightarrow{AC} - \overrightarrow{BC}$
B. $\overrightarrow{AB} = \overrightarrow{BC} - \overrightarrow{BD}$
C. $\overrightarrow{AC} + \overrightarrow{BD} = 2\overrightarrow{BC}$
D. $\overrightarrow{CD} + \overrightarrow{BD} = \overrightarrow{AD}$

Solution: Consider the following parallelogram.



✓ $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$

$\Rightarrow \overrightarrow{AB} = \overrightarrow{AC} - \overrightarrow{BC}$ .....is true.

✓ $\overrightarrow{AB} = \overrightarrow{BC} - \overrightarrow{BD}$

$\Rightarrow \overrightarrow{BC} = \overrightarrow{AB} + \overrightarrow{BD} = \overrightarrow{AD}$

$\Rightarrow \overrightarrow{BC} = \overrightarrow{AD}$..... Opposite side of a parallelogram

✓ $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$ and $\overrightarrow{BD} = \overrightarrow{BA} + \overrightarrow{AD}$

$\overrightarrow{AC} + \overrightarrow{BD} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{BA} + \overrightarrow{AD}$

$= \overrightarrow{BC} + \overrightarrow{AD}$

$= \overrightarrow{BC} + \overrightarrow{BC} = 2\overrightarrow{BC}$

✓ $\overrightarrow{CD} + \overrightarrow{BD}$ can never be equal to $\overrightarrow{AD}$

∴ ***Answer D***

51. Let A = (0, 1), B = (2,7) and C = (a, b) be points in the coordinate plane. What is (a, b) so that $\overrightarrow{AC} = \overrightarrow{AB} + 2\overrightarrow{CA}$?

A. $\left(\frac{1}{3}, 2\right)$ B. (1,3) C. (2,3) D. $\left(\frac{2}{3}, 3\right)$

Solution: $2\overrightarrow{CA} = 2(0 - a, 1 - b)$

$= 2(-a, 1 - b)$

$= (-2a, 2 - 2b)$

$\overrightarrow{AB} = (2 - 0, 7 - 1)$

$= (2,6)$

$\overrightarrow{AC} = (a - 0, b - 1)$

$= (a, b - 1)$

$\therefore \overrightarrow{AC} = \overrightarrow{AB} + 2\overrightarrow{CA}$

$(a, b - 1) = (2, 6) + (-2a, 2 - 2b)$

$\Rightarrow a = 2 - 2a$ and $b - 1 = 6 + 2 - 2b$

$\Rightarrow 3a = 2$ and $3b = 9$

$\Rightarrow a = \frac{2}{3}$ and $b = 3$

Therefore, $(a, b) = \left(\frac{2}{3}, 3\right)$

∴ ***Answer D***

52. If $\vec{U} = 2\vec{i} - \vec{j}$, $\|\vec{V}\| = 5$ and $\vec{U}$ and $\vec{V}$ are perpendicular vectors, then $\|\vec{U} + 3\vec{V}\|$ is equal to:

A. $\sqrt{55}$ B. $9\sqrt{5}$ C. $\sqrt{230}$ D. 20

Solution:

$$\|\vec{U} + 3\vec{V}\| = \sqrt{(\vec{U} + 3\vec{V}) \cdot (\vec{U} + 3\vec{V} \cdot)}$$

$$= \sqrt{\|U^2\| + 6\|\vec{U}\|\,\|\vec{V}\| \cos 90^0 + 9\|V^2\|}$$

$$= \sqrt{\sqrt{5}^2 + 0 + 9(5)^2} = \sqrt{230}$$

∴ ***Answer C***

53. A translation takes $x^2 + (y + 1)^2 = 5$ to $(x - 2)^2 + y^2 = 5$. What is the image of (1, 3) under this transformation?

A. (3,4) B. (3,2) C. (-1,2) D. (-2, 3)

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Solution: The center of $x^2 + (y+1)^2 = 5$ is (0,-1) and the center for $(x-2)^2 + y^2 = 5$ is (2, 0).

So (0,-1) $\xrightarrow{T}$ (2, 0)

That is the translation vector T = (2 – 0, 0 – (-1)) = (2, 1)

Therefore (1,3) $\xrightarrow[T]{(2,1)}$ (1+2, 3+1) = (3,4)

∴ The image of (1, 3) under this transformation is (3,4)

∴ ***Answer A***

54. Let ℓ be a line given by $x - 2y = 0$. A parametric vector equation of the line that passes through (5,0) and perpendicular to ℓ is:

A. (x, y) = (5,0) + t(1,2), t∈(-∞,∞)
C. (x,y) = (5 + t,-2t), t∈(-∞,∞)
B. (x,y) = (5,0) + t(1,-2), t≥0
D. (x,y) = (5−t,2t), t ≥ 0

Solution: ℓ: $x - 2y = 0$

$\Rightarrow 2y = x$

$\Rightarrow y = \frac{1}{2}x$

Hence a line perpendicular to ℓ could be y = -2x.

Since (1,-2) lies on y = -2x, the parametric vector equation required is (x, y) = (5, 0) + t(1,-2): t∈ℝ

That is (x, y) = (5+t, -2t), t∈ℝ

∴ A parametric vector equation of the line that passes through (5,0) and perpendicular to ℓ is (5 + t, -2t), t∈ℝ

∴ ***Answer C***

1. Which one of the following is the domain of $g(x) = x - \frac{(x^2+1)}{x^4+x}$?

A. ℝ\{0}
B. ℝ\{-1}
C. ℝ\{0,1}
D. ℝ\{-1,0}

Solution: $g(x) = x - \frac{(x^2+1)}{x^4+x}$

$g(x) = \frac{x(x^4+x) - (x^2+1)}{x^4+x}$

Domain of g(x) = ℝ\{$x^4 + x = 0$}

$\Rightarrow x^4 + x = 0$

$\Rightarrow x(x^3 + 1) = 0$

$\Rightarrow x = 0, x^3 + 1 = 0 \Rightarrow x^3 = -1$

$\Rightarrow x = -1$

∴ ***Answer D***

2. Which of the following is a simplified form of

$\frac{t-\frac{1}{t}}{1+\frac{t}{1-t}} \div \frac{t^2-2t+1}{t}$, $t \neq 0$ and $t \neq 1$?

A. t + 1
B. t − 1
C. 1 − t
D. −t − 1

Solution: $\frac{t-\frac{1}{t}}{1+\frac{t}{1-t}} \div \frac{t^2-2t+1}{t} = \frac{\frac{t^2-1}{t}}{\frac{(1-t)+t}{1-t}} \times \frac{t}{(t-1)^2}$

$= \frac{(t-1)(t+1)}{t} \times \frac{1-t}{1} \times \frac{t}{(t-1)^2}$

$= \frac{-(t-1)(t+1)}{t} \frac{(t-1)}{1} \times \frac{t}{(t-1)^2}, t \neq 0$

$= -\frac{(t-1)^2(t+1)}{(t-1)^2}, t \neq 1$

$= -t - 1$

∴ ***Answer D***

3. What are the respective values of A,B and C, so that $\frac{x-1}{x^3+x} = \frac{A}{x} + \frac{Bx+C}{x^2+1}$?

A. -1,1,1
B. -1,1,2
C. 1,-2,0
D. 3,1,2

Solution: $\frac{x-1}{x^3+x} = \frac{x-1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+c}{x^2+1}$

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$\Rightarrow x-1 = A(x^2+1) + (Bx+C)x$

$= Ax^2 + A + Bx^2 + Cx$

$= (A+B)x^2 + Cx + A$

$\Rightarrow A + B = 0, C = 1, A = -1$

$\Rightarrow A = -B$

$\Rightarrow B = 1$

$\therefore A = -1, B = 1, C = 1$

$\therefore$ ***Answer A***

4. What is the solution set of $1 - \frac{5}{x^2+4} = \frac{x-1}{x^2-x} - \frac{1}{x}$?

A. {1,-1} B. {2} C. {-1} D. {1,2,-1}

Solution: $1 - \frac{5}{x^2+4} = \frac{x-1}{x^2-x} - \frac{1}{x}$

$\Rightarrow 1 - \frac{5}{x^2+4} = \frac{1}{x} - \frac{1}{x}, x \neq 1$

$\Rightarrow 1 - \frac{5}{x^2+4} = 0, x \neq 0$

$\Rightarrow \frac{(x^2+4)-5}{x^2+4} = 0$

$\Rightarrow \frac{x^2-1}{x^2+4} = 0$

$\Rightarrow x^2 - 1 = 0 \Rightarrow x = 1, x = -1$

Solution set $= \{-1\}$

$\therefore$ ***Answer C***

5. Which of the following is NOT true about the graph of $g(x) = \frac{x^2-1}{x^2+1}$?

A. The range of g is $(-\infty, 1)$.

B. g is an even function.

C. The line $y = 1$ is a horizontal asymptote.

D. As $x \to -\infty$, $g(x) \to 1$.

Solution: $g(x) = \frac{x^2-1}{x^2+1}$. Let $g(x) = y$, then $y = \frac{x^2-1}{x^2+1}$

$\Rightarrow y(x^2+1) = x^2 - 1$

$\Rightarrow yx^2 + y = x^2 - 1$

$\Rightarrow yx^2 - x^2 = -y - 1$

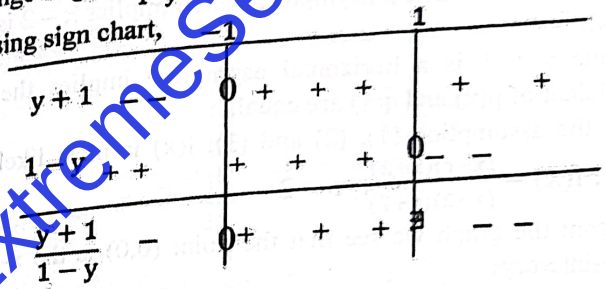
$\Rightarrow x^2(y-1) = -y - 1$

$\Rightarrow x^2 = \frac{-y-1}{y-1} = \frac{-(y+1)}{y-1} = \frac{y+1}{1-y}$

$\Rightarrow x = \pm\sqrt{\frac{y+1}{1-y}}$

✓ Range of g(x): $\frac{y+1}{1-y} \geq 0$

Using sign chart,



$\therefore$ Range of $g(x) = [-1, 1)$

✓ $g(-x) = \frac{(-x)^2-1}{(-x)^2+1} = \frac{x^2-1}{x^2+1} = g(x)$

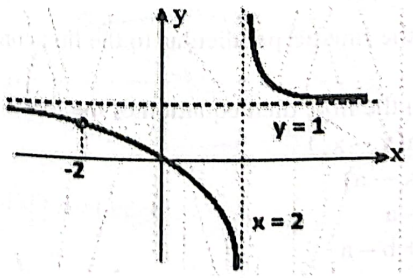
$\therefore$ g(x) is an even function.

✓ Since $g(x) = \frac{x^2-1}{x^2+1} = \frac{p(x)}{q(x)}$ and p(x) and g(x) have equal degree then the line $y = \frac{a}{b}$ which $\frac{\text{leading coefficient of p(x)}}{\text{leading coefficient of q(x)}}$ is the horizontal asymptote.

✓ $g(x) = \frac{x^2-1}{x^2+1} = \frac{\frac{x^2}{x^2} - \frac{1}{x^2}}{\frac{x^2}{x^2} + \frac{1}{x^2}} = \frac{1-\frac{1}{x^2}}{1+\frac{1}{x^2}}$. As $x \to -\infty$, $g(x) \to 1$.

$\therefore$ ***Answer A***

6. Which of the following functions could most likely be drawn as in the figure below?



A. $f(x) = \frac{x}{x-2}$

B. $f(x) = \frac{x^2+2x}{x^2-4}$

C. $f(x) = \frac{x+1}{x-2}$

D. $f(x) = \frac{x^3+2x}{x^2-4}$

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Solution: Let $f(x) = \frac{p(x)}{q(x)}$

From the figure we see that:

1. at x = -2, there is a hole which implies x + 2 is a factor of both p(x) and q(x).
2. at x = 2, there is vertical asymptote which implies x – 2 is a factor of q(x).
3. the line y = 1 is a horizontal asymptote implies the leading coefficient of p(x) and q(x) are equal.

From the assumption (1), (2) and (3), f(x) is most likely of the form : $f(x) = \frac{(x-r)(x+2)}{(x-2)(x+2)}, r \neq 2$

but from the graph we see that the point (0,0) is the x-intercept and y-intercept

$$\Rightarrow f(0) = \frac{(0-r)(0+2)}{(0-2)(0+2)}$$

$$\Rightarrow 0 = \frac{-2r}{4} \Rightarrow r = 0$$

$$\therefore f(x) = \frac{x(x+2)}{(x-2)(x+2)} = \frac{x^2+2x}{x^2-4}$$

∴ Answer B

7. Let (a,b) and (b,a) be points such that a ≠ b. Which of the following is the equation of the line through (a, b) and perpendicular to the line containing the given points?

A. y = x + b – a
B. y = x – a – b
C. y = -x + b + a
D. y = -x –a + b

Solution: (a, b) and (b, a), a≠b are given points.

Then, the slope of a line that passes through the given points is

$$m = \frac{a-b}{b-a} = -1.$$

Therefore, the slope of the line perpendicular to the line containing the given points is 1.

Since (a, b) is a point on the line, then equation of the line is

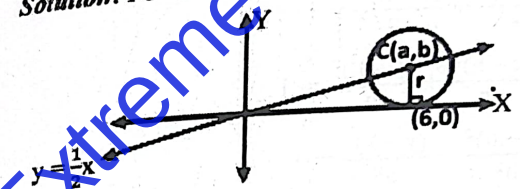
$$y - y_1 = m(x - x_1)$$
$$y - b = 1(x - a)$$
$$y - b = x - a$$
$$y = x + b - a$$

∴ Answer A

8. The equation of the circle tangent to the x-axis at (6,0) whose center is on the line $y = \frac{1}{2}x$ is

A. $(x-6)^2 + y^2 = 9$
B. $(x-6)^2 + (y-3)^2 = 36$
C. $(x-6)^2 + y^2 = 36$
D. $(x-6)^2 + (y-3)^2 = 9$

Solution: Point of tangency = (6, 0) and center on the line $y = \frac{1}{2}x$



Since r ⊥to the x –axis, then a = 6 and (a, b) is m then $y = \frac{1}{2}x$

$$\Rightarrow b = \frac{1}{2}a = \frac{1}{2} \times 6 = 3$$

∴ center = (6,3) and r = |b – 0| = |3 – 0| = 3

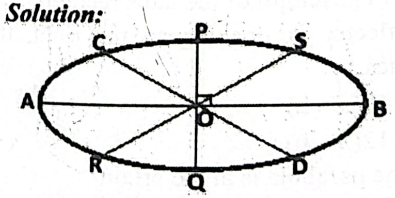
∴ Equation of the circle is $(x-6)^2 + (y-3)^2 = 9$

∴ Answer D

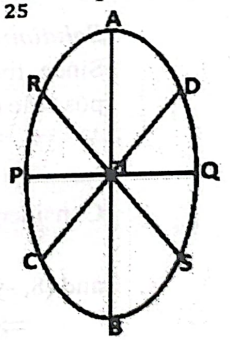
9. A mirror has an elliptic edge. If the shortest and longest distances between any two points on the edge of the mirror through the geometric center of the mirror are respectively 6cm and 10cm, which of the following is a possible equation of the edge of the mirror?

A. $\frac{x^2}{9} + \frac{(y-1)^2}{5} = 1$
B. $\frac{(x-1)^2}{25} + \frac{y^2}{9} = 1$
C. $\frac{(x+1)^2}{36} + \frac{y^2}{100} = 1$
D. $\frac{(x+1)^2}{36} + \frac{(y-1)^2}{25} = 1$

Solution:



Elliptical edge



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The segment $\overline{AB}$, $\overline{PQ}$, $\overline{CD}$, $\overline{RS}$ are chords that passes through the center. But $\overline{AB}$ and $\overline{PQ}$ are respectively the longest and the shortest chords.

Then, $AB = 2a = 10, \Rightarrow a = 5$

$PQ = 2b = 6, \Rightarrow b = 3$

$\Rightarrow$ The equation of the ellipse becomes a form of

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

$$\Rightarrow \frac{(x-h)^2}{25} + \frac{(y-k)^2}{9} = 1$$

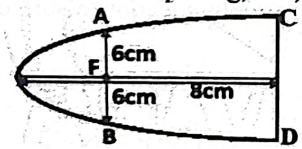
or $$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$$

$$\Rightarrow \frac{(x-h)^2}{9} + \frac{(y-k)^2}{25} = 1.$$

$\therefore$ From the given alternative the possible equation of the edge of the mirror is $\frac{(x-1)^2}{25} + \frac{y^2}{9} = 1$

$\therefore$ ***Answer B***

10. A cross-section of a parabolic reflector is shown in the figure below. The opening at the focus, AB, is 12cm. What is the diameter of the opening, CD, 8cm from the vertex?



A. $8\sqrt{6}$ cm
B. 12 cm
C. 9 cm
D. $6\sqrt{8}$ cm

Solution: $AB = 12 = 4p$ (The length of the latus rectum).

Since the parabolic reflector is opening to the right, then the possible equation becomes :

$\Rightarrow (y-k)^2 = 4p(x-h)$

$\Rightarrow (y-k)^2 = 12(x-h)$

Consider the vertex of the parabola is at the origin

$\Rightarrow y^2 = 12x$

and (8, -y) and (8, y) and the respective coordinates of D and C

$\Rightarrow CD = |y-(-y)| = 2y$

Since (8, y) on the parabola, then $y^2 = 12 \times 8$

$y^2 = 96 = 4\sqrt{6}$

$\Rightarrow CD = 2y = 2 \times 4\sqrt{6} = 8\sqrt{6}$

$\therefore$ ***Answer A***

11. The equation $3x^2 - 6x = y^2$ represents:
A. an ellipse with center at (1,0).
B. a hyperbola with center at (1,0).
C. an ellipse with center at (-1,0)
D. a hyperbola with one of its foci at (1,0).

Solution: $3x^2 - 6x = y^2$

$\Rightarrow 3(x^2 - 2x) - y^2 = 0$

$\Rightarrow 3\left(x^2 - 2x + \left(\frac{-2}{2}\right)^2\right) - y^2 = 3$

$\Rightarrow 3(x-1)^2 - y^2 = 3$

$\Rightarrow \frac{3(x-1)^2}{3} - \frac{y^2}{3} = 1$

$\Rightarrow \frac{(x-1)^2}{1} - \frac{y^2}{3} = 1$

It is a hyperbola with center (1,0)

$\therefore$ ***Answer B***

12. Which one of the following points is outside the sphere

$x^2 + y^2 + z^2 - 4z = 0$?

A. (0, -1, 1) B. (-2, 0, 2) C. (1, -1, 3) D. (-1, 2, 0)

Solution: Let $x^2 + y^2 + z^2 - 4z = 0$ is an equation of the sphere and (a, b, c) be any given point. When we substitute (a, b, c) in the given equation one of the condition is satisfied i.e

if $a^2 + b^2 + c^2 - 4c = 0$..... (a, b, c) is on the sphere.

if $a^2 + b^2 + c^2 - 4c < 0$..... (a, b, c) is inside the sphere.

if $a^2 + b^2 + c^2 - 4c > 0$..... (a, b, c) is outside the sphere.

Using this relation the point (-1, 2, 0) is outside the sphere i.e $(-1)^2 + 2^2 + 0^2 - 4 \times 0 = 5 > 0$.

$\therefore$ ***Answer D***

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13. Let $\{a_n\}_{n=1}^{\infty}$ be a sequence such that $a_{n+1} = -na_n$, $a_1 = 1$, n = 1, 2, 3, ... Which of the following is NOT true about the sequence?

A. $a_3 = 2$ C. $a_n > 0$ if n is odd

B. $a_4 = -3$ D. $a_n < 0$ if n is even

Solution: $a_{n+1} = -n\,a_n$, $a_1 = 1$, n = 1, 2, 3, ...

$\Rightarrow a_{1+1} = a_2 = -1.a_1 = -1$

$\Rightarrow a_{2+1} = a_3 = -2.a_2 = -2.-1 = 2$

$\Rightarrow a_{3+1} = a_4 = -3.a_3 = -3 \times 2 = -6$

$\Rightarrow a_{4+1} = a_5 = -4.a_4 = -4 \times (-6) = 24$ etc

Then some of the terms of the sequence are: 1, -1, 2, -6, 24, ...

Thus, $a_3 = 2$, $a_4 = -6$, $a_n > 0$ if n is odd and $a_n < 0$ if n is even.

(Hence a_n refers the sequence)

∴ ***Answer B***

14. Let $S_n = 4n^2$, n = 1, 2, 3, ... be the sum of the first n terms of a sequence $\{a_n\}_{n=1}^{\infty}$. Which of the following is the third term a_3?

A. 8 B. 12 C. 16 D. 20

Solution: $s_n = 4n^2$, n = 1,2,3, ...

$\Rightarrow \underbrace{a_1 + a_2 + \ldots + a_{n-1}}_{S_{n-1}} + a_n = 4n^2$

$\Rightarrow s_{n-1} + a_n = s_n$

$\Rightarrow a_n = s_n - s_{n-1}$

$\Rightarrow a_3 = s_3 - s_2$

$= 4.(3)^2 - 4.(2)^2$

$= 36 - 16$

$= 20$

∴ ***Answer D***

15. Consider the sequence $\{a_n\}_{n \geq 1}$, where the first term is $a_1 = 2$, and $a_n = 3a_{n-1}$ for all $n \geq 2$. Then, $\sum_{i=1}^{20} a_i$ is equal to

A. $3^{20} - 1$ B. $1 - \frac{1}{3^{20}}$ C. $1 - \frac{1}{2^{21}}$ D. $3^{-20} - 1$

Solution: $a_1 = 2$, $a_n = 3a_{n-1}$, $n \geq 2$

$\Rightarrow \frac{a_n}{a_{n-1}} = 3$

It is a geometric sequence with common ratio r = 3

$\Rightarrow \sum_{i=1}^{20} a_i = S_{20} = \frac{G_1(r^{20}-1)}{r-1}$

$= \frac{2(3^{20}-1)}{3-1}$

$= 3^{20} - 1$

∴ ***Answer A***

16. Suppose that every year the amount of atoms of a certain radioactive substance is half of its previous year. If there were 6.4×10^8 atoms of the substance during the initial observation, how many atoms of the substance are present by the end of 20 years after the initial observation?

A. 3.2×10^{-20} B. $2^{-20} \times 10^8$ C. $2^{-14} \times 10^9$ D. $2^{-14} \times 10^7$

Solution: Initial observation $a_0 = 6.4 \times 10^8$

After one year $a_1 = \frac{1}{2} a_0 = 3.2 \times 10^8$

After two years $a_2 = \frac{1}{2} a_1 = 1.6 \times 10^8$

After 20 years?

Since the sequence is geometric with common ratio $r = \frac{1}{2}$ then

$G_n = r^{n-1} G_1$, $n \in \mathbb{N}$

$\Rightarrow G_{20} = \left(\frac{1}{2}\right)^{20-1} \times 3.2 \times 10^8$

$= 2^{-19} \times 3.2 \times 10^8$

$= 2^{-19} \times 32 \times 10^7$

$= 2^{-19} \times 2^5 \times 10^7$

$= 2^{-14} \times 10^7$ atoms

or when we consider

$G_1 = a_0 = 6.4 \times 10^8$

$\Rightarrow G_{21} = \left(\frac{1}{2}\right)^{20} \times 6.4 \times 10^8$

$= 2^{-20} \times 64 \times 10^7$

$= 2^{-20} \times 2^6 \times 10^7$

$= 2^{-14} \times 2^6 \times 10^7$

$= 2^{-14} \times 10^7$ atoms

∴ ***Answer D***

17. Which of the following is equal to $\lim_{n\to\infty} \frac{n + \cos 3n}{2-n}$?

A. -1 B. 0 C. 1 D. 3

Solution: $\lim_{n\to\infty} \frac{n+\cos 3n}{2-n} = \lim_{n\to\infty} \frac{n}{2-n} + \lim_{n\to\infty} \frac{\cos 3n}{2-n}$

$= \lim_{n\to\infty} \frac{\frac{n}{n}}{\frac{2}{n} - \frac{n}{n}} + \lim_{n\to\infty} \frac{\cos 3n}{2-n}$

$= -1 + \lim_{n\to\infty} \frac{\cos 3n}{2-n}$

You know that $\lim_{n\to\infty} \cos 3n$ is occilating between -1 and 1 and $\lim_{n\to\infty} 2-n$ is $-\infty$.

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$\Rightarrow \lim_{n\to\infty} \frac{\cos 3n}{2-n} = 0$

$\therefore \lim_{n\to\infty} \frac{n+\cos 3n}{2-n} = -1$

∴ Answer A

18. Let $f(x) = \frac{\sqrt{x}-2}{\frac{4}{x}+1}$. Then, $\lim_{x\to 4^+} f(x)$ is equal to:

A. 0 B. $\frac{1}{4}$ C. 1 D. $-\infty$

Solution: $f(x) = \frac{\sqrt{x}-2}{\frac{-4}{x}+1}$

$\Rightarrow f(x) = \frac{\sqrt{x}-2}{\frac{-4+x}{x}} = \frac{x(\sqrt{x}-2)}{x-4}$

$\Rightarrow \lim_{x\to 4^+} f(x) = \lim_{x\to 4^+} \frac{x(\sqrt{x}-2)}{x-4}$

$= \lim_{x\to 4^+} x \times \lim_{x\to 4^+} \frac{\sqrt{x}-2}{x-4}$

$= 4\times \lim_{x\to 4^+} \frac{\sqrt{x}-2}{x-4} \times \frac{\sqrt{x}+2}{\sqrt{x}+2}$

$= 4\times \lim_{x\to 4^+} \frac{x-4}{(x-4)(\sqrt{x}+2)}$

$= 4\times \lim_{x\to 4} \frac{1}{\sqrt{x}+2}$

$= 4\times\frac{1}{4} = 1$

∴ Answer C

19. $\lim_{x\to 2} \frac{6x-12}{x^2-x-2}$ is equal to:

A. 0 B. 2 C. 1 D. ∞

Solution: $\lim_{n\to 2} \frac{6x-12}{x^2-x-2} = \lim_{n\to 2} \frac{6(x-2)}{(x+1)(x-2)}$

$= \lim_{n\to 2} \frac{6}{x+1} = \frac{6}{3} = 2$

∴ Answer B

20. Which of the following is equal to $\lim_{x\to\pi} \frac{x\sin 2x}{x-\pi}$?

A. 2π B. π C. 0 D. 1

Solution: $\lim_{x\to\pi} \frac{x\sin 2x}{x-\pi} = \lim_{x\to\pi} x \times \lim_{x\to\pi} \frac{\sin 2x}{x-\pi}$

$= \pi \times \lim_{x\to\pi} \frac{\sin 2x}{x-\pi}$

Let $u = x - \pi$

As $x \to \pi$, then $u \to 0$

$\Rightarrow \lim_{x\to\pi} \frac{\sin 2x}{x-\pi} = \lim_{u\to 0} \frac{\sin 2(u+\pi)}{u}$

$= \lim_{u\to 0} \frac{\sin(2u+2\pi)}{u}$

$= \lim_{u\to 0} \frac{(\sin 2u\cos 2\pi + \cos 2u \sin 2\pi)}{u}$

$= \lim_{u\to 0} \frac{\sin 2u}{u}$ (since $\cos 2\pi = 1$, $\sin 2\pi = 0$)

$= \lim_{u\to 0} \frac{2\sin 2u}{2u}$

$= 2\lim_{u\to 0} \frac{\sin 2u}{2u} = 2\times 1 = 2$

$\therefore \lim_{x\to\pi} \frac{x\sin 2x}{x-\pi} = \pi \lim_{x\to\pi} \frac{\sin 2x}{x-\pi} = 2\pi$

∴ Answer A

21. The left hand limit, $\lim_{x\to 0^-} \frac{|x|-2xe^x}{x}$ is equal to

A. 0 B. -2 C. -3 D. doesn't exist

Solution: $\lim_{x\to 0^-} \frac{|x|-2xe^x}{x}$

Since $|x| = \begin{cases} x, x \geq 0 \\ -x, x < 0 \end{cases}$, then

$\lim_{x\to 0^-} \frac{|x|-2xe^x}{x} = \lim_{x\to 0^-} \frac{-(x)-2xe^x}{x} = \lim_{x\to 0^-} \frac{-x(1+2e^x)}{x}$

$= -\lim_{x\to 0^-} (1+2e^x)$

$= -(1+2e^0)$

$= -3$

∴ Answer C

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22. Let $f(x)=\begin{cases}3a-x, \text{if } x<1\\ b+1, \text{if } x=1\\ ax, \text{if } x>1\end{cases}$

What is the value of b if f is continuous at $x=1$?

A. $-\frac{1}{2}$ B. $\frac{1}{2}$ C. -1 D. 3

Solution: $f(x)=\begin{cases}3a-x, \text{if } x<1\\ b+1, \text{if } x=1\\ ax, \text{if } x>1\end{cases}$

f is continuous at $x=1$ implies $\lim_{x\to1^-} f(x) = \lim_{x\to1^+} f(x) = f(1)$

$\Rightarrow \lim_{x\to1^-} 3a-x = \lim_{x\to1^+} ax = b+1$

From the 1st two equality first find the value of a

i.e $\lim_{x\to1^-} 3a-x = \lim_{x\to1^+} ax$

$3a-1=a$

$2a=1$

$\therefore a=\frac{1}{2}$

$\Rightarrow \lim_{x\to1^+} ax = b+1$

$\Rightarrow \lim_{x\to1^+} \frac{1}{2}x = b+1$

$\Rightarrow \frac{1}{2}=b+1$

$b=\frac{1}{2}-1$

$\therefore b=-\frac{1}{2}$

∴ Answer A

23. On which of the following intervals is f continuous, where

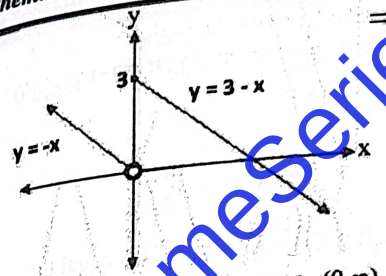
$f(x)=\begin{cases}\frac{x-x^2}{x-1}, & x<0\\ 3-x, & x\geq 0\end{cases}$

A. $(-\infty, 0]$ B. [-2, 1] C. [-1, 2] D. $[0, \infty]$

Solution: $f(x)=\begin{cases}\frac{x-x^2}{x-1}, & x<0\\ 3-x, & x\geq 0\end{cases}$

since, $\frac{x-x^2}{x-1}=\frac{x(1-x)}{x-1}=\frac{-x(x-1)}{x-1}=-x,\ x<0$

$\Rightarrow f(x)=\begin{cases}-x, & x<0\\ 3-x, & x\geq 0\end{cases}$



$\lim_{x\to0^+} f(x) = f(0)$ and for any $t\in(0,\infty)$, $\lim_{x\to t} f(x) = f(t) = 3-t$

$\therefore$ f is continuous on $[0, \infty)$

∴ Answer D

24. If $f(x)=x^5+2x$, then $\lim_{h\to0}\frac{f(1+h)-f(1)}{h}$ is equal to

A. 2 B. 7 C. 5 D. ∞

Solution: $f(x)=x^5+2x$

By definition of derivative, $f'(x)=\lim_{h\to0}\frac{f(x+h)-f(x)}{h}$

Then, $\lim_{h\to0}\frac{f(1+h)-f(1)}{h}$ implies $f'(1)$

$\Rightarrow f'(x)=5x^4+2$

$\Rightarrow f'(1)=7$

∴ Answer B

25. Let $f(x)=e^{3x-1}$ and $g(x)=\tan x$. If $F(x)=xf(x)+f(g(x))$, which of the following is equal to $F'(0)$?

A. $3e^{-1}$ B. $4e^{-1}$ C. $\frac{1}{e}\left(1+\frac{1}{e}\right)$ D. $\frac{1}{e}\left(1+\frac{3}{e}\right)$

Solution: $f(x)=e^{3x-1}$

$\Rightarrow f'(x)=(e^{3x-1})'=e^{3x-1}\times(3x-1)'=3e^{3x-1}$

$g(x)=\tan x$

$\Rightarrow g'(x)=(\tan x)'=\sec^2 x$

$F(x)=xf(x)+\{f(g(x))$

$\Rightarrow F'(x)=(xf(x))'+(f(g(x)))'$

$=x'f(x)+xf'(x)+f'(g(x))\times g'(x)$

$=f(x)+xf'(x)+f'(g(x))\,g'(x)$

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$= e^{3x-1} + 3xe^{3x-1} + 3e^{3\tan x-1} \times \sec^2 x$

$\Rightarrow F'(0) = e^{3.0-1} + 3.0.e^{3.0-1} + 3e^{3\tan 0-1} \times \sec^2 0$

$= e^{-1} + 0 + 3e^{3.0-1} \times 1$

$= e^{-1} + 3e^{-1}$

$= 4e^{-1}$

∴ ***Answer B***

26. Let $f(x) = g(\sqrt{4x+1})$. If $g'(3) = \frac{1}{2}$, then $f'(2)$ is equal to:

A. $\frac{2}{3}$ B. $\frac{1}{3}$ C. $\frac{1}{2}\sqrt{5}$ D. $\frac{1}{2}\sqrt{13}$

Solution:

$f(x) = g(\sqrt{4x+1})$. It is a composition of $g(x)$, $\sqrt{x}$ and $4x+1$.

$\Rightarrow f'(x) = g'(\sqrt{4x+1}) \times (\sqrt{4x+1})'$

$= g'(\sqrt{4x+1}) \times \frac{1}{2\sqrt{4x+1}} \times (4x+1)'$

$\Rightarrow f'(x) = g'(\sqrt{4x+1}) \times \frac{2}{\sqrt{4x+1}}$

$\Rightarrow f'(2) = g'(\sqrt{4\times 2+1}) \times \frac{2}{\sqrt{4\times 2+1}}$

$= g'(\sqrt{9}) \times \frac{2}{\sqrt{9}}$

$= g'(3) \times \frac{2}{3}$

$= \frac{1}{2} \times \frac{2}{3}$

$= \frac{1}{3}$

∴ ***Answer B***

27. Which of the following is the second derivative of $f(x) = \ln(x^2+1)$?

A. $\frac{1-2x^2}{x^2+1}$ B. $\frac{-2(x^2-1)}{x^2+1}$ C. $\frac{-2(x^2-1)}{(x^2+1)^2}$ D. $\frac{2-4x^2}{(x^2+1)^2}$

Solution:

$f(x) = \ln(x^2+1)$ .It is a composition of $y = \ln x$ and $y = x^2+1$

$f'(x) = \ln'(x^2+1) \times (x^2+1)'$

$f'(x) = \frac{1}{x^2+1} \times 2x$

$f'(x) = \frac{2x}{x^2+1}$

and $f''(x) = \left[\frac{2x}{x^2+1}\right]'$

$= \frac{(2x)'(x^2+1) - 2x(x^2+1)'}{(x^2+1)^2}$ ……Quotient rule

$= \frac{2(x^2+1) - 2x \times 2x}{(x^2+1)^2}$

$= \frac{2x^2+2-4x^2}{(x^2+1)^2}$

$= \frac{2-2x^2}{(x^2+1)^2}$

$= \frac{-2(x^2-1)}{(x^2+1)^2}$

∴ ***Answer C***

28. Let $f(x) = x|x|$, for all $x \in (-\infty, \infty)$. Which of the following is true?

A. f is not differentiable at $x = 0$.

B. $f'(x) = 2|x|$ at each $x \in (-\infty, \infty)$.

C. f is differentiable only on $(0, \infty)$

D. $f'(x) = 2x$ at each $x \in (-\infty, \infty)$.

Solution: Let $f(x) = x|x|$, $x \in (-\infty, \infty)$

Since $|x| = \begin{cases} x, x \geq 0 \\ -x, x < 0 \end{cases}$, then $x|x| = \begin{cases} x^2, x \geq 0 \\ -x^2, x < 0 \end{cases}$

$f'(x) = \begin{cases} 2x, & x > 0 \\ -2x, & x < 0 \end{cases}$

This implies f is not differentiable at $x = 0$

$\Rightarrow f'(x) = 2|x|$, $x \in \mathbb{R} \setminus \{0\}$

i.e f is differentiable on $(-\infty, 0)$ and $(0, \infty)$

∴ ***Answer A***

29. Let $f(x) = x^4 - 6x^2 + 1$ on $[-2, 3]$.

Which of the following is a local maximum value of f ?

A. f(0) B. $f(\sqrt{3})$ C. f(3) D. $f(-\sqrt{3})$

Solution: $f(x) = x^4 - 6x^2 + 1$, $[-2, 3]$

$\Rightarrow f'(x) = 4x^3 - 12x$

$\Rightarrow f'(x) = 4x[x^2 - 3]$

Since the graph of f(x) has a local maximum at the critical points, then the critical points are $f'(x) = 0$

$4x[x^2 - 3] = 0$

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$x = 0$ or $x^2 - 3 = 0$

$x = \pm\sqrt{3}$

$\Rightarrow f(0) = 0,\ f(-\sqrt{3}) = (-\sqrt{3})^4 - 6(-\sqrt{3})^2 + 1$

$= 9 - 18 + 1$

$= -8$

and $f(\sqrt{3}) = (\sqrt{3})^4 - 6(\sqrt{3})^2 + 1$

$= 9 - 18 + 1$

$= -8$

$\therefore$ Local maximum $= f(0) = 0$

But the absolute maximum lies either at the end points or at the critical points.

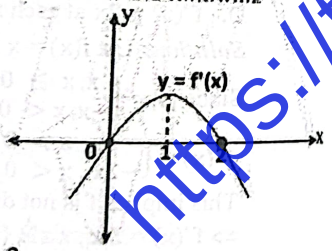
$f(-2) = (-2)^4 - 6(-2)^2 + 1 = 16 - 24 + 1 = -7$

$f(3) = 3^4 - 6(3)^2 + 1 = 81 - 54 + 1 = 28$

$\therefore$ Absolute maximum $= f(3) = 28$

$\therefore$ ***Answer A***

30. Suppose $f(x)$ is differentiable at all $x \in (-\infty, \infty)$ and the graph of $y = f'(x)$ is as shown in the figure below. Which of the following is true?

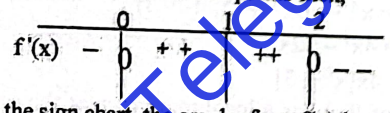


A. $f(x)$ is increasing on $(-\infty, 1)$.
B. $f(x)$ attains its maximum value at $x = 2$.
C. $f(x)$ attains its maximum value at $x = 1$.
D. $f(x)$ is decreasing on $(0, 2)$.

Solution: From the graph of $f'(x)$ we see the following relations.

$f'(0) = f'(2) = 0$

$\Rightarrow x = 0$ and $x = 2$ are critical points then,



From the sign chart, the graph of $y = f(x)$ is

- decreasing on $(-\infty, 0]$ and $[2, \infty)$

- increasing on $[0,2]$.
- has a local minimum at $x = 0$
- has an inflection point at $x = 1$
- has a local maximum at $x = 2$

$\therefore$ ***Answer B***

31. Let $f(x) = x^4 - \frac{16}{3}x^3 + 6x^2$ for all $x \in (-\infty, \infty)$. What is the range of f?

A. $(-\infty, \infty)$ B. $[0, \infty)$ C. $[-9, \infty)$ D. $\left(\frac{5}{3}, -9\right]$

Solution: $f(x) = x^4 - \frac{16}{3}x^3 + 6x^2,\ x \in (-\infty, \infty)$

$f'(x) = 4x^3 - 16x^2 + 12x$

$f'(x) = 4x[x^2 - 4x + 3]$

$f'(x) = 4x[x-1][x-3]$

$\Rightarrow$ critical points (numbers): $f'(x) = 0$

$\Rightarrow 4x(x-1)(x-3) = 0$

$\Rightarrow x = 0,\ x = 1,\ x = 3.$

i.e the graph of f has an upward turning at the points

$(0, f(0)),\ (1, f(1))$ and $(3, f(3))$

$\Rightarrow f(0) = 0,\ f(1) = 1^4 - \frac{16}{3} \times 1^3 + 6 \times 1^2 = 1 - \frac{16}{3} + 6 = \frac{5}{3}$

$\Rightarrow f(3) = 3^4 - \frac{16}{3} \times 3^3 + 6 \times 3^2 = 81 - 16 \times 9 + 54 = -9$

$\Rightarrow$ The minimum point is $(3, -9)$.

$\therefore$ Range of $f = [-9, \infty)$

$\therefore$ ***Answer C***

32. A farmer wants to make three identical rectangular enclosures (same width and same length) a long a straight river, as in the diagram shown below. If he has a fencing material of 1,200 meters and the sides along the river need no fence, what should be

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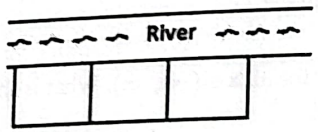
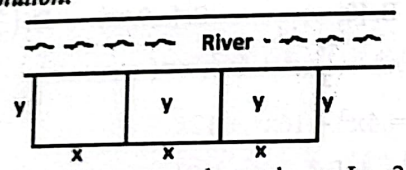
the dimensions (length l and width w) of each enclosure if the total area of the enclosures is to be maximized?
(Take the sides parallel to the river as the length of the enclosures)



A. $l = 100$ m, w = 225 m
B. $l = 300$ m, w = 75 m
C. $l = 200$ m, w = 200 m
D. $l = 200$ m, w = 150 m

Solution:



Length of the rectangular enclosure L = 3x
Width of the rectangular enclosure W = y
P = 3x + 4y = 1200m = length of fencing material
A(rectangular enclosure) = L × W

$$= 3x \times y$$
$$= 3x \times \left[\frac{1200 - 3x}{4}\right]$$
$$= \frac{3}{4}x \times 1200 - \frac{9}{4}x^2$$
$$A(x) = 900x - \frac{9}{4}x^2$$

To maximize area, $A'(x) = 900 - \frac{9}{2}x$

$$A'(x) = 0$$
$$\Rightarrow 900 - \frac{9}{2}x = 0$$
$$\Rightarrow 900 = \frac{9}{2}x$$
$$\Rightarrow x = 200\text{m}$$
$$\Rightarrow 3x + 4y = 1200\text{m}$$
$$600 + 4y = 1200\text{m}$$
$$4y = 600\text{m}$$
$$\Rightarrow y = 150\text{ m}$$

Dimensions of each enclosure is 200m by 150m.

∴ ***Answer D***

33. The radius of a right circular cylinder of height 10 cm increases at a rate of 3 cm/min. What is the rate of change of the volume of the cylinder when the radius is 8 cm?

A. 80π cm^3/min
B. 320π cm^3/min
C. 480π cm^3/min
D. 640π cm^3/min

Solution:

$\frac{dr}{dt} = 3\text{cm/min}$

$\frac{dv}{dt} = ?$ when r = 8

$V_{cylinder} = Bh$ where B = base area

$V = \pi r^2 h$ (The variables are v and r)

⇒ Differentiating both sides w.r.t time

$$\frac{dv}{dt} = 2\pi hr\frac{dr}{dt} = 2\pi hr\frac{dv}{dt}$$
$$\frac{dv}{dt} = 2\pi \times 10\text{cm} \times 8\text{cm} \times 3\text{cm/min}$$
$$\frac{dv}{dt} = 480\pi\text{cm}^3\text{/min}$$

∴ ***Answer C***

34. Which of the following is an antiderivative of $f(x) = \frac{3}{x\sqrt{x}}$?

A. $\frac{6}{\sqrt{x}} + c$ B. $-6\sqrt{x} + c$ C. $-\frac{6}{\sqrt{x}} + 1$ D. $-\frac{3}{2\sqrt{x}} + 3$

Solution: Let F(x) be the antiderivative of $f(x) = \frac{3}{x\sqrt{x}}$

$$\Rightarrow F(x) = \int \frac{3}{x\sqrt{x}}\,dx$$
$$= 3\int \frac{1}{x \times x^{1/2}}\,dx$$
$$= 3\int \frac{1}{x^{3/2}}\,dx$$
$$= 3\int x^{-3/2}dx$$
$$= \frac{3x^{-3/2+1}}{\frac{-3}{2}+1} + c$$

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$= \frac{3x^{\frac{1}{2}}}{\frac{1}{2}} + c$... Where C is any arbitrary constant

$= \frac{-6}{\sqrt{x}} + 1$

∴ *Answer C*

35. Which of the following is equal to $\int \frac{x^2-1}{x^3+x} dx$?

A. $\ln|x^2+1| + \ln|x| + c$ C. $\ln(x^2+1) - \ln|x| + c$

B. $\frac{1}{2}\ln(x^2+1) - \ln|x| + c$ D. $2\ln(x^2+1) - \ln|x| + c$

Solution: $\int \frac{x^2-1}{x^3+x} dx = \int \frac{x^2-1}{x(x^2+1)} dx$

$\frac{x^2-1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+c}{x^2+1}$ ..... Partial fraction

$= \frac{(Ax^2+A) + Bx^2 + Cx}{x(x^2+1)}$

$x^2 - 1 = (A+B)x^2 + Cx + A$

$\Rightarrow A + B = 1, C = 0, A = -1$

$\Rightarrow B = 2$

$\therefore \frac{x^2-1}{x(x^2+1)} = \frac{-1}{x} + \frac{2x}{x^2+1}$

$\Rightarrow \int \frac{x^2-1}{x(x^2+1)} dx = \int \frac{-1}{x} dx + \int \frac{2x}{x^2+1} dx$

$= -\int \frac{1}{x} dx + 2\int \frac{x}{x^2+1} dx$

$= -\ln|x| + C_1 + 2\int \frac{x}{x^2+1} dx$

Using substitution, let $u = x^2 + 1$

$\frac{du}{dx} = 2x \Rightarrow \frac{du}{2x} = dx$

$\Rightarrow 2\int \frac{x}{x^2+1} dx = 2\int \frac{x}{u} \times \frac{du}{2x} = \int \frac{1}{u} du = \ln|u| + C_2$

$= \ln|x^2+1| + C_2$

$\Rightarrow \int \frac{x^2-1}{x^3+x} dx = -\ln|x| + C_1 + \ln|x^2+1| + C_2$

$= \ln(x^2+1) - \ln|x| + C$, where C is any arbitrary constant.

∴ *Answer C*

36. $\int (2 + \sin^2 x)\cot x\, dx$ is equal to

A. $2\csc x - \sec^2 x + c$ C. $2\ln|\sin x| - \cos^2 x + c$

B. $2\ln|\sin x| + \frac{1}{2}\sin^2 x + c$ D. $2\csc x + \sin x \cos x + c$

Solution: $\int (2 + \sin^2 x)\cot x\, dx = \int 2\cot x dx + \int \sin^2 x \cot x\, dx$

$= 2\int \frac{\cos x}{\sin x} dx + \int \sin^2 x \times \frac{\cos x}{\sin x} dx$

$= 2\int \frac{\cos x}{\sin x} dx + \int \sin x \times \cos x dx$

Using substitution, let $u = \sin x$, then $\frac{du}{dx} = \cos x$

$= 2\int \frac{\cos x}{u} \times \frac{du}{\cos x} + \int u \times \cos x \cdot \frac{du}{\cos x}$

$= 2\int \frac{1}{u} du + \int u\, du$

$= 2\ln|u| + \frac{1}{2}u^2 + C$

$= 2\ln|\sin x| + \frac{1}{2}\sin^2 x + C$

∴ *Answer B*

37. Which of the following is equal to $\int_0^1 \frac{dx}{e^{-x}(1+e^x)}$?

A. $\ln(e-1)$ B. $\ln(1+e) - \ln 2$ C. $\ln(2+e)$ D. $\ln 2$

Solution: $\int_0^1 \frac{dx}{e^{-x}(1+e^x)} = \int_0^1 \frac{e^x}{1+e^x} dx$

Using integration by substitution, let $u = 1 + e^x$

$\frac{du}{dx} = e^x$

$\Rightarrow dx = \frac{du}{ex}$ and if $x = 0$, then $u = 2$

if $x = 1$, then $u = 1 + e$

$\Rightarrow \int_0^1 \frac{e^x}{1+e^x} dx = \int \frac{e^x}{u}\frac{du}{e^x} = \int_2^{1+e} \frac{1}{u} du$

$= \ln|u| \Big|_2^{1+e}$

$= \ln(1+e) - \ln 2$

∴ *Answer B*

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38. $\int_1^2 x\left(\frac{1}{x^2}+2\ln x\right)dx$ is equal to

A. $2\ln 2-1$ B. $5\ln 2+\frac{5}{2}$ C. $2\ln 2-\frac{5}{2}$ D. $5\ln 2-\frac{3}{2}$

Solution: $\int_1^2 x\left(\frac{1}{x^2}+2\ln x\right)dx=\int_1^2\frac{1}{x}dx+\int_1^2 2x\ln x\,dx$

$=\ln x\Big|_1^2+2\int_1^2 x\ln x\,dx$

$=(\ln 2-\ln 1)+2\int_1^2 x\ln x\,dx$

$=\ln 2+2\int_1^2 x\ln x\,dx$

Using integration by parts, Let $u=\ln x$, then $du=\frac{1}{x}dx$

$dv=xdx$ then $v=\int dv=\int xdx=\frac{x^2}{2}$

$\int x\ln x\,dx=\int udv=uv-\int vdu$

$=\frac{1}{2}x^2\ln x-\int\frac{1}{2}x^2\times\frac{1}{x}dx$

$=\frac{1}{2}x^2\ln x-\frac{1}{2}\int x\,dx$

$=\frac{1}{2}x^2\ln x-\frac{1}{2}\times\frac{x^2}{2}$

$=\frac{1}{2}x^2\ln x-\frac{1}{4}x^2$

$\Rightarrow 2\int_1^2 x\ln x\,dx=2\left[\frac{1}{2}x^2\ln x-\frac{1}{4}x^2\right]\Big|_1^2=\left[x^2\ln x-\frac{x^2}{2}\right]\Big|_1^2$

$=(4\ln 2-2)-\left(1\ln 1-\frac{1}{2}\right)$

$=4\ln 2-2+\frac{1}{2}$ .....since $\ln 1=0$

$=4\ln 2-\frac{3}{2}$

$\therefore\int_1^2 x\left(\frac{1}{x^2}+2\ln x\right)dx=\ln 2+4\ln 2-\frac{3}{2}$

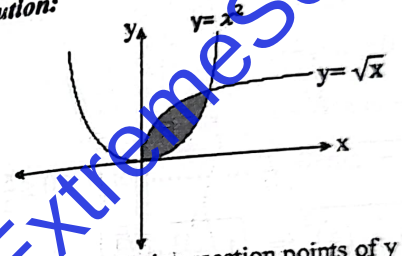
$=5\ln 2-\frac{3}{2}$

∴ Answer D

39. The area of the region bounded between the graph of $y=x^2$ and $y=\sqrt{x}$ is

A. 3 sq. unit B. $\frac{1}{3}$ sq. unit C. $\frac{2}{3}$ sq. unit D. 1sq. unit

Solution:



Let's first find the intersection points of $y=x^2$ and $y=\sqrt{x}$

$\Rightarrow x^2=\sqrt{x}$

$\Rightarrow x^4=x$ .....squaring both sides

$\Rightarrow x^4-x=0$

$\Rightarrow x(x^3-1)=0$

$\Rightarrow x=0, x^3-1=0$

$\Rightarrow(x-1)(x^2+x+1)=0$

$\Rightarrow x=1$ (x^2+x+1 can not be zero in real № system)

$\Rightarrow$ A(R) $=\int_0^1(f(x)-g(x))dx$ where $f(x)=\sqrt{x}$(the upper function) and $g(x)=x^2$ (the lower function)

$\Rightarrow A(R)=\int_0^1\left(\sqrt{x}-x^2\right)dx=\int_0^1\sqrt{x}\,dx-\int_0^1 x^2\,dx$

$=\int_0^1 x^{\frac{1}{2}}dx-\int_0^1 x^2\,dx$

$=\frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1}\Big|_0^1-\frac{x^{2+1}}{2+1}\Big|_0^1$

$=\frac{2}{3}x^{3/2}\Big|_0^1-\frac{x^3}{3}\Big|_0^1$

$=\frac{2}{3}-\frac{1}{3}=\frac{1}{3}$sq. unit

∴ Answer B

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40. If $\theta = \arcsin\left(\frac{1}{2}\right)$ then $\tan(2\theta)$ is equal to

A. $\frac{1}{\sqrt{3}}$ B. 1 C. $\sqrt{3}$ D. $\frac{\sqrt{3}}{2}$

Solution: $\theta = \arcsin\left(\frac{1}{2}\right)$

$\Rightarrow \sin\theta = \frac{1}{2}, 0 \le \theta \le \frac{\pi}{2}$

Since $\sin^2\theta + \cos^2\theta = 1$

$\Rightarrow \cos\theta = \sqrt{1 - \sin^2\theta}$ .....For $0 \le \theta \le \frac{\pi}{2}$

$= \sqrt{1 - \left(\frac{1}{2}\right)^2} = \sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}$

Then, $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}}$

$\Rightarrow \tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$

$= \frac{2 \times \frac{1}{\sqrt{3}}}{1-\left(\frac{1}{\sqrt{3}}\right)^2} = \frac{\frac{2}{\sqrt{3}}}{1-\frac{1}{3}} = \frac{\frac{2}{\sqrt{3}}}{2/3}$

$= \frac{2}{\sqrt{3}} \times \frac{3}{2} = \frac{3}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \sqrt{3}$

∴ ***Answer C***

41. What is the amplitude and period, respectively, of the graph of $f(x) = 2 - 3\sin\left(\frac{x+\pi}{4}\right)$?

A. $2, \frac{\pi}{4}$ B. $-3, 8\pi$ C. $3, \frac{\pi}{2}$ D. $3, 8\pi$

Solution: $f(x) = 2 - 3\sin\left(\frac{x+\pi}{4}\right)$

First re-write the function in the form $y = a\sin(kx + b) + c$

$f(x) = -3\sin\left(\frac{x+\pi}{4}\right) + 2$

Amplitude $A = |a| = |-3| = 3$ and Period $P = \frac{2\pi}{K} = \frac{2\pi}{\frac{1}{4}} = 8\pi$

∴ ***Answer D***

42. Which one of the following is a simplified form of $(1 + \tan^2 x)\sin\left(x - \frac{3}{2}\pi\right)$?

A. $\sec x$ B. $\cos x$ C. $\sec^2 x$ D. $-\cos x$

Solution: Using the identity $\sec^2 x = 1 + \tan^2 x$ and the difference formula, $\sin\left(x - \frac{3}{2}\pi\right) = \sin x \cos\frac{3}{2}\pi - \cos x \sin\frac{3}{2}\pi$

$= \cos x$

$(1 + \tan^2 x)\sin\left(x - \frac{3}{2}\pi\right) = \sec^2 x \cos x = \sec x$

∴ ***Answer A***

43. Which of the following is the solution set of $\sqrt{3}\sin 2x = \cos 2x$ on $[0,\pi]$?

A. $\left\{\frac{\pi}{6}, \frac{\pi}{12}\right\}$ B. $\left\{\frac{\pi}{6}, \frac{\pi}{3}\right\}$ C. $\left\{\frac{\pi}{12}, \frac{7}{12}\pi\right\}$ D. $\left\{\frac{\pi}{6}, \frac{2}{3}\pi\right\}$

Solution: $\sqrt{3}\sin 2x = \cos 2x$, $x \in [0,\pi]$

Let $\theta = 2x$, then $\theta \in [0, 2\pi]$

$\Rightarrow \sqrt{3}\sin 2x = \cos 2x$

$\Rightarrow \sqrt{3}\sin\theta = \cos\theta$

$\frac{\sqrt{3}\sin\theta}{\sqrt{3}\cos\theta} = \frac{\cos\theta}{\sqrt{3}\cos\theta} \Rightarrow \tan\theta = \frac{1}{\sqrt{3}}$

$\Rightarrow \theta = \tan^{-1}\frac{1}{\sqrt{3}}, \theta \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

$\Rightarrow \theta = \frac{\pi}{6}, \theta = \frac{7\pi}{6}$ and $\theta \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

Since $\theta = 2x$

$\frac{\pi}{6} = 2x \Rightarrow x = \frac{\pi}{12}$ and $\frac{7\pi}{6} = 2x \Rightarrow x = \frac{7\pi}{12}$

solution set $= \left\{\frac{\pi}{12}, \frac{7}{12}\pi\right\}$.

∴ ***Answer C***

44. A 1 m high pole is fixed vertically on the ground at a point $50\sqrt{3}$ m far from the foot of a building. If the angle of elevation from the top of the pole to the top of the building is 30°, what is the height of the building?

A. 50 m B. 51 m C. 150 m D. 151 m

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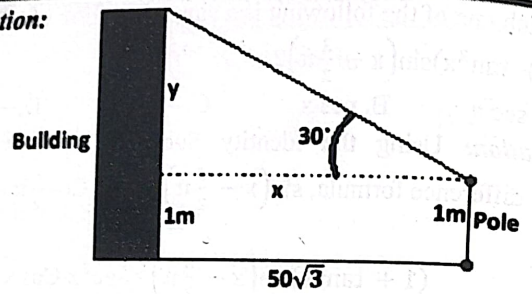
Solution:



Let the height of the building is h which is (y + 1)m

$\Rightarrow \tan 30° = \frac{y}{x} = \frac{y}{50\sqrt{3}}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{y}{50\sqrt{3}}$

$\Rightarrow y = 50m$

Height h = 51m

$\therefore$ ***Answer B***

45. If $\vec{A} = (3,-3)$ and $\vec{B} = (1,-3)$, what is the unit vector in the direction of $\vec{C} = 3\vec{A} - \vec{B}$?

A. $\left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$ C. $\left(\frac{4}{5}, \frac{-3}{5}\right)$

B. $\left(\frac{1}{\sqrt{10}}, \frac{-3}{\sqrt{10}}\right)$ D. $\left(\frac{-6}{10}, \frac{8}{10}\right)$

Solution: $\vec{A} = (3,-3)$, $\vec{B} = (1,-3)$

Then, $\vec{C} = 3\vec{A} - \vec{B}$

$= 3(3, -3) - (1, -3)$

$= (9, -9) - (1, -3)$

$= (9-1, -9+3)$

$= (8, -6)$

And $|\vec{C}| = \sqrt{8^2 + (-6)^2} = \sqrt{64 + 36} = 10$

A unit vector in the direction of $\vec{C} = 3\vec{A} - \vec{B}$ is

$$\frac{\vec{C}}{|\vec{C}|} = \frac{(8,-6)}{10} = \left(\frac{8}{10}, \frac{-6}{10}\right) = \left(\frac{4}{5}, \frac{-3}{5}\right)$$

$\therefore$ **Answer C**

46. If $\vec{u}$ is vector in the direction as vector $\vec{v} = (-3, 4)$ having length 3 times the length of $\vec{v}$, what is the length of the vector $\vec{u} - \frac{3}{2}\vec{v} - 2\vec{j}$, where $\vec{j}$ is the unit vector in the direction of the positive y-axis?

A. $\frac{1}{2}\sqrt{185}$ B. $\frac{1}{2}\sqrt{145}$ C. $\frac{21}{2}$ D. $\frac{17}{2}$

Solution:

$\vec{u}$ is in the direction of $\vec{v} = (-3,4)$ and $|\vec{u}| = 3|\vec{v}|$ implies $\vec{u} \parallel \vec{v}$.

Then, $\vec{u} = 3(-3, 4) = (-9, 12)$ and $j = (0,1)$

$\Rightarrow \vec{u} - \frac{3}{2}\vec{v} - 2j = (-9, 12) - \frac{3}{2}(-3, 4) - 2(0, 1)$

$= (-9, 12) + \left(\frac{9}{2}, -6\right) + (0,-2)$

$= \left(-9 + \frac{9}{2} + 0,\ 12 - 6 - 2\right)$

$= \left(\frac{-9}{2}, 4\right)$

$\Rightarrow \left|\vec{u} - \frac{3}{2}\vec{v} - 2j\right| = \sqrt{\left(\frac{-9}{2}\right)^2 + 4^2} = \sqrt{\frac{81}{4} + 16} = \frac{\sqrt{81+64}}{2} = \frac{1\sqrt{145}}{2}$

$\therefore$ ***Answer B***

47. If θ is the angle between the vectors $\vec{A} = 2\vec{i} - \vec{j}$ and $\vec{B} = \vec{i} + \vec{j}$, then $\cos\theta$ is equal to

A. $\frac{1}{\sqrt{10}}$ B. $\frac{1}{2}$ C. $\frac{\sqrt{3}}{2}$ D. $\frac{2}{\sqrt{5}}$

Solution: $\vec{A} = 2i - j$, $\vec{B} = i + j$

Let θ be the angle between $\vec{A}$ and $\vec{B}$, then

$$\cos\theta = \frac{\vec{A}.\vec{B}}{|\vec{A}||\vec{B}|} = \frac{(2i-j).(i+j)}{\sqrt{2^2+(-1)^2}\sqrt{1^2+1^2}}$$

$= \frac{2-1}{\sqrt{5}\sqrt{2}} \ldots\ldots\ldots\ldots$ (since $i.i = 1$, $i.j = 0$)

$= \frac{1}{\sqrt{10}}$

$\therefore$ ***Answer A***

48. Which of the following is the reflection of the parabola $y = x^2 + 1$ in the line $y = -x$?

A. $y = -x^2 - 1$ C. $x = y^2 + 1$

B. $y = x^2 - 1$ D. $x = -y^2 - 1$

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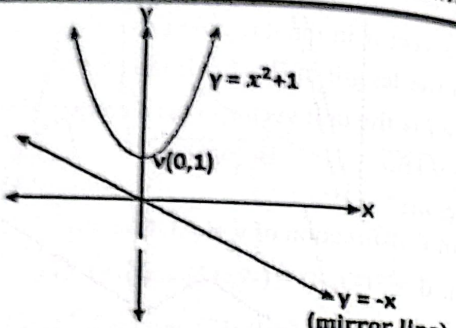
Solution:



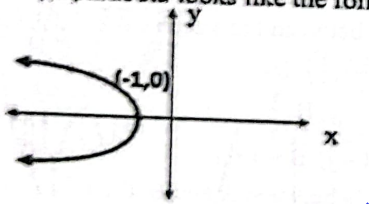
$V(0,1) \xrightarrow{y=-x} V'(-1,0)$ (i.e $(x, y) \xrightarrow{y=-x} (-y,-x)$)

Take any two points on the parabola and reflect along the line $y = -x$

$P(1,2) \xrightarrow{y=-x} P'(-2,-1)$

$Q(-1, 2) \xrightarrow{y=-x} Q'(-2,1)$

Then the image parabola looks like the following



Therefore, equation of the image parabola is

$(y-k)^2 = -4p(x-h)$

$\Rightarrow y^2 = -4p(x+1)$

Since the point (-2, 1) is on the parabola, then

$1^2 = -4p(-2+1)$

$1 = 4P$

$\Rightarrow P = \frac{1}{4}$

$\therefore y^2 = -4\left(\frac{1}{4}\right)(x+1)$

$y^2 = -x - 1$

$x = -y^2 - 1$

∴ Answer D

49. Consider the circle given by $x^2 + y^2 = 2$ and the line l given by parametric vector equation $(x, y) = (2,0) + t(-1,1)$. Which of the following is true?

A. l is tangent to the circle at $\left(\frac{1}{2}, \frac{\sqrt{7}}{2}\right)$.

B. l intersects with the circle at two distinct points.

C. l and the circle have no common point.

D. The distance from the center of the circle to l is $\sqrt{2}$.

Solution: Circle: $x^2 + y^2 = 2$

Center (0, 0) and radius $= \sqrt{2}$

Line ℓ: $(x, y) = (2,0) + t(-1,1)$ ... Vector equation of ℓ.

Let's convert to symmetric equation of ℓ

$\Rightarrow (x, y) = (2 - t, t)$

$\Rightarrow x = 2 - t, y = t$ .....Parametric equation of ℓ

$\Rightarrow x = 2 - y$

$\Rightarrow y = -x + 2$ .....symmetric equation of ℓ.

Intersection point of ℓ and the circle is $x^2 + (-x + 2)^2 = 2$

$\Rightarrow x^2 + (x^2 - 4x + 4) = 2$

$\Rightarrow 2x^2 - 4x + 2 = 0$

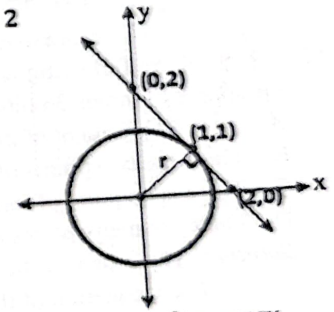
$\Rightarrow x^2 - 2x + 1 = 0$

$\Rightarrow (x - 1)^2 = 0$

$\therefore x = 1$

$\Rightarrow y = -x + 2$

$\therefore y = 1$



∴ Intersection point (1,1) which is the point of tangency.

Then $r = \sqrt{(1-0)^2 + (1-0)^2} = \sqrt{2}$ which is the distance from the center of the circle to the line ℓ.

∴ Answer D

50. If the midpoint of a line segment is M (-3,3,4) and one of the endpoints of the segment is at A(0,-1,3), what is the coordinates of its other endpoint?

A. (-6, 7, 5) B. (-3, 2, 7) C. (0, 1, 3) D. (6, -7, 5)

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Solution: Consider the end points of the segment are A(0,-1,3) and B(x, y, z). Then, the midpoint of the line segment is

$$M = \left(\frac{x+0}{2}, \frac{y-1}{2}, \frac{z+3}{2}\right)$$

$$\Rightarrow (-3, 3, 4) = \left(\frac{x}{2}, \frac{y-1}{2}, \frac{z+3}{2}\right)$$

$$\Rightarrow -3 = \frac{x}{2} \Rightarrow x = -6,\ 3 = \frac{y-1}{2} \Rightarrow y = 7, \text{ and } 4 = \frac{z+3}{2} \Rightarrow z = 5$$

$\therefore$ B(−6, 7, 5)

$\therefore$ ***Answer A***

51. If A(-1, 1, 0) and B(1, 3, 2) are endpoints of a diameter of a sphere, then what is the radius of the sphere?

A. 3 B. 6 C. $\sqrt{3}$ D. $\sqrt{12}$

Solution: A(-1, 1, 0) and B(1, 3, 2) are end points of a diameter

$$\Rightarrow d = \sqrt{(1-(-1))^2 + (3-1)^2 + (2-0)^2}$$

$$= \sqrt{4+4+4}$$

$$= \sqrt{12}$$

Since $r = \frac{1}{2}d = \frac{1}{2}\sqrt{12} = \frac{1}{2} \times 2\sqrt{3} = \sqrt{3}$

$\therefore$ ***Answer C***

52. Which of the following is true about the points of intersection of the sphere of radius 3 units with center at (-1, 3, 2) and the z-axis?

A. (0, 0, 3) is a point of intersection.
B. There are two points of intersection.
C. There is no point of intersection.
D. There are many points of intersection.

Solution: The center of the sphere C(-1,3,2)and r = 3.

Then, equation of the sphere becomes

$$(x+1)^2 + (y-3)^2 + (z-2)^2 = 3^2$$

$$\Rightarrow (x+1)^2 + (y-3)^2 + (z-2)^2 = 9$$

The Z-axis has coordinates (0,0,z) where $z \in \mathbb{R}$.

Thus, the intersection point is $(0+1)^2 + (0-3)^2 + (z-2)^2 = 9$

$$\Rightarrow 1 + 9 + (Z-2)^2 = 9$$

$$\Rightarrow 10 + (Z-2)^2 = 9$$

$$\Rightarrow (Z-2)^2 = 9 - 10$$

$$\Rightarrow (Z-2)^2 = -1$$

Since there is no real number whose square is negative. Therefore, the sphere and the z-axis have no point of intersection.

$\therefore$ ***Answer C***

53. If $(\neg P \vee Q) \Rightarrow (R \wedge \neg R)$ is true, then which of the following is necessarily true?

A. $R \Rightarrow Q$ B. $(Q \Rightarrow P) \wedge R$ C. $P \wedge \neg R$ D. $(Q \wedge R) \Rightarrow P$

Solution:

$(\neg P \vee Q) \Rightarrow (R \wedge \neg R)$ is True implies the following conditions are satisfied.

i) $\neg P \vee Q$ is T and $R \wedge \neg R$ is T
ii) $\neg P \vee Q$ is F and $R \wedge \neg R$ is T
iii) $\neg P \vee Q$ is F and $R \wedge \neg R$ is F

But the condition $R \wedge \neg R$ T is never and ever. So only we consider the 3rd condition.

$\neg P \vee Q$ is F means both $\neg P$ and Q are F.

Thus P is T and Q is F.

And $R \wedge \neg R$ is F means R has the possibility either T or F.

✓ $R \Rightarrow Q$
$T(F) \Rightarrow F$
$F(T)$

✓ $(Q \Rightarrow P) \wedge R$
$(F \Rightarrow T) \wedge T(F)$
$T \wedge T(F)$
T(F)

✓ $P \wedge \neg R$
$T \wedge T(F)$
T(F)

✓ $(Q \wedge R) \Rightarrow P$
$(F \wedge T(F)) \Rightarrow T$
$F \Rightarrow T$
T

$\therefore$ A, B and C are not necessarily true.

$\therefore$ ***Answer D***

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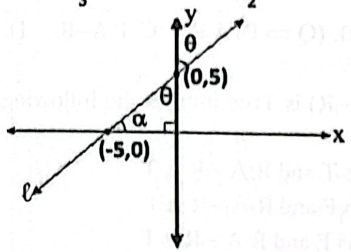
1. Line ℓ passes through (0, 5) and (-5,0). What is the angle between the y-axis and ℓ in radian measure ?

A. $\frac{\pi}{4}$ B. $\frac{\pi}{3}$ C. $\frac{\pi}{2}$ D. $\frac{3}{4}\pi$

Solution:



Let θ be the angle between ℓ and the y-axis.

$m_\ell = \frac{5-0}{0-(-5)} = \frac{5}{5} = 1$

let α be the angle between ℓ and the x-axis

$\Rightarrow \tan\alpha = \frac{m_1+m_2}{1-m_1m_2} = \frac{m_\ell + m_x}{1-m_\ell m_x} = \frac{1+0}{1-1\times 0} = 1$

$\Rightarrow \tan\alpha = 1$

$\alpha = \tan^{-1}(1) = \frac{\pi}{4}$

Using angle sum theorem

$\alpha + \theta + \frac{\pi}{2} = \pi$

$\frac{\pi}{4} + \theta + \frac{\pi}{2} = \pi$

$\theta = \pi - \frac{3}{4}\pi = \frac{\pi}{4}$. Since the question is the angle from the y-axis to ℓ, then $\pi - \frac{\pi}{4} = \frac{3}{4}\pi$

∴ Answer D

2. Consider a circle whose center is on the x-axis. If a line given by $y = x$ is tangent to the circle at point (2, 2), what is the equation of the circle?

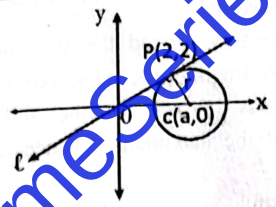
A. $x^2 + y^2 = 8$

B. $(x-2)^2 + y^2 = 4$

C. $(x-4)^2 + y^2 = 8$

D. $(x-1)^2 + y^2 = 5$

Solution:



Let the center of the circle be c(a, b). Then b = 0, and equation of a circle becomes $(x-a)^2 + y^2 = r^2$

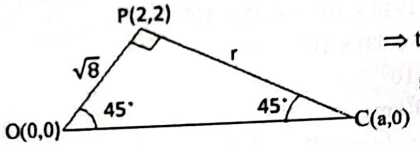
But r is the distance from the center to the point of contact p(2,2),

then $r^2 = (a-2)^2 + 4$

$\Rightarrow (x-a)^2 + y^2 = (a-2)^2 + 4$ ——————Equation1

Since the angle between the x-axis and the line ℓ is 45°, and the distance from O to P is $\sqrt{8}$.i.e

$OP = \sqrt{(2-0)^2 + (2-0)^2} = \sqrt{8}$



$\Rightarrow \tan 45° = \frac{r}{\sqrt{8}} \Rightarrow r = \sqrt{8}$

and $r^2 = (a-2)^2 + 4$

$8 = (a-2)^2 + 4$

∴ Equation of the circle becomes $\Rightarrow a = 4$

∴ $(x-4)^2 + y^2 = 8$

∴ Answer C

3. What is the vertex and the equation of the directrix, respectively, of the parabola $x + y^2 + 2y + 1 = 0$?

A. (0,-1), $x = \frac{-1}{4}$

B. (-1,0), $y = \frac{-1}{4}$

C. (0,-1), $x = \frac{1}{4}$

D. (-1,0), $y = \frac{1}{4}$

Solution: $x + y^2 + 2y + 1 = 0$

$y^2 + 2y = -x - 1$

$y^2 + 2y + 1 = -x - 1 + 1$

$(y+1)^2 = -x$

$(y-(-1))^2 = -(x-0)$. It is in the form of $(y-k)^2 = -4p(x-h)$

It is horizontal parabola opening to the left.

Vertex = (0,-1), $-4p = -1 \Rightarrow p = \frac{1}{4}$

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∴ **Answer C**

4. The orbit of Mercury around the sun forms an ellipse with eccentricity 0.206, length of the major axis 1.16×10^8 km and the sun at one focus. Which of the following is the best approximation of the maximum distance from Mercury to the sun?

A. 7.596×10^7km
B. 5.695×10^7km
C. 8.695×10^7km
D. 6.995×10^7km

Solution: $e = 0.206$

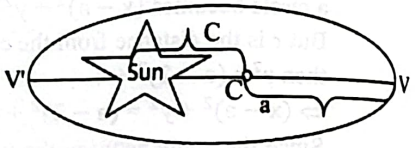
$2a = 1.16 \times 10^8$

$\Rightarrow a = 0.58 \times 10^8$

$e = \frac{c}{a} \Rightarrow 0.206$

$\Rightarrow c = 0.206 \times 0.58 \times 10^8$

$\Rightarrow c = 0.11948 \times 10^8$



The maximum distance from the vertex to the focus (sun) is

$d = c + a = 0.11948 \times 10^8 + 0.58 \times 10^8$

$= (0.11948 + 0.58) \times 10^8$

$= 0.69948 \times 10^8$

$= 6.995 \times 10^7$km

∴ **Answer D**

5. Suppose proposition $p \Rightarrow \neg q$ is false (F), which of the following is true?

A. $\neg q \wedge (p \Rightarrow q)$
B. $\neg p \vee (q \Rightarrow \neg p)$
C. $(\neg q \vee p) \Leftrightarrow q$
D. $(p \wedge q) \Leftrightarrow \neg p$

Solution:

$p \Rightarrow \neg q$ is False if and only if p is T and $\neg q$ is F, then, q = T.

A) $\neg q \wedge (p \Rightarrow q)$
$F \wedge (T \Rightarrow T)$
$F \wedge T$
F

B) $\neg p \vee (q \Rightarrow \neg p)$
$F \vee (T \Rightarrow F)$
$F \vee F$
F

C) $(\neg q \vee p) \Leftrightarrow q$
$(F \vee T) \Leftrightarrow T$
$T \Leftrightarrow T$
T

D) $(p \wedge q) \Leftrightarrow \neg p$
$(T \wedge T) \Leftrightarrow F$
$T \Leftrightarrow F$
F

∴ **Answer C**

6. If $x < 0$, then the simplest form of $f(x) = \frac{x - |x|}{x}$ is equal to:

A. 2x
B. 2
C. -2
D. 0

Solution: $f(x) = \frac{x-|x|}{x}$

Since $|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$

$\Rightarrow$ For $x < 0$, $f(x) = \frac{x-(-x)}{x} = \frac{2x}{x} = 2$

∴ **Answer B**

7. If $f(x) = \frac{\sqrt{x+2}}{x+2}$ and $g(x) = \frac{1}{x} - 2$, then f(g(x)) is equal to:

A. $\sqrt{x} - 2$
B. $\sqrt{x} + 2$
C $\sqrt{x}$
D. $\frac{\sqrt{x}}{x}$

Solution: $f(x) = \frac{\sqrt{x+2}}{x+2}$, $g(x) = \frac{1}{x} - 2$

$\Rightarrow f(g(x)) = \frac{\sqrt{g(x)+2}}{g(x)+2} = \frac{\sqrt{\left(\frac{1}{x} - 2\right) + 2}}{\left(\frac{1}{x} - 2\right)+2} = \frac{\frac{1}{\sqrt{x}}}{\frac{1}{x}} = \frac{1}{\sqrt{x}} \times \frac{x}{1}$

$= \frac{x}{\sqrt{x}} \times \frac{\sqrt{x}}{\sqrt{x}} = \sqrt{x}$

∴ **Answer C**

8. If $f(x) = \ln\left(\frac{x}{x-1} + 2\right)$, for $x > 1$, then which one of the following is the inverse of f ?

A. $g(x) = \frac{e^x - 2}{e^x - 3}$
B. $g(x) = \frac{e^x - 2}{e^x + 1}$
C. $g(x) = \frac{e^x}{e^x + 1} - 2$
D. $g(x) = \frac{x}{e^{x-1}} - 2$

Solution: $f(x) = \ln\left(\frac{x}{x-1} + 2\right)$, $x > 1$

Let $f(x) = y$

$\Rightarrow y = \ln\left(\frac{x}{x-1} + 2\right)$

$\Rightarrow X = \ln\left(\frac{y}{y-1} + 2\right)$ .....Interchange x and y.

$\Rightarrow$ solve for y

$e^x = \frac{y}{y-1} + 2$

$e^x (y - 1) = y + 2(y - 1)$

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$$e^x y - e^x = y + 2y - 2$$

$$ye^x - 3y = e^x - 2$$

$$y(e^x - 3) = e^x - 2$$

$$y = \frac{e^x - 2}{e^x - 3} = f^{-1}(x)$$

∴ ***Answer A***

9. Which of the following is a simplified form of

$$\frac{4-(a-2)^2}{a^4+2a^2+1} \div \frac{1-\frac{1}{a^2+1}}{a} \; ?$$

A. $\frac{4-a}{a}$ B. $-\frac{a}{a^2+1}$ C. $-\frac{1}{a^2(a^2+1)}$ D. $\frac{4-a}{a^2+1}$

Solution: $\frac{4-(a-2)^2}{a^4+2a^2+1} \div \frac{1-\frac{1}{a^2+1}}{a} = \frac{4-(a-2)^2}{(a^2+1)^2} \div \frac{(a^2+1)-1}{a(a^2+1)}$

$$= \frac{4-(a-2)^2}{(a^2+1)^2} \times \frac{a(a^2+1)}{a^2}, \; a \neq 0$$

$$= \frac{4-(a-2)^2}{a^2+1} \times \frac{1}{a} = \frac{4-(a^2-4a+4)}{a(a^2+1)}$$

$$= \frac{4a-a^2}{a(a^2+1)} = \frac{a(4-a)}{a(a^2+1)} = \frac{4-a}{a^2+1}, \; a \neq 0$$

∴ ***Answer D***

10. If $f(x) = \frac{x-1}{(x-2)^2(x+1)}$, which of the following is true about f ?

A. Its graph has an oblique asymptote.
B. The graph of f does not meet its asymptote.
C. As $x \to -1^+$, $f(x) \to -\infty$.
D. As $x \to 2^-$, $f(x) \to -\infty$.

Solution: $f(x) = \frac{x-1}{(x-2)^2(x+1)}$

Since the degree of the numerator is less than the degree of the denominator, then y = 0 is the horizontal asymptote. Therefore, the graph has no oblique asymptote. (i.e HA and OA does not occur at the same time.)

let y = f(x)

$$y = \frac{x-1}{(x-2)^2(x+1)}$$

$$0 = \frac{x-1}{(x-2)^2(x+1)}$$

$$\Rightarrow x - 1 = 0$$

$$x = 1$$

Therefore, the graph of f(x) intersects in its HA at the point (1,0).

$$\text{As } x \to -1^+, \; f(x) \to \frac{-1^+-1}{(-1^+-2)^2(-1^++1)} = \frac{-}{+\times 0^+} = -\infty$$

$$\text{As } x \to 2^-, \; f(x) \to \frac{2^--1}{(2^--2)^2(2^-+1)} = \frac{+}{0^+(+)} = \infty$$

∴ ***Answer C***

11. Let $A = \begin{pmatrix} -2 & 0 & x \\ 2y & x+y & -4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -y \\ 0 & 3 \\ 1-x & 2 \end{pmatrix}$ such that $A + 2B^T = 0$. Then which of the following is the value of y?

A. 0 B. $-\frac{13}{2}$ C. -8 D. any real number

Solution: $A = \begin{bmatrix} -2 & 0 & x \\ 2y & x+y & -4 \end{bmatrix}$, $B = \begin{pmatrix} 1 & -y \\ 0 & 3 \\ 1-x & 2 \end{pmatrix}$

$$A + 2B^T = 0$$

$$\Rightarrow \begin{pmatrix} -2 & 0 & x \\ 2y & x+y & -4 \end{pmatrix} + 2\begin{pmatrix} 1 & 0 & 1-x \\ -y & 3 & 2 \end{pmatrix} = 0$$

$$\Rightarrow \begin{pmatrix} -2 & 0 & x \\ 2y & x+y & -4 \end{pmatrix} + 2\begin{pmatrix} 2 & 0 & 2-2x \\ -2y & 6 & 4 \end{pmatrix} = 0$$

$$\Rightarrow \begin{pmatrix} -2+2 & 0+0 & x+2-2x \\ 2y-2y & x+y+6 & -4+4 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 0 & 0 & 2-x \\ 0 & x+y+6 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow 2 - x = 0 \qquad x + y + 6 = 0$$

$$x = 2 \qquad 2 + y + 6 = 0$$

$$y = -8$$

∴ ***Answer C***

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12. Let A and B be 3 × 3 matrices such that

$$A=\begin{bmatrix}2 & 0 & 0\\ 1 & 5 & 0\\ 0 & -1 & \frac{1}{2}\end{bmatrix} \text{ and } |B|=\frac{1}{10}$$

Which one of the following is equal to $|2AB^T|$?

A. 1 B. 4 C. 100 D. 400

Solution: $A=\begin{bmatrix}2 & 0 & 0\\ 1 & 5 & 0\\ 0 & -1 & \frac{1}{2}\end{bmatrix}$ and $|B|=\frac{1}{10}$

$$\Rightarrow |2AB^T| = 2^3|AB^T| = 8|A||B^T| = 8|A||B|$$

$$\text{But } |A| = 2\begin{vmatrix}5 & 0\\ -1 & \frac{1}{2}\end{vmatrix} - 0\begin{vmatrix}1 & 0\\ 0 & \frac{1}{2}\end{vmatrix} + 0\begin{vmatrix}1 & 5\\ 0 & -1\end{vmatrix}$$

$$= 2\left(5\times\frac{1}{2}\right) = 5$$

$$\Rightarrow 8|A||B| = 8\times 5\times\frac{1}{10} = 4$$

∴ ***Answer B***

13. A three-digited library identification card is to be printed from the numbers 0,1,2,3,4,5 in such a way that the first is non-zero and no number is to be repeated. How many such cards can be printed?

A. 100 B. 120 C. 150 D. 180

Solution:

H T U ←digit

5	5	4

№ of possibilities

By the fundamental Principle of counting, the total number of cards becomes 5×5×4 = 100

∴ ***Answer A***

14. A student needs to select 3 books from 3 different mathematics, 3 different physics and 1 history book. What is the probability that one of them is mathematics and the other two are either physics or history books?

A. $\frac{3}{35}$ B. $\frac{9}{35}$ C. $\frac{15}{35}$ D. $\frac{18}{35}$

Solution:

Subject	Mathematics	Physics	History
№ of books	3	3	1

$$\Rightarrow \text{№ of sample space} = {}^7C_3 = \frac{7!}{3!(7-3)!} = \frac{7!}{3!4!}$$

$$= \frac{7\times 6\times 5\times 4!}{3\times 2\times 4!} = 35$$

№ of events = $3C_1.4C_2$

$$= \frac{3!}{1!2!}\times\frac{4!}{2!2!} = \frac{3\times 2!}{2!}\times\frac{4\times 3\times 2!}{2\times 2!} = 18$$

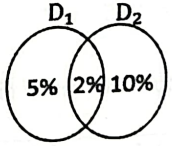
$$P(E) = \frac{n(E)}{n(S)} = \frac{18}{35}$$

∴ ***Answer D***

15. Items produced by a certain company are subjected to two kinds of defects D_1 and D_2. Out of the total production, if 5% have defect D_1, 10% have defect D_2 and 2% have both defects, then what is the probability for an item to have defect D_2, given that it has defect D_1.

A. 0.2 B. 0.05 C. 0.1 D. 0.4

Solution:



$$\Rightarrow P(D_1 \cap D_2) = P(D_1)\times P(D_2|D_1) = P(D_2)\times P(D_1|D_2)$$

Since probability of D_2 given that D_1 means $P(D_2|D_1)$

$$\Rightarrow P(D_2|D_1) = \frac{P(D1\cap D2)}{P(D_1)} = \frac{\frac{2}{17}}{\frac{5}{17}} = \frac{2}{5} = 0.4$$

∴ ***Answer D***

16. You are given a data on the age of students, in a primary school.

Age	8	10	11	12	13
Number of students	5	15	8	10	2

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Which of the following is NOT true about the data?

A. The median is 10.5 C. The mean is 10.5

B. The mode is 10 D. The range is 5

Solution:

Age	8	10	11	12	13
№ of students	5	15	8	10	2

Mode = most frequent value = 10

Range = maximum value – maximum value

$= 13 - 8 = 5$

$$\text{Mean} = \frac{8\times5+10\times15+11\times8+12\times10+13\times2}{5+15+8+10+2}$$

$$= \frac{40+150+88+120+26}{40}$$

$$= \frac{424}{40} = 10.6$$

Total number of students = 40

$$\Rightarrow \text{Median} = \frac{\left(\frac{40}{2}\right)^{th} + \left(\frac{40}{2}+1\right)^{th}}{2}$$

$$= \frac{20^{th}\text{item} + 21^{th}\text{item}}{2}$$

$$= \frac{10+11}{2}$$

$$= \frac{21}{2} = 10.5$$

∴ ***Answer C***

17. The following is a table of simple frequency distribution of a data with variable x.

x	1	3	4	5	7
Frequency	2	5	6	5	2

The standard deviation of the data is equal to:

A. $\sqrt{3}$ B. 3 C. $\sqrt{2.3}$ D. $\sqrt{3.6}$

Solution:

X	1	3	4	5	7
Frequency	2	5	6	5	2

$$\sigma = \sqrt{\frac{\sum fi\,(xi-\bar{x})^2}{\sum fi}}, \quad \bar{X} = \frac{1\times2+3\times5+4\times6+5\times5+7\times2}{2+5+6+5+2}$$

$$= \frac{2+15+24+25+14}{20}$$

$$= 4$$

x	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$fi(x_i - \bar{x})^2$
1	-3	9	18
3	-1	1	5
4	0	0	0
5	1	1	5
7	3	9	18
			Sum = 46

$$\Rightarrow \sigma = \sqrt{\frac{46}{20}} = \sqrt{\frac{23}{10}} = \sqrt{2.3}$$

∴ ***Answer C***

18. Suppose e(x) ≡ x is even; p(x) ≡ x is prime; d(x) ≡ x is divisible by 2. Which one of the following has the truth value F on the set of natural numbers?

A.$(\exists x)\,[e(x) \wedge p(x)]$ C. $(\exists x)\,[e(x) \wedge \neg\, d(x)]$

B.$(\forall x)\,[e(x) \Rightarrow d(x)]$ D. $(\forall x)\,[e(x) \vee \neg\, d(x)]$

Solution: e(x) ≡ x is even

p(x) ≡ x is prime

d(x) ≡ x is divisible by 2.

✓ $(\exists x)[e(x) \wedge p(x)]$ ..... True. x = 2.

✓ $(\forall x)(e(x) \Rightarrow d(x))$..... For every x, if x is even, then x is divisible by 2.....True.

✓ $(\exists x)[e(x) \wedge \neg d(x)]$..... There exist at least one x such that x is even and x is not divisible by 2. Which is false.

✓ $(\forall x)(e(x) \vee \neg d(x)) \equiv (\forall x)(\neg d(x) \vee e(x))$

$\equiv (\forall x)[d(x) \Rightarrow e(x)]$

✓ For every x, if x is divisible by 2; then x is even which is True.

∴ ***Answer C***

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19. Consider the following argument form. Production is high if rain continues. Rain does not continue. Therefore, either production is low or rain continues.

Let p: production is low

q: rain continues

The following table is also given about p and q.

Rows	p	q	¬p	¬q	¬p⇒q	q⇒¬p	p∨q
1	T	T	F	F	T	F	T
2	T	F	F	T	T	T	T
3	F	T	T	F	T	T	T
4	F	F	T	T	F	T	F

Which of the following is necessarily true?

A. The argument form is valid due to row 2.

B. The argument form is valid due to rows 2 and 3.

C. The argument form is invalid due to row 4.

D. The argument form is invalid due to rows 1 and 3.

Solution: Let P: production is low

q: rain continues.

Argument form: $\dfrac{\text{production is high if rain continues}\quad q\Rightarrow\neg p}{\text{rain does not continue}\quad \neg q}$

$\therefore$ either production is low or rain continues $\equiv p\vee q$

$\neg q = T$ .....Premise

$q = F$ .... Definition of $\neg$

$q \Rightarrow \neg p = T$ ..... Premise

$\neg p = T(F)$ .....Definition of $\Rightarrow$

$p = F(T)$ ..... Definition $\neg$

$\therefore$ If $p = F$, the conclusion is invalid

$\therefore$ The argument is invalid or using truth table

p	q	¬p	¬q	q⇒¬p	p∨q
T	T	F	F	F	T
T	F	F	T	T	T
F	T	T	F	T	T
F	F	T	T	T	F

$\therefore$ Invalid argument.

$\therefore$ ***Answer C***

20. A school has three classrooms for grade 11, namely, 11A, 11B and 11C. The number of students in these classrooms is 28, 20 and 22, respectively. All the students took an examination and the average score of the students of 11A, 11B and 11C is 60, 70 and 70, respectively. What is the average score in this examination for all grade 11 students?

A. 66 B. 66.67 C. 65 D. 65.67

Solution:

Section	11A	11B	11C
No. of students	28	20	22
Average($\bar{x}$)	60	70	70

$\Rightarrow$ For 11A, $\bar{x}_A = \dfrac{\sum x_A}{28}$

$60 = \dfrac{\sum x_A}{28}$

$\sum x_A = 1680$

For11 B, $\bar{x}_B = \dfrac{\sum x_B}{20}$

$70 = \dfrac{\sum x_B}{20}$

$\sum x_B = 1400$

For 11C, $\bar{x}_C = \dfrac{\sum x_C}{22}$

$70 = \dfrac{\sum x_C}{22}$

$\sum x_C = 1540$

$\Rightarrow$ For all grade 11 students

$$\bar{x} = \frac{\sum x}{n} = \frac{\sum x_A + \sum x_B + \sum x_C}{28 + 20 + 22}$$

$$= \frac{1680+1400+1540}{70}$$

$$= \frac{4620}{70} = 66$$

$\therefore$ ***Answer A***

21. If $\begin{pmatrix} \alpha & 2 & \beta \\ 2 & 1 & 3 \\ -1 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -5 \\ 5 \\ 0 \end{pmatrix}$ and the determinant of the coefficient matrix is -5, then the value of x is equal to

A. 3 B. $\alpha+\beta$ C. -5α D. 5

Solution:

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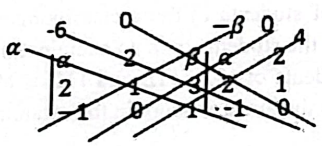
$$\begin{pmatrix} \alpha & 2 & \beta \\ 2 & 1 & 3 \\ -1 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -5 \\ 5 \\ 0 \end{pmatrix}$$



$\Rightarrow (\alpha - 6) - (4 - \beta) = -5$

$\Rightarrow \alpha + \beta - 10 = -5$

$\therefore \alpha + \beta = 5$

$-x + z = 0$

$z = x$

$2x + y + 3z = 5$

$2x + y + 3x = 5$

$5x + y = 5$ ..... *

$\alpha x + 2y + \beta z = -5$

$\alpha x + 2y + \beta x = -5$

$(\alpha+\beta)x + 2y = -5$

$5x + 2y = -5$ ..... **

From * and **

$$\begin{cases} 5x + y = 5 \\ 5x + 2y = -5 \end{cases}$$

$-y = 10$

$y = -10$

By substituting in either of equation $x = 3$

∴ Answer A

22. Which one of the following is the simplest form of $\frac{4-3i}{3+4i} + \overline{1-2i}$?

A. 1+i B. 1+3i C. 2−i D. 1 − 3i

Solution: $\frac{4-3i}{3+4i} + \overline{1-2i} = \frac{4-3i}{3+4i} \times \frac{3-4i}{3-4i} + 1+2i$

$= \frac{12-16i-9i+12i^2}{3^2+4^2} + 1 + 2i$

$= \frac{-25i}{25} + 1 + 2i$

$= -i + 1 + 2i$

$= i + 1$

∴ Answer A

23. Let $a_n = n + \cos n\pi$, n=1,2,3,... be the n^{th} term of a sequence. Then which of the following is true?

A. The fifth term is 6 C. $a_7 + a_8 = 15$

B. The 10^{th} term is 9 D. $a_7 - a_6 = 1$

Solution: $a_n = n + \cos n\pi$, n = 1,2,3,...

A) $a_5 = 5 + \cos 5\pi$

$= 5 + (-1) = 4$

B) $a_{10} = 10 + \cos 10\pi$

$= 10 + \cos 2(5\pi)$

$= 10 + 0 = 10$

C) $a_7 = 7 + \cos 7\pi$

$= 7 + \cos(6\pi + \pi)$

$= 7 + \cos\pi$

$= 6$

$a_8 = 8 + \cos 8\pi$

$= 8 + \cos 2(4\pi)$

$= 8 + 1 = 9$

$a_7 + a_8 = 6 + 9 = 15$

$a_6 = 6 + \cos 6\pi$

$= 6 + \cos 2(3\pi)$

$= 6 + 1 = 7$

D) $\therefore a_7 - a_6 = 6 - 7 = -1$

∴ Answer C

24. What is the sum of the series $\sum_{n=0}^{\infty} 2^n 3^{2-n}$?

A. 6 B. 18 C. 27 D. ∞

Solution: $\sum_{n=0}^{\infty} 2^n 3^{2-n}$

$2^n 3^{2-n} = 2^n \times 3^2 \times 3^{-n}$

$= \frac{2^n}{3^n} \times 9$

$= 9\left(\frac{2}{3}\right)^n$ which is a convergent geometric series

with $r = \frac{2}{3}$ and $G_1 = 9$

$\Rightarrow S_\infty = \frac{G_1}{1-r} = \frac{9}{1-\frac{2}{3}} = \frac{9}{\frac{1}{3}} = 27$

∴ Answer C

25. If $\{A_n\}$ is an arithmetic sequence with first term $A_1 = 5$ and fifth term $A_5 = 21$, then the partial sum $\sum_{n=1}^{20} A_n$ is equal to

A. 760 B. 780 C. 860 D. 870

Solution: $A_1 = 5, A_5 = 21$

Since the sequence is Arithmetic, then

$A_n = A_1 + (n-1)\,d$

$\Rightarrow A_5 = A_1 + 4d$

$\Rightarrow 21 = 5 + 4d$

$\Rightarrow d = 4$ .....The common difference

and $S_n = \frac{n}{2}[2A_1 + (n-1)d]$

$S_{20} = \frac{20}{2}[2\times 5 + (20-1)\times 4] = 860$

∴ Answer C

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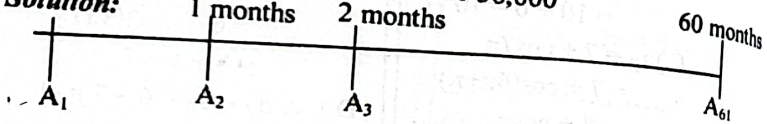
26. A business association gets a net profit of Birr 3,000 at the end of each month. Just after the fifth year, its amount was Birr 230,000. What was the starting amount of the business?

A. Birr 47,000 C. Birr 53,000

B. Birr 50,000 D. Birr 56,000

Solution:



Since the common difference d = 3000 and the five years have $12 \times 5 = 60$ months.

$\Rightarrow A_{61} = A_1 + 60d$

$230,000 = A_1 + 60 \times 3000$

$230,000 = A_1 + 180,000$

$A_1 = 230,000 - 180,000$

$= 50,000$ **Birr**

∴ Answer B

27. Which of the following is equal to $\lim_{n\to\infty} \frac{3\sqrt{n}-6n+5}{4n+1}$?

A. $\frac{-3}{2}$ B. 0 C. $\frac{3}{4}$ D. $\frac{5}{4}$

Solution: $\lim_{n\to\infty} \frac{3\sqrt{n}-6n+5}{4n+1}$

Dividing each term by n

$$\Rightarrow \lim_{n\to\infty} \frac{\frac{3\sqrt{n}}{n} - \frac{6n}{n} + \frac{5}{n}}{\frac{4n}{n} + \frac{1}{n}} = \lim_{n\to\infty} \frac{\frac{3}{\sqrt{n}} - 6 + \frac{5}{n}}{4 + \frac{1}{n}}$$

$$= \frac{\lim_{n\to\infty} \frac{3}{\sqrt{n}} - \lim_{n\to\infty} 6 + \lim_{n\to\infty} \frac{5}{n}}{\lim_{n\to\infty} 4 + \lim_{n\to\infty} \frac{1}{n}}$$

$$= \frac{-6}{4} = \frac{-3}{2}$$

∴ Answer A

28. What is the solution set of the following system of equations

$$\begin{cases} x + y + 2z = 1 \\ x + 2y + z = 2 \\ -2x - 2y - 4z = -2 \end{cases} ?$$

A. $\{(0,1,0)\}$ C. $\{(-3k, k+1, k) | k \in (-\infty,\infty)\}$

B. $(-\infty,\infty)$ D. $\{(3k, k-1, k) | k \in (-\infty,\infty)\}$

Solution: $\begin{cases} x + y + 2z = 1 \\ x + 2y + z = 2 \\ -2x - 2y - 4z = -2 \end{cases}$

The augmented matrix of the system is

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 1 & 2 & 1 & 2 \\ -2 & -2 & -4 & -2 \end{array}\right)$$

Using row operation, convert to echelon form.

i.e $R_2 \to R_2 - R_1$ and $R_3 \to 2R_1 + R_3$

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

From the last row we see that the system has an infinite number of solutions.

$\Rightarrow x - z = 1$ ..... From 2nd row

$y = z + 1$

and $x + y + 2z = 1$

$x + (z+1) + 2z = 1$

$x + 3z = 0$

$x = -3z$

Let's say $z = k \in (-\infty,\infty)$ be a parameter

Solution Set = $\{(-3k, k+1, k)\}$

∴ Answer C

29. Let z be a complex number. Which of the following is the solution set of $z^3 - iz = 0$?

A. $\left\{\pm\frac{1}{\sqrt{2}}(1+i)\right\}$ C. $\{\pm\sqrt{2}(1+i)\}$

B. $\left\{0, \pm\frac{1}{\sqrt{2}}(1-i)\right\}$ D. $\left\{0, \pm\frac{1}{\sqrt{2}}(1+i)\right\}$

Solution: $z^3 - iz = 0$

$z(z^2 - i) = 0$

$z = 0$, or $z^2 - i = 0 \Rightarrow z^2 = i \Rightarrow z = \pm\sqrt{i}$

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Using De Moviers theorem let's find what $\sqrt{i}$ mean?

$i = 0+i$

$\theta = 90°, |i| = 1$

$i = 1(\cos 90° + i\sin 90°)$

$\Rightarrow \sqrt{i} = i^{\frac{1}{2}} = [1(\cos 90° + i\sin 10°)]^{\frac{1}{2}}$

$= 1^{1/2}\left(\cos\frac{90°}{2} + i\sin\frac{10°}{2}\right)$

$= 1(\cos 45° + i\sin 45°)$

$= \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$

$\therefore z = \pm\frac{\sqrt{2}}{2}(1+i)$

$Z = \pm\frac{1}{\sqrt{2}}(1+i)$

Solution Set $= \left\{0, \pm\frac{1}{\sqrt{2}}(1+i)\right\}$

∴ Answer D

30. In the set of complex numbers, the solution set of $x^2 - 2x + 5 = 0$ is

A. $\varnothing$ C. $\{1+2i, 1-2i\}$

B. $\{2+i, 2-i\}$ D. $\{2+4i, 2-4i\}$

Solution: $x^2 - 2x + 5 = 0$

Since it is a quadratic equation, we apply the quadratic formula.

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$= \frac{2 \pm \sqrt{(-2)^2 - 4(1)(5)}}{2}$$

$$= \frac{2 \pm \sqrt{4-20}}{2}$$

$$= \frac{2 \pm \sqrt{-16}}{2}$$

$$= \frac{2 \pm \sqrt{16}\sqrt{-1}}{2}$$

$$= \frac{2 \pm 4i}{2} = 1 \pm 2i$$

Solution Set $= \{1+2i, 1-2i\}$

∴ Answer C

31. If $f(x) = \pi^2$, then $f'(1)$ is equal to

A. 2π B. π^2 C. 1 D. 0

Solution: $f(x) = \pi^2$ which is a constant function.

$\Rightarrow f'(x) = 0$

$\Rightarrow f'(1) = 0$

∴ Answer D

32. Let $f(x) = \frac{6x}{x+a}$. For what value of a is $f'(a) = 1$?

A. $\frac{1}{3}$ B. $\frac{2}{3}$ C. $\frac{3}{2}$ D. 3

Solution: $f(x) = \frac{6x}{x+a}$

$$f'(x) = \frac{(6x)'(x+a) - (x+a)'(6x)}{(x+a)^2}$$

$$= \frac{6(x+a) - 6x}{(x+a)^2} = \frac{6x+6a-6x}{(x+a)^2} = \frac{6a}{(x+a)^2}$$

$$\Rightarrow f'(a) = \frac{6a}{(a+a)^2} = \frac{6a}{(2a)^2} = \frac{6a}{4a^2} = \frac{6}{4a}$$

$$\Rightarrow 1 = \frac{6}{4a} \Rightarrow 4a = 6$$

$$\Rightarrow a = \frac{3}{2}$$

∴ Answer C

33. Which one of the following is necessarily true about a function f(x)?

A. If f is continuous at $x = a$, then f is differentiable at $x = a$.

B. If f is not differentiable at $x = a$, then $\lim_{x\to a^-} f(x) \neq \lim_{x\to a^+} f(x)$.

C. If f is differentiable at $x = a$, then $\lim_{x\to a^-} f(x) = f(a) = \lim_{x\to a^+} f(x)$.

D. If the derivative $f'(a) = 0$, then f attains its maximum value at $x = a$.

Solution: If a function f is differentiable at a, then it is continuous at a. If f is continuous at a, then $\lim_{x\to a} f(x) = f(a)$

But $\lim_{x\to a} f(x) = \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x)$

∴ If f is differentiable at a, then $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = f(a)$

∴ Answer C

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34. If $g(x) = \frac{f(x)}{x+1} + (f(x))^2$, $f(1) = 8$ and $f'(1) = 2$, then $g'(1)$ is equal to

A. 36 B. 31 C. 25 D. 16

Solution: $g(x) = \frac{f(x)}{x+1} + (f(x))^2$, $f(1) = 8$, $f'(1) = 2$.

$$g'(x) = \frac{f'(x)(x+1) - f(x)(x+1)'}{(x+1)^2} + 2f(x) \times f'(x)$$

$$\Rightarrow g'(1) = \frac{f'(1)(1+1) - f(1)}{(1+1)^2} + 2f(1) \times f'(1)$$

$$= \frac{2 \times 2 - 8}{4} + 2 \times 8 \times 2$$

$$= 31$$

∴ ***Answer B***

35. If $f(x) = xe^{3x} - \cos(2x)$, then $f''(0)$ is equal to

A. 0 B. 2 C. 6 D. 10

Solution: $f(x) = xe^{3x} - \cos(2x)$

$f'(x) = x'e^{3x} + x(e^{3x})' + 2\sin(2x)$

$f'(x) = e^{3x} + 3xe^{3x} + 2\sin 2x$

$f''(x) = (e^{3x})' + (3x)'e^{3x} + 3x(e^{3x})' + 4\cos 2x$

$f''(x) = 3e^{3x} + 3e^{3x} + 9xe^{3x} + 4\cos 2x$

$f''(x) = 6e^{3x} + 9xe^{3x} + 4\cos 2x$

$f''(0) = 6e^0 + 9 \times 0 \times e^0 + 4\cos 0$

$f''(0) = 6 + 4$

$f''(0) = 10$

∴ ***Answer D***

36. $\lim_{x \to 0} \frac{x\csc 3x}{x+1}$ is equal to:

A. $\frac{1}{3}$ B. 0 C. 1 D. 3

Solution: $\lim_{x \to 0} \frac{x\csc 3x}{x+1} = \lim_{x \to 0} \frac{x}{x+1} \times \frac{1}{\sin 3x}$ (i.e $\csc x = \frac{1}{\sin x}$)

$$= \lim_{x \to 0} \frac{1}{x+1} \times \frac{3x}{\sin 3x} \times \frac{1}{3}$$

$$= \left(\lim_{x \to 0} \frac{1}{x+1} \times \lim_{x \to 0} \frac{3x}{\sin 3x}\right)\frac{1}{3} = 1 \times 1 \times \frac{1}{3} = \frac{1}{3}$$

∴ ***Answer A***

37. $\lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^{2x+4}$ is equal to

A. e B. e^2 C. e^4 D. ∞

Solution: $\lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^{2x+4} = \lim_{x \to \infty} \left[\left(1 + \frac{1}{x}\right)^{2x} \times \left(1 + \frac{1}{x}\right)^4\right]$

$$= \lim_{x \to \infty} \left[\left(1 + \frac{1}{x}\right)^x\right]^2 \times \lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^4$$

$$= \left[\lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^x\right]^2 \times 1$$

$$= e^2$$

∴ ***Answer B***

38. To which of the following is $\lim_{x \to 1^-} \frac{x^4 - 1}{|x-1|}$ equals to?

A. 4 B. 0 C. -1 D. -4

Solution: $\lim_{x \to 1^-} \frac{x^4 - 1}{|x-1|}$

Since $|x-1| = \begin{cases} x-1, & \text{if } x \geq 1 \\ -(x-1), & \text{if } x < 1 \end{cases}$

Then, $\lim_{x \to 1^-} \frac{x^4-1}{|x-1|} = \lim_{x \to 1^-} \frac{x^4-1}{-(x-1)}$

$$= \lim_{x \to 1^-} \frac{(x^2-1)(x^2+1)}{-(x-1)}$$

$$= \lim_{x \to 1^-} \frac{(x-1)(x+1)(x^2+1)}{-(x-1)}$$

$$= -\lim_{x \to 1^-} (x+1)(x^2+1)$$

$$= -(1+1)(1+1)$$

$$= -4$$

∴ ***Answer D***

39. Let $f(x) = \begin{cases} \frac{a}{x^2+3}, & \text{if } x \leq 1 \\ \frac{x-1}{\sqrt{x}-1}, & \text{if } x > 1 \end{cases}$

What is the value of a if f is continuous at $x = 1$?

A. 0 B. 2 C. 4 D. 8

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Solution: $f(x) = \begin{cases} \frac{a}{x^2+3}, & \text{if } x \le 1 \\ \frac{x-1}{\sqrt{x}-1}, & \text{if } x > 1 \end{cases}$

f is continuous at x = 1 implies $\lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x) = f(1)$

$$\Rightarrow \lim_{x\to 1^+} \frac{x-1}{\sqrt{x}-1} = \frac{a}{1^2+3}$$

$$\Rightarrow \lim_{x\to 1^+} \frac{x-1}{\sqrt{x}-1} \times \frac{\sqrt{x}+1}{\sqrt{x}+1} = \frac{a}{4}$$

$$\Rightarrow \lim_{x\to 1^+} \frac{x\sqrt{x}+x-\sqrt{x}-1}{x-1} = \frac{a}{4}$$

$$\Rightarrow \lim_{x\to 1^+} \frac{\sqrt{x}(x-1)+(x-1)}{x-1} = \frac{a}{4}$$

$$\Rightarrow \lim_{x\to 1^+} \frac{(\sqrt{x}+1)(x-1)}{x-1} = \frac{a}{4}$$

$$\Rightarrow 2 = \frac{a}{4} \Rightarrow a = 8$$

∴ Answer D

40. Which of the following is equal to $\int \frac{x}{x^2+2x+1} dx$?

A. $\ln(x^2+2x+1)+c$
C. $\ln(x^2+2x+1)-\frac{1}{x+1}+c$
B. $\ln|x+1|-\frac{1}{x+1}+c$
D. $\ln(x^2+2x+1)+\frac{1}{x+1}+c$

Solution: $\int \frac{x}{x^2+2x+1} dx = \int \frac{x}{(x+1)^2} dx$

Using substitution, let $u = x+1 \Rightarrow \frac{du}{dx} = 1$

$$\Rightarrow x = u-1$$

$$\Rightarrow \int \frac{x}{(x+1)^2} dx = \int \frac{u-1}{u^2} du = \int \frac{u}{u^2} du - \int \frac{1}{u^2} du$$

$$= \int \frac{1}{u} du - \int \frac{1}{u^2} du$$

$$= \ln|u| - \int u^{-2} du$$

$$= \ln|u| - \left[\frac{u^{-2+1}}{-2+1}\right] + c$$

$$= \ln|u| - \left[\frac{u^{-1}}{-1}\right] + c$$

$$= \ln|u| + \frac{1}{u} + c$$

$$= \ln|x+1| + \frac{1}{x+1} + c$$

∴ No Answer

41. $\int \frac{e^{2x}-4x}{xe^{2x}} dx$ is equal to

A. $\ln|x| + 2e^{-2x} + c$
C. $\ln|x| + 4e^{-2x} + c$
B. $\ln|x| - 4e^{-2x} + c$
D. $\frac{1}{x^2} + 2e^{-2x} + c$

Solution: $\int \frac{e^{2x}-4x}{xe^{2x}} dx = \int \frac{e^{2x}}{xe^{2x}} dx - \int \frac{4x}{xe^{2x}} dx$

$$= \int \frac{1}{x} dx - \int \frac{4}{e^{2x}} dx$$

$$= \ln|x| - 4\int \frac{1}{e^{2x}} dx$$

$$= \ln|x| - 4\int e^{-2x} dx$$

Using substitution, let $u = -2x$ and $\frac{du}{dx} = -2$

$$\Rightarrow \int e^{-2x} dx = \int e^u \times \frac{du}{-2} = \frac{1}{2} e^u + c$$

$$\Rightarrow \int e^{-2x} dx = \frac{1}{2} e^{-2x} + c$$

$$\Rightarrow \int \frac{e^{2x}-4x}{xe^{2x}} dx = \ln|x| - 4\left[-\frac{1}{2} e^{-2x}\right] + c$$

$$\Rightarrow \int \frac{e^{2x}-4x}{xe^{2x}} dx = \ln|x| + 2e^{-2x} + c$$

∴ Answer A

42. Which of the following is equal to $\int_0^1 \frac{(x-1)^2}{x+1} dx$?

A. $-\frac{3}{2} + \ln 16$
C. $-\frac{5}{2} + 4\ln 2$
B. $-\frac{5}{2} + \ln 15$
D. $\frac{5}{2} + \ln 16$

Solution: $\int_0^1 \frac{(x-1)^2}{x+1} dx$

Let $u = x+1 \Rightarrow \frac{du}{dx} = 1$ and $x = u-1$

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$\Rightarrow$ As $x = 0$, then $u = 1$, As $x = 1$, $u = 2$

$$\int_0^1 \frac{(x-1)^2}{x+1}dx = \int_1^2 \frac{(u-2)^2}{u}du = \int_1^2 \frac{u^2-4u+4}{u}du$$

$$= \int_1^2 \frac{u^2}{u}du - \int_1^2 \frac{4u}{u}du + \int_1^2 \frac{4}{u}du$$

$$= \int_1^2 u\,du - \int_1^2 4\,du + 4\int_1^2 \frac{1}{u}du$$

$$= \frac{u^2}{2}\Big|_1^2 - 4u|_1^2 + 4\ln|u|\Big|_1^2$$

$$= \left(2 - \frac{1}{2}\right) - (8-4) + 4(\ln 2 - \ln 1)$$

$$= \frac{3}{2} - 4 + 4\ln 2$$

$$= \frac{-5}{2} + 4\ln 2$$

$\therefore$ ***Answer C***

43. If $f(x) = \begin{cases} \sin x, & x \le 0 \\ 3x\sqrt{x^2+1}, & x > 0 \end{cases}$, which of the following is equal to $\int_{\frac{-\pi}{2}}^{\sqrt{8}} f(x)dx$?

A. 17 B. 25 C. 27 D. $3\sqrt{8}$

Solution: $f(x) = \begin{cases} \sin x, & x \le 0 \\ 3x\sqrt{x^2+1}, & x > 0 \end{cases}$

$$\Rightarrow \int_{\frac{-\pi}{2}}^{\sqrt{8}} f(x)dx = \int_{\frac{-\pi}{2}}^{0} f(x)dx + \int_0^{\sqrt{8}} f(x)dx$$

$$= \int_{\frac{-\pi}{2}}^{0} \sin x\,dx + \int_0^{\sqrt{8}} 3x\sqrt{x^2+1}\,dx$$

$$= -\cos x\Big|_{\frac{-\pi}{2}}^{0} + \int_0^{\sqrt{8}} 3x\sqrt{x^2+1}\,dx$$

Let $u = x^2 + 1$, As $x = 0 \Rightarrow u = 1$

$\frac{du}{dx} = 2x$ As $x = \sqrt{8}$, $u = 9$

$$\Rightarrow \int_0^{\sqrt{8}} 3x\sqrt{x^2+1}\,dx = \int_1^9 3x\sqrt{u}\frac{du}{2x}$$

$$= \frac{3}{2}\int_1^9 \sqrt{u}\,du$$

$$= \frac{3}{2}\int_1^9 u^{1/2}du = \frac{3}{2}\frac{u^{3/2}}{\frac{3}{2}}\Big|_1^9$$

$$= u^{3/2}\Big|_1^9$$

$$\Rightarrow \int_{\frac{-\pi}{2}}^{\sqrt{8}} f(x)dx = -\cos x|_{\frac{-\pi}{2}}^{0} + u^{3/2}\Big|_1^9$$

$$= -\cos 0 - \left(-\cos\left(\frac{-\pi}{2}\right)\right) + 9^{3/2} - 1^{3/2}$$

$$= -1 + 0 + (3^2)^{3/2} - 1$$

$$= 25$$

$\therefore$ ***Answer B***

44. At what value(s) of x does $f(x) = \frac{1}{3}x^3 - \frac{1}{2}x^2 - 2x + 1$ have a local maximum?

A. $x = 2$ B. $x = -1$ C. $x = 5$ D. $x = 2, x = -1$

Solution: $f(x) = \frac{1}{3}x^3 - \frac{1}{2}x^2 - 2x + 1$

The critical points are: $f'(x) = x^2 - x - 2$

$$f'(x) = 0$$

$$x^2 - x - 2 = 0$$

$$\Rightarrow x = -1, x = 2$$

$$\Rightarrow f(-1) = \frac{1}{3}(-1)^3 - \frac{1}{2}(-1)^2 - 2(-1) + 1$$

$$= -\frac{1}{3} - \frac{1}{2} + 2 + 1$$

$$= -\frac{5}{6} + 3 = \frac{13}{6} \text{.....Local maximum}$$

and $f(2) = \frac{1}{3}(2)^3 - \frac{1}{2}(2)^2 - 2(2) + 1$

$$= \frac{8}{3} - 2 - 4 + 1$$

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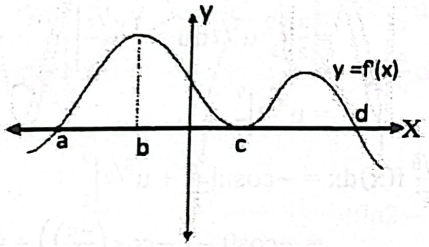
$= \frac{-7}{3}$ .....Local minimum

∴ ***Answer B***

45. Look at the following graph of f '(x).



Which of the following is true about f ?

A. f is increasing on [c, d].
B. f is decreasing on [b, c].
C. f has a local minimum at d.
D. f has a local extreme value at c.

Solution: From the graph of f '(x) we see the following relations

	a	b	c	d	
f '(x)	–	+	+	+	–

i) The graph of f(x) is decreasing on $(-\infty,a]$ and $[d,\infty)$

ii) The graph of f(x) is increasing on [a, d].

From i) and ii) f(x) has a local minimum at x = a and a local maximum at x = d.

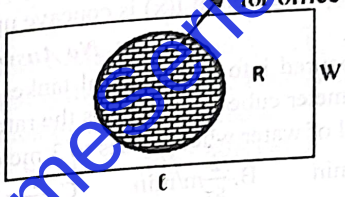
Since f '(c) = 0 and the graph of f(x) does not change sign at x = c, then x = c is an inflection point.

∴ ***Answer A***

46. A rectangular field of length ℓ and width w meters for $w < \ell$ has perimeter 400 meters. If a circular region of area w^2 is to be reserved for office purpose, what should be the length of the field so that the area of the remaining region is maximum?

A. 500 meters B. 100 meters C. 120 meters D. 150 meters

Solution:



Circular region reserved for office purpose .

$p = 2\ell + 2w$

$400 = 2(\ell + w)$

$\ell + w = 200$ .....*

$A(R) = A_{rectangle} - A_{circle}$

$A(R) = \ell w - w^2$

$A(R) = (200 - w)w - w^2$

$A(R) = 200w - w^2 - w^2$

$A(R) = 200w - 2w^2$

$A'(R) = 200 - 4w = 0$

$\Rightarrow w = 50m$

Since $\ell + w = 200m$

$\ell = \mathbf{150m}$

∴ ***Answer D***

47. On which of the following intervals is the graph of $f(x) = \ln(x^2 + 1)$ concave upward?

A. $(-\infty, -1] \cup [1,\infty)$ B. $[0,\infty)$ C. $(-\infty, 0]$ D. $[-1,1]$

Solution: $f(x) = \ln(x^2 + 1)$

$$f'(x) = \frac{1}{x^2+1} \times (x^2 + 1)' = \frac{2x}{x^2+1}$$

$$f''(x) = \frac{(2x)'(x^2+1) - 2x(x^2+1)'}{(x^2+1)^2}$$

$$f''(x) = \frac{2(x^2+1)-4x^2}{(x^2+1)^2} = \frac{2-2x^2}{(x^2+1)^2}$$

$f''(x) = 0$ if $2 - 2x^2 = 0$

$\Rightarrow x = \pm 1$

Using concavity test, if $f''(x) > 0$, $\forall x \in I$, then the graph of f is concave upward on I .

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$f'(x)$: $-$ | 0 | $+$ | 0 | $-$ (sign change at -1 and 1)

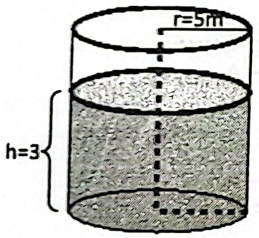
$\therefore$ The graph of $f(x)$ is concave upward on $(-1,1)$

$\therefore$ ***No Answer***

48. Water is poured into a cylindrical tanker of radius 5 meters at a rate of 10 meter cube/min. What is the rate of change of the height of the level of water when it rises to 3 meters?

A. $\frac{3}{25\pi}$m/min B. $\frac{2}{5\pi}$m/min C. $\frac{1}{2\pi}$m/min D. $\frac{3}{5\pi}$m/min

Solution:



$\frac{dv}{dt} = 10m^3/min$, $\frac{dh}{dt} = ?$

$V_{cylinder} = Bh$, where B = base area

$V = \pi r^2 h$

$V = 25\pi h$

$\Rightarrow \frac{dv}{dt} = 25\pi \times \frac{dh}{dt}$ (v and h are related variables)

$\Rightarrow 10m^3/min = 25\pi m^2 \times \frac{dh}{dt}$

$\Rightarrow \frac{10}{25\pi}m/min = \frac{dh}{dt}$

$\Rightarrow \frac{dh}{dt} = \frac{2}{5\pi}$ m/min

$\therefore$ ***Answer B***

49. Which of the following is an antiderivative of $f(x) = \tan x$?

A. $\frac{1}{2}\tan^2 x$ B. $\sec^2 x$ C. $\ln(\cos x)$ D. $\ln(\sec x)$

Solution: Let $f(x)$ be the antiderivative of $f(x) = \tan x$

$\Rightarrow F(x) = \int f(x)dx$

$\Rightarrow F(x) = \int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx$

Let $u = \cos x$

$\frac{du}{dx} = -\sin x$

$\Rightarrow F(x) = \int \frac{\sin x}{u} - \frac{du}{\sin x} = -\int \frac{1}{u}du$

$= -\ln|\cos x| + c$

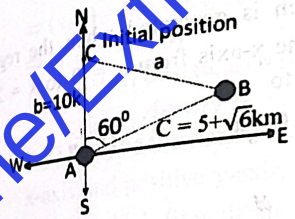
$F(x) = \ln(\sec x) + c$

$\therefore$ ***Answer D***

50. A patrol boat on a sea sailed from its station 10 km to the south and then changed its course and sailed $5 + \sqrt{6}$ km in the direction N 60°E. Then, what is the distance the boat should travel in order to return to its station by the shortest route?

A. $4\sqrt{6}$ km B. 9 km C. $3\sqrt{6}$ km D. 7 km

Solution:



Using cosine law

$a^2 = b^2 + c^2 - 2bc\cos A$

$= (10)^2 + (5+\sqrt{6})^2 - 2(10)(5+\sqrt{6})\cos 60°$

$= 100 + 25 + 10\sqrt{6} + 6 - 20(5+\sqrt{6})\frac{1}{2}$

$= 131 + 10\sqrt{6} - 50 - 10\sqrt{6}$

$a^2 = 81$

$a = \mathbf{9\ km}$

$\therefore$ ***Answer B***

51. If a sphere with center C(0,1,1) intersects the z-axis at P(0,0,3), then the radius of the sphere is equal to

A. 5 B. 3 C $\sqrt{3}$ D. $\sqrt{5}$

Solution: Center C(0,1,1)

Sphere: $(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$

$x^2 + (y-1)^2 + (z-1)^2 = r^2$

Intersection point of the sphere and the z – axis is P(0,0,3) means the point P satisfies both the equation of the sphere and the z-axis.

$\Rightarrow 0^2 + (0-1)^2 + (3-1)^2 = r^2$

$\Rightarrow 1 + 4 = r^2$

$\Rightarrow r = \sqrt{5}$

$\therefore$ ***Answer D***

52. Suppose A and B are the end points of a diameter of the sphere whose equation is $x^2 + y^2 + (z+2)^2 = 1$. If A = (1,0,-2), then B is equal to

A. (0,1,-2) B. (0,0,-1) C. (-1,0,-2) D. (0,-1,-2)

Solution: $x^2 + y^2 + (z+2)^2 = 1$

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End points of the diameter A = (1,0,-2) and B = (a, b, c)

Center of the sphere $C = (0, 0, -2) = \left(\frac{1+a}{2}, \frac{0+b}{2}, \frac{-2+c}{2}\right)$

$0 = \frac{1+a}{2} \Rightarrow a = -1$

$0 = \frac{0+b}{2} \Rightarrow b = 0$

$-2 = \frac{-2+c}{2} \Rightarrow c = -2$

$\therefore B = (-1,0,-2)$

∴ Answer C

53. The volume of the solid which is generated when the region bounded by $y = \sqrt{x+1}$ and the x-axis from x = 0 to x = 2 is rotated about the x-axis is equal to

A. 4π B. 3π C. $\frac{4}{3}\pi$ D. $\frac{3}{4}\pi$

Solution:

$V = \int_0^2 \pi(f(x))^2 dx$

$= \int_0^2 \pi(\sqrt{x+1})^2 dx$

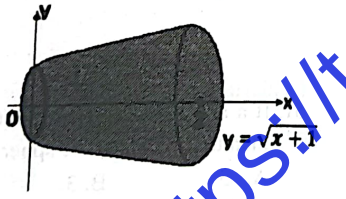
$= \pi \int_0^2 (x+1)\, dx$

$= \pi \int_0^2 x\, dx + \pi \int_0^2 1\, dx$

$= \pi \frac{x^2}{2}\Big|_0^2 + \pi x\Big|_0^2$

$= 2\pi + 2\pi$

$= 4\pi$



∴ Answer A

54. Which of the following is a vector that lies on the line through (0, 0) and (2,4)?

A. $\vec{u} = (2,1)$ C. $\vec{u} = (\frac{1}{2},2)$

B. $\vec{u} = (-1,-2)$ D. $\vec{u} = (-2,-6)$

Solution:

Equation of a line that passes through the points (0,0) and (2,4) is

$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

$\Rightarrow y - 0 = \frac{4-0}{2-0}(x - 0)$

$\Rightarrow y = 2x$

Let $\vec{V}$ be on ℓ

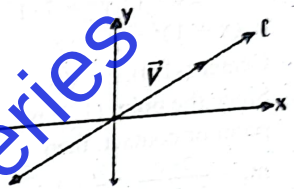
$\Rightarrow \vec{V} \parallel \ell$

Let $\vec{V} = (a, b)$ and (x, y) is on ℓ

$\Rightarrow (x, y) = k(a, b), k \in \mathbb{R}$

$\Rightarrow (1,2) = k(a, b)$

If $k = \frac{1}{2}$ then $(a, b) = (-1,-2)$



Answer B

55. Let $\vec{i}$ and $\vec{j}$ be the standard unit vectors in the directions of positive x-axis and positive y-axis, respectively, and $\overrightarrow{AB}$ be a vector from the point A(-1, 0) to the point B(2, 2). If $\vec{v} = 3\overrightarrow{AB} - 3\vec{i} + 2\vec{j}$, then the unit vector in the direction of $\vec{v}$ is equal to

A. $\left(\frac{3}{5}, \frac{4}{5}\right)$ B. $\left(\frac{-3}{5}, \frac{-4}{5}\right)$ C. $\left(\frac{-3}{5}, \frac{4}{5}\right)$ D. $\left(\frac{3}{5}, \frac{-4}{5}\right)$

Solution: A(-1,0), B = (2,2)

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = (2, 2) - (-1,0) = (2+1, 2-0) = (3,2)$

$\vec{V} = 3\overrightarrow{AB} - 3i + 2j$

$= 3(3, 2) - 3(1, 0) + 2(0, 1)$

$= (9, 6) - (3, 0) + (0, 2)$

$= (6, 8)$

$|\vec{V}| = \sqrt{6^2 + 8^2} = 10$

Unit vector in the direction of $\vec{V}$ is $\left(\frac{6}{10}, \frac{8}{10}\right) = \left(\frac{3}{5}, \frac{4}{5}\right)$

∴ Answer A

56. Which of the following is a vector equation of the line tangent to the circle $x^2 + y^2 + 2x - 7 = 0$ at (1, 2)?

A. $(x, y) = (0,3) + \lambda(-1,2)$ C. $(x,y) = (0,3) + \lambda(1,-1)$

B. $(x, y) = (1,2) + \lambda(2,-1)$ D. $(x, y) = (1,2) + \lambda(-1,2)$

Solution: $x^2 + y^2 + 2x - 7 = 0$

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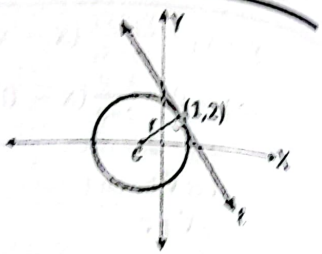
$x^2 + 2x + y^2 = 7$

$x^2 + 2x + 1 + y^2 = 7 + 1$

$(x + 1)^2 + y^2 = 8$

Center $(-1,0)$, $r = \sqrt{8}$



Since the point (1,2) is point of contact, then

$m_r = \frac{2-0}{1-(-1)} = \frac{2}{2} = 1$

and $r \perp \ell$, then $m_\ell = -1$

$\therefore$ Equation of ℓ: $y - y_1 = m_\ell(x - x_1)$

$y - 2 = -1(x - 1)$

$y = -x + 1 + 2$

$y = -x + 3$

Thus, the points (0, 3) and (1,2) are on ℓ, then

$\vec{v} = (1, 2) - (0,3) = (1,-1)$

$\therefore$ The vector equation of ℓ is

$(x, y) = (0, 3) + t(1,-1)$ or

$(x, y) = (1, 2) + t(1,-1)$

$\therefore$ ***Answer C***

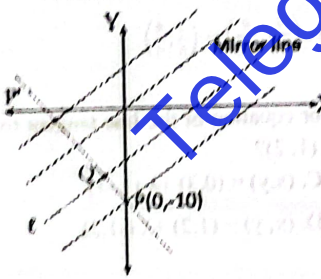
57.Let ℓ be the line whose equation is $2x - y = 10$. Which one of the following is the equation of the image of ℓ after a reflection in the line $y = 2x - 5$ followed by a rotation through the angle of 90° about the origin?

A. $x + 2y = 0$ B. $2x + y = 0$ C. $x + 2y = 5$ D. $x - 2y = 5$

Solution: ℓ: $2x - y = 10$

$\Rightarrow y = 2x - 10$

Reflection line: $y = 2x - 5$



First find the image of the point P(0,-10) about the line $y = 2x - 5$

Equation of the line passing through P(0,-10) and $\perp$lar to ℓ is

$y - y_1 = m(x - x_1)$

$y - (-10) = \frac{-1}{2}(x - 0)$

$y + 10 = \frac{-1}{2}x$

$y = \frac{-1}{2}x - 10$

Then, find the point of the intersection of the two

$\begin{cases} y = 2x - 5 \\ y = \frac{-1}{2}x - 10 \end{cases} \Rightarrow 2x - 5 = \frac{-1}{2}x - 10$

$4x - 10 = -x - 20$

$5x = -20 + 10$

$x = -2$ and

$y = 2x - 5$

$y = -9$

$\therefore Q(-2,-9)$

Since $PQ = QP'$, then $(-2,-9) = \left(\frac{a+0}{2}, \frac{b-10}{2}\right)$, where $P'(a, b)$

$\Rightarrow \frac{a}{2} = -2 \Rightarrow a = -4$

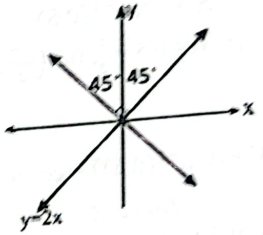
$\Rightarrow \frac{b-10}{2} = -9 \Rightarrow b = -8$

$\therefore P'(-4,-8)$ is the image point. Since the image line is passes through P' and parallel to ℓ, then its equation is

$y - y_1 = m(x - x_1)$

$\Rightarrow y + 8 = 2(x + 4)$

$\Rightarrow y = 2x$



Now rotate $y = 2x$, 90°, the image line becomes

(i.e the point (x, y) is mapped to the point $(x', y')$where

$x' = x\cos\theta - y\sin\theta$

$y' = x\sin\theta + y\cos\theta$

$y = \frac{1}{-2}x$

$\Rightarrow 2y + x = 0$

$\therefore$ ***Answer A***

58.Which one of the following is a simplified form of

$\csc\left(\frac{\pi}{2} - x\right)\cos x - \sin^2 x$?

A. $2\cos x$ B. $\cos^2 x$ C. $2\sin x$ D. $\sin 2x$

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Solution: $\csc x = \frac{1}{\sin x}$

$$\Rightarrow \csc\left(\frac{\pi}{2} - x\right)\cos x - \sin^2 x = \frac{1}{\sin\left(\frac{\pi}{2} - x\right)}\cos x - \sin^2 x$$

$$= \frac{\cos x}{\sin\left(\frac{\pi}{2} - x\right)} - \sin^2 x$$

$$= \frac{\cos x}{\sin\frac{\pi}{2}\cos x - \sin x \cos\frac{\pi}{2}} - \sin^2 x$$

$$= \frac{\cos x}{\cos x - 0} - \sin^2 x$$

$$= 1 - \sin^2 x$$

$$= \cos^2 x$$

∴ ***Answer B***

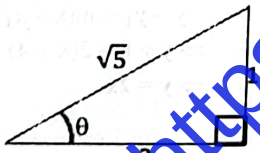
59. $\cos(\tan^{-1}(0.5))$ is equal to:

A. $\frac{1}{\sqrt{5}}$ B. $\frac{1}{\sqrt{3}}$ C. $\frac{2}{\sqrt{5}}$ D. $\frac{2}{\sqrt{3}}$

Solution: $\cos(\tan^{-1}(0.5))$

Let $\theta = \tan^{-1}(0.5)$

$\tan\theta = 0.5 = \frac{1}{2}, \theta \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$



$\Rightarrow \cos(\tan^{-1}(0.5)) = \cos\theta = \frac{2}{\sqrt{5}}$

∴ ***Answer C***

60. If x and y are non-negative integers, which of the following is **NOT** true?

A. $(\forall x)(\exists y)(y > x^2 - 1)$ C. $(\exists y)(\forall x)(y \leq x^2 - 1)$

B. $(\exists x)(\forall y)(y > x^2 - 1)$ D. $(\exists y)(\exists x)(y \leq x^2 - 1)$

Solution: $x, y \in \{0,1,2,3,\ldots\}$

A) $(\forall x)(\exists y)(y > x^2 - 1)$ – For every x, there is a corresponding y such that $y > x^2 - 1$ .....Which is true.

B) $(\exists x)(\forall y)(y > x^2 - 1)$ – There is at least one x such that $y > x^2 - 1$ for every y.

i.e $y+1 > x^2 \Rightarrow x^2 < y + 1$

$x < \sqrt{y+1}$

C) $(\exists y)(\forall x)(y \leq x^2 - 1)$ – There is at least one y for every x such that $y \leq x^2 - 1$

Let $x = 0$, then $y \leq -1$ which is not a non-negative integer.

∴ It is false

D) $(\exists y)(\exists x)(y \leq x^2 - 1)$ – There is some y and some x such that $y \leq x^2 - 1$ .....which is true.

∴ ***Answer C***

61. Let $a_n = n^2 - n$ when $n \in \mathbb{N}$ (set of natural numbers). Which one of the following is true, when k is an arbitrarily chosen natural number and m is an integer?

A. a_n is not a multiple of 2 for some $n \in \mathbb{N}$ because $a_1 = 0$.

B. a_n is a multiple of 2 for all $n \in \mathbb{N}$ because $a_1 = 0$ and if $a_k = 2m$, then $a_{k+1} = 2(m+1)$.

C. a_n is a multiple of 2 for all $n \in \mathbb{N}$ because $a_1 = 0$ and if $a_k = 2m$, then $a_{k+1} = 2(m+k)$.

D. a_n is a multiple of 3 for all $n \in \mathbb{N}$ because $a_1 = 0$ and if $a_k = 3m$, then $a_{k+1} = 3(m+k-1)$.

Solution: $a_n = n^2 - n = n(n-1)$, $n \in \mathbb{N}$ which is a product of two consecutive non-negative integers.

Some of the terms are:

$a_1 = 0, a_2 = 2, a_3 = 6, a_4 = 12, a_5 = 20,$

This implies, $a_n = 2m$, $m \in \{0,1,2,\ldots\}$

Let's take k any arbitrarily chosen national number

Say $k = 4$, then $a_k = a_4 = 12 = 2\times6$, there $m = 6$

$$\Rightarrow a_{k+1} = a_5 = 20 = 2\times10 = 2(6+4)$$
$$= 2(m+k)$$

∴ ***Answer C***

62. Consider the following statement:

$\frac{x^2+2}{x} \neq 0$ for every real number x.

To show this, a person constructed the following proof.

"**Proof:** Take $x = 1$. Then, $\frac{x^2+2}{x} = \frac{1^2+2}{1} = 3 \neq 0$.

In the same way, if we take $x = n$ for any real number n we get $\frac{x^2+2}{x} = \frac{n^2+2}{n} \neq 0$.

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It follows that $\frac{x^2+2}{x} \neq 0$, for every real number x."

Which one of the following is true about this proof?

A. The proof is correct by the principle of induction.
B. The proof is correct by the method of exhaustion.
C. The proof is correct and it uses the method of direct proof.
D. It is not a valid proof since its argument cannot lead to the conclusion.

Solution: Since the argument cannot lead to the conclusion, then it is not a valid proof.

∴ ***Answer D***

63. Let ℓ_1 and ℓ_2 be two lines in space intersecting at the origin, (0, 0, 0). If ℓ_1 and ℓ_2 pass through point A(1,1,0) and B(0,1,1), respectively, then the angle between ℓ_1 and ℓ_2 is equal to

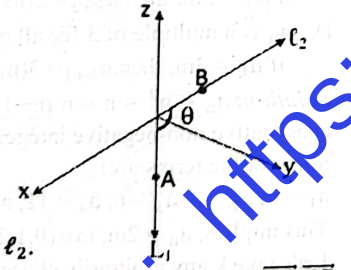
A. 30° B. 45° C. 60° D. 90°

Solution:

✓ A(1,1,0) – A point on the xy-plane .

✓ B(0,1,1) –A point on the yz-plane .

Consider the vector $\overrightarrow{OA} = (1,1,0)$ which lies on ℓ, and $\overrightarrow{OB}$ which lies on ℓ_2.



Let θ be the angle between ℓ_1 and ℓ_2 then $\cos\theta = \frac{\overrightarrow{OA}.\overrightarrow{OB}}{|\overrightarrow{OA}|.|\overrightarrow{OB}|}$

$\cos\theta = \frac{(1,1,0).(0,1,1)}{\sqrt{1^2+1^2}\sqrt{1^2+1^2}} = \frac{1}{\sqrt{2}\times\sqrt{2}} = \frac{1}{2} \Rightarrow \theta = 60°$

But each line measures 45° from the y-axis.

∴ ***Answer C***

64. Let $\vec{V} = 3\vec{i} - 4\vec{k}$, where $\vec{i}$ and $\vec{k}$ are the standard unit vectors in the directions of positive x-axis and positive z-axis, respectively, and $\overrightarrow{AB}$ be a vector from the point A(0,1,2) to a point B in space. If $\overrightarrow{AB}$ is parallel to $\vec{V}$ and $\|\overrightarrow{AB}\| = 10$, then point B is at

A. (-6, -1, 10) B. (6, -1, -10) C. (6,1,-6) D. (-6,-1,6)

Solution: $\vec{V} = 3i - 4k = (3, 0, -4)$

Point A(0, 1, 2) and let point B(a, b, c)

$\Rightarrow \overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$

$= (a, b, c) - (0, 1, 2)$

$= (a, b-1, c-2)$

$\overrightarrow{AB} \parallel \vec{V}$ means $\overrightarrow{AB} = t\vec{V}, t \in \mathbb{R}$

$\Rightarrow (a, b-1, c-2) = t(3, 0, -4)$

$\Rightarrow (a, b-1, c-2) = (3t, 0, -4t)$

$\Rightarrow a = 3t,\ b-1 = 0,\ c-2 = -4t$

$b = 1,\ c = 2 - 4t$

$\|\overrightarrow{AB}\| = 10 \Rightarrow \|\overrightarrow{AB}\|^2 = 100$

$(b-1)^2 + (c-2)^2 = 100$

$9t^2 + 16t^2 = 100$

$25t^2 = 100$

$t = \pm 2$

If t = 2, then B = (3t, 1, 2− 4t) = (6, 1, −6)

If t = -2, then B = (−6, 1, 10)

∴ ***Answer C***

65. Let $\vec{a} = -\vec{i} + 3\vec{k}$ and $\vec{b} = -\vec{i} + \vec{j}$ be vectors in the space. Which one of the following is the cosine of the angle between $\vec{a}$ and $\vec{a} - \vec{b}$?

A. $\frac{9}{10}$ B. $\frac{3}{5}$ C. $\frac{3}{\sqrt{10}}$ D. $-\frac{9}{10}$

Solution: $\vec{a} = -\vec{i} + 3k = (-1, 0, 3)$

$\vec{b} = -\vec{i} + j = (-1, 1, 0)$

$\vec{a} - \vec{b} = (-1, 0, 3) - (-1, 1, 0)$

$= (-1+1, 0-1, 3-0)$

$= (0, -1, 3)$

$\Rightarrow \cos\theta = \frac{\vec{a}.(\vec{a} - \vec{b})}{|\vec{a}||\vec{a} - \vec{b}|}$

$= \frac{(-1,\ 0,\ 3)\times(0,-1,3)}{\sqrt{(-1)^2 + 0^2 + 3^2}\sqrt{0^2 + (-1)^2 + 3^2}}$

$= \frac{-1\times0 + 0\times-1 + 3\times3}{\sqrt{10}\times\sqrt{10}}$

$= \frac{9}{10}$

∴ ***Answer A***

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1. What is the 50^{th} term of the sequence 3,10, 17, 24, 31, ... ?

A. 310 B. 346 C. 510 D. 531

Solution: Terms of the sequence: 3,10,17,24,31, ... then there is a common difference d = 7 between each successive terms. Therefore, the sequence is an AP.

$\Rightarrow A_n = A_1 + (n-1)d$

$\Rightarrow A_{50} = 3 + (50-1)\times 7$

$= 3 + 49 \times 7$

$= 3 + 343$

$= 346$

$\therefore$ ***Answer B***

2. Suppose that A and B are 3×3 matrices, I is the identity matrix of order 3 such that AB = 2I. If det B = $|B| = 6$. What is det (A^T)?

A. $\frac{1}{3}$ B. $\frac{4}{3}$ C. 12 D. 48

Solution: O(A) = 3×3

O(B) = 3×3 and O(1) = 3×3

AB = 2I and |B| = 6

$\Rightarrow |AB| = |2I|$

$\Rightarrow |A||B| = 2^3|I|$, Since $|kA| = k^n|A|$, where O(A) = n

$\Rightarrow |A| \times 6 = 8$

$\Rightarrow |A| = \frac{8}{6} = \frac{4}{3}$

Since $|A| = |A^T|$, then $|A^T| = \frac{4}{3}$

$\therefore$ ***Answer B***

3. Suppose $A = \begin{pmatrix}1 & 2\\ 2 & 3\end{pmatrix}$. If X is a 2 × 2 matrix such that $AX - A^T = 2A$, then what is the value of X?

A. $\begin{pmatrix}3 & 1\\ 1 & 3\end{pmatrix}$ B. $\begin{pmatrix}3 & 3\\ 3 & 3\end{pmatrix}$ C. $\begin{pmatrix}3 & 6\\ 6 & 9\end{pmatrix}$ D. $\begin{pmatrix}3 & 0\\ 0 & 3\end{pmatrix}$

Solution: $A = \begin{bmatrix}1 & 2\\ 2 & 3\end{bmatrix}$ let $x = \begin{bmatrix}a & c\\ b & d\end{bmatrix}$ such that $Ax - A^T = 2A$.

$\Rightarrow \begin{bmatrix}1 & 2\\ 2 & 3\end{bmatrix}\times\begin{bmatrix}a & c\\ b & d\end{bmatrix} - \begin{bmatrix}1 & 2\\ 2 & 3\end{bmatrix} = 2\begin{bmatrix}1 & 2\\ 2 & 3\end{bmatrix}$, since $A = A^T$.

$\Rightarrow \begin{bmatrix}a+2b & c+2d\\ 2a+3b & 2c+3d\end{bmatrix} - \begin{bmatrix}1 & 2\\ 2 & 3\end{bmatrix} = \begin{bmatrix}2 & 4\\ 4 & 6\end{bmatrix}$

$\Rightarrow \begin{bmatrix}a+2b-1 & c+2d-2\\ 2a+3b-2 & 2c+3d-3\end{bmatrix} = \begin{bmatrix}2 & 4\\ 4 & 6\end{bmatrix}$

$\Rightarrow \begin{matrix}a+2b-1=2, & c+2d-2=4\\ 2a+3b-2=4 & 2c+3d-3=6\end{matrix}$

Solving simultaneously,

$\begin{cases}a+2b=3\\ 2a+3b=6\end{cases}$, $\begin{cases}c+2d=6\\ 2c+3d=9\end{cases}$. we get a = 3 and c = 0, b = 0 and d = 3

$\therefore X = \begin{bmatrix}3 & 0\\ 0 & 3\end{bmatrix}$

$\therefore$ ***Answer D***

4. Which one of the following is the simplest form of $|3+4i| - \frac{25i}{3+4i}$?

A. 5 − 5i B. 5 + 5i C. 1 + 3i D. 1 − 3i

Solution: $|3+4i| - \frac{25i}{3+4i} = \sqrt{3^2+4^2} - \frac{25i}{3+4i}$

$= 5 - \frac{25i}{3+4i}$

$= \frac{15+20i-25i}{3+4i}$

$= \frac{15-5i}{3+4i}$

$= 5\left(\frac{3-i}{3+4i}\times\frac{3-4i}{3-4i}\right)$

$= \frac{5}{25}(3-i)(3-4i)$

$= \frac{1}{5}(9-12i-3i+4i^2)$

$= \frac{1}{5}(5-15i) = 1-3i$

$\therefore$ ***Answer D***

5. If $Z = \cos\left(\frac{\pi}{10}\right) + i\sin\left(\frac{\pi}{10}\right)$, then what is the value of z^5?

A. $\frac{\pi}{2}+\frac{\pi}{2}i$ B. $\frac{1}{2}+\frac{1}{2}i$ C. i D. 1 + i

Solution: $Z = \cos\left(\frac{\pi}{10}\right) + i\sin\left(\frac{\pi}{10}\right)$

Using Demovires theorem

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$Z^5 = \cos\left(5 \times \frac{\pi}{10}\right) + i \sin\left(5 \times \frac{\pi}{10}\right)$

$= \cos\frac{\pi}{2} + i \sin\frac{\pi}{2}$

$= i$

∴ ***Answer C***

6. What is the value of $|x| + 2x$ if $X < 0$?

A. $-3x$ B. $3x$ C. $-x$ D. x

Solution: $|x| + 2x$ if $x < 0$

$= -x + 2x$

$= x$

Counter example, Let $x = -3$, then $|x| + 2x = |-3| + 2(-3)$

$x = 3 - 6$

$x = -3$

∴ ***Answer D***

7. On which one of the following intervals does $f(x) = x^4 + 4x$ increase?

A. $(-\infty, -1]$ B. $(-\infty, 0]$ C. $[-1,\infty)$ D. $(-\infty,\infty)$

Solution: $f(x) = x^4 + 4x$

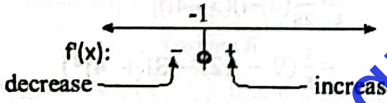
$\Rightarrow f'(x) = 4x^3 + 4$

$\Rightarrow f'(x) = 4(x^3 + 1)$

Critical number: $f'(x) = 0$

$\Rightarrow x^3 + 1 = 0$

$\Rightarrow x = -1$



∴ $f(x)$ increases on $[-1,\infty)$

∴ ***Answer C***

8. If $f(x) = e^{2x} + x - 3\cos(x)$, then what is $f''(x)$?

A. $e^{2x} + 1 - 3\sin(x)$ C. $4e^{2x} - 3\cos(x)$

B. $e^{2x} + 1 + 3\sin(x)$ D. $4e^{2x} + 3\cos(x)$

Solution: $f(x) = e^{2x} + x - 3\cos(x)$

$\Rightarrow f'(x) = (e^{2x} + x - 3\cos x)'$

$\Rightarrow f'(x) = (e^{2x})' + x' - (3\cos x)'$

$\Rightarrow f'(x) = e^{2x} \times (2x)' + 1 - 3(\cos x)'$

$\Rightarrow f'(x) = 2e^{2x} + 1 + 3\sin x$

Similarly, $f''(x) = [2e^{2x} + 3\sin x + 1]' = (2e^{2x})' + (3\sin x)' + 1'$

$f''(x) = 4e^{2x} + 3\cos x$

∴ ***Answer D***

9. If $f(x) = x^2 + 2\ln x$, then what is $\lim_{h\to 0} \frac{f(2+h) - f(2)}{h}$?

A. 5 B. 4 C. 2 D. 0

Solution: $f(x) = x^2 + 2\ln x$

$\Rightarrow \lim_{h\to 0} \frac{f(2+h) - f(2)}{h} = f'(2)$

$\Rightarrow$ since $f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h}$

$\Rightarrow f'(x) = 2x + \frac{2}{x}$

$\Rightarrow f'(2) = 2(2) + \frac{2}{2} = \frac{10}{2} = 5$

∴ ***Answer A***

10. If $\{a_n\}$ is a sequence such that $a_1 = 2$, and $a_{n+1} = a_n + 4$ for all $n \geq 1$, then $\sum_{n=1}^{35} a_n$ is equal to:

A. 2460 B. 2458 C. 2450 D. 2442

Solution: $a_1 = 2$, $a_{n+1} = a_n + 4$, $\forall n \geq 1$

$\Rightarrow a_2 = a_{1+1} = a_1 + 4 = 2 + 4 = 6$

$\Rightarrow a_3 = a_{2+1} = a_2 + 4 = 6 + 4 = 10$

$\Rightarrow a_4 = a_{3+1} = a_3 + 4 = 10 + 4 = 14$

Some of the terms of the sequence are 2, 6,10, 14, …

$\Rightarrow$ The sequence is AP with $d = 4$

$\Rightarrow s_n = \frac{n}{2}[2a_1 + (n-1)d]$

$\Rightarrow s_{35} = \frac{35}{2}[2 \times 2 + (35-1)4]$

$= \frac{35}{2}[4 + 34 \times 4]$

$= \frac{35}{2}[140]$

$= 35 \times 70$

$= 2450$

∴ ***Answer C***

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11. What is the value of $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{\frac{-x}{2}}$?

A. $\frac{1}{\sqrt{e}}$ B. $\sqrt{e}$ C. e^{-2} D. ∞

Solution: $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{\frac{-x}{2}}$

Since $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{mx}=e^{m}$

$\Rightarrow\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{\frac{-x}{2}}=e^{-\frac{1}{2}}=\frac{1}{\sqrt{e}}$

∴ ***Answer A***

12. Among students who took a quiz, 15 students scored 6, 20 students scored 7,10 students scored 8 and 5 students scored 10. What is the average score of the students?

A. 7.8 B. 7.5 C. 7.2 D. 7.0

Solution:

V	6	7	8	10
f	15	20	10	5

$$\bar{x}(\text{Average})=\frac{\sum_{i=1}^{4}fixi}{\sum_{i=1}^{4}fi}=\frac{6\times15+7\times20+8\times10+10\times5}{15+20+10+5}$$

$$=\frac{90+140+80+50}{50}$$

$$=\frac{360}{50}$$

$$=7.2$$

∴ ***Answer C***

13. For real numbers x and y, which one of the following statements is true?

A. $(\forall x)(\exists y)(x^2+y+1=0)$

B. $(\exists x)(\forall y)(x^2+y+1=0)$

C. $(\exists y)(\forall x)(x^2+y+1=0)$

D. $(\forall y)(\exists x)(x^2+y+1=0)$

Solution: Note the following

$\forall_x\exists_y P(x,y)$: For every x there is a corresponding y that satisfy the property P.

$\Rightarrow(\forall_x)(\exists_y)(x^2+y+1=0)$

$\Rightarrow x^2+y+1=0$

$\Rightarrow y=-1-x^2$ if $x\in\mathbb{R}$, then there is $y\in\mathbb{R}$.

$\therefore(\forall_x)(\exists_y)$: $x^2+y+1=0$ is T.

$(\forall_y)(\exists_x)(x^2+y+1=0)$

$x^2=-1-y$. If $y\in\mathbb{R}$, then x is not necessarily in $\mathbb{R}$.

$\therefore(\forall_y)(\exists_x)$ $x^2+y+1=0$ is F.

∴ ***Answer A***

14. Which one of the following functions has **NO** vertical asymptote?

A. $f(x)=\ln(x+1)$ C. $f(x)=\frac{x^2-9}{x-3}$

B. $f(x)=\frac{x^2+1}{x^3+8}$ D. $f(x)=\frac{x-1}{x^2-x}$

Solution: If $f(x)=\frac{p(x)}{q(x)}$ is a rational function, then $x=a$ is its vertical asymptote iff $q(a)=0$ and $p(a)\neq0$.

$f(x)=\frac{x^2-9}{x-3}=\frac{(x-3)(x+3)}{x-3}=x+3,\ x\neq3$ has no vertical asymptote.

∴ ***Answer C***

15. Let p, q and r be propositions such that $p\Rightarrow(r\vee\neg q)$ is false. Then, which one of the following propositions is true?

A. $p\Rightarrow r$ B. $\neg r\Rightarrow q$ C. $\neg p\Leftrightarrow q$ D. $q\Leftrightarrow r$

Solution: $P\Rightarrow(r\vee\neg q)$ is F iff P is T and $(r\vee\neg q)$ is F

$\Rightarrow r\vee\neg q$ is F iff both r and $\neg q$ are F.

∴ P = T, r = F and q = T .

∴ $\neg r\Rightarrow q=\neg F\Rightarrow T=T$.

∴ ***Answer B***

16. If $f(x)=\frac{1}{e^x+1}$, then which of the following is equal to $f^{-1}(x)$?

A. $\ln(1-x)-\ln(x)$ B. $e^{-x}+1$ C. $\ln\left(\frac{1}{x+1}\right)$ D. $\frac{1}{e^{-x}+1}$

Solution: $f(x)=\frac{1}{e^x+1}$

Let $y=\frac{1}{e^x+1}$

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$\Rightarrow x = \frac{1}{e^y+1}$ .....Interchange x and y

$\Rightarrow xe^y + x = 1$..... Solve for y

$\Rightarrow e^y + 1 = \frac{1}{x}$

$\Rightarrow e^y = \frac{1}{x} - 1 = \frac{1-x}{x}$

$\Rightarrow \ln e^y = \ln\left[\frac{1-x}{x}\right]$

$\Rightarrow y = \ln\left(\frac{1-x}{x}\right)$

$\Rightarrow f^{-1}(x) = \ln\left(\frac{1-x}{x}\right)$

$\Rightarrow f^{-1}(x) = \ln(1-x) - \ln x$

∴ ***Answer A***

17. If $x^2 - 6x + y^2 + k = 0$ is equation of a circle with radius 2, then what is the value of k?

A. 13 B. 5 C. 4 D. −4

Solution: $x^2 - 6x + y^2 + k = 0$ ..... Circle with radius = 2

$\Rightarrow x^2 - 6x + 9 + y^2 + k = 9$

$\Rightarrow (x-3)^2 + y^2 = 9 - k$

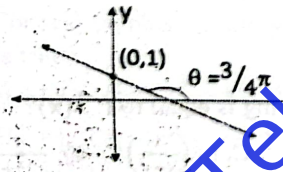
$\Rightarrow 9 - k = r^2 = 4$

$k = 5$

∴ ***Answer B***

18. If a line with angle of inclination of $\frac{3}{4}\pi$ passes through (0, 1), which one of the following is the equation of the line?

A. $y = -x + 1$ B. $y = x + 1$ C. $y = -x - 1$ D. $y = x - 1$



Solution:

$m = \tan\frac{3}{4}\pi = -1$

→ point-slope equation of a line becomes

$y - y_1 = m(x - x_1)$

$y - 1 = -1(x - 0)$

$y - 1 = -x$

$y = -x + 1$

∴ ***Answer A***

19. If $g(x) = xf(x) - \sqrt{f(x)}$ and $f(2) = f'(2) = 4$, then which of the following is equal to $g'(2)$?

A. 11 B. 8 C. 2 D. 0

Solution: $g(x) = xf(x) - \sqrt{f(x)}$, $f(2) = f'(2) = 4$.

$\Rightarrow g'(x) = [xf(x)]' - \left[\sqrt{f(x)}\right]'$

$\Rightarrow g'(x) = x'f(x) + xf'(x) - \frac{1}{2\sqrt{f(x)}} \times f'(x)$

$\Rightarrow g'(2) = f(2) + 2 \times f'(2) - \frac{1}{2\sqrt{f(2)}} \times f'(2)$

$= 4 + 2 \times 4 \frac{1}{2\sqrt{4}} \times 4$

$= 11$

∴ ***Answer A***

20. What is the sum of the series $\sum_{n=1}^{\infty} \frac{2^{2n+1}}{5^{n-1}}$?

A. 40 B. 20 C. 10 D. 8

Solution: $\sum_{n=1}^{\infty} \frac{2^{2n+1}}{5^{n-1}}$

Let $a_n = \frac{2^{2n+1}}{5^{n-1}}$, then $a_{n+1} = \frac{2^{2(n+1)+1}}{5^{n+1-1}} = \frac{2^{2n+3}}{5^n}$.

$\Rightarrow \frac{a_{n+1}}{a_n} = \frac{2^{2n+3}}{5^n} \times \frac{5^{n-1}}{2^{2n+1}} = \frac{2^{2n} \times 2^3}{5^n} \times \frac{5^n \times 5^{-1}}{2^{2n} \times 2} = \frac{4}{5}$.

∴ The sequence $\frac{2^{2n+1}}{5^{n-1}}$ is a GP with common ratio $r = \frac{4}{5}$

$\Rightarrow S_\infty = \sum_{n=1}^{\infty} G_n = \frac{G_1}{1-r} = \frac{2^3}{1 - \frac{4}{5}} = \frac{8}{1/5} = 40$

∴ ***Answer A***

21. Which one of the following is equal to $\lim_{n\to\infty} \frac{1-n-3n^2}{6n^2+1}$?

A. $\frac{1}{6}$ B. $\frac{-1}{2}$ C. $\frac{-1}{6}$ D. $-\infty$

Solution: $\lim_{n\to\infty} \frac{1-n-3n^2}{6n^2+1} = \lim_{n\to\infty} \frac{\frac{1}{n^2} - \frac{n}{n^2} - \frac{3n^2}{n^2}}{\frac{6n^2}{n^2} + \frac{1}{n^2}}$

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$= \lim_{n\to\infty} \frac{\frac{1}{n^2} - \frac{1}{n} - 3}{6 + \frac{1}{n^2}} = -\frac{3}{6} = -\frac{1}{2}$

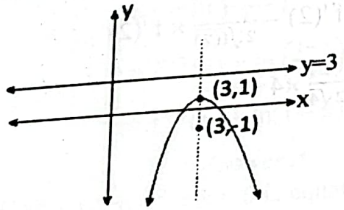
∴ ***Answer B***

22. A parabola with focus at (3,-1) has directrix y = 3. Which one of the following is the equation of the parabola?

A. $(x-3)^2 = -4(y+1)$ C. $(x-3)^2 = 4(y+1)$

B. $(x-3)^2 = -8(y-1)$ D. $(x-3)^2 = 8(y-1)$

Solution: F(3,-1), directrix: y = 3



It is a y-parabola opening downwards whose vertex is (3, 1) and p = 2

⇒ Its equation has a form $(x-h)^2 = -4p(y-k)$

⇒ $(x-3)^2 = -8(y-1)$

∴ ***Answer B***

23. The volume V of a melting ice cube after t seconds is $V = 2000 - 40t + 0.2t^2$(in cm^3). How fast is the volume changing when t = 40 seconds?

A. $24cm^3/sec$ C. $15cm^3/sec$

B. $15cm^3/sec$ D. $-24cm^3/sec$

Solution: $V = (2000 - 40t + 0.2t^2)\ cm^3$

$\frac{dv}{dt} = -40 + 0.4t$ (The rate of change of volume w.r.t time)

$\Rightarrow \left.\frac{dv}{dt}\right|_{t=40\ se} = -40 + 0.4 \times 40$

$= -24 cm^3/sec$

∴ ***Answer D***

24. The population of certain country is currently 80 million with growth rate of 2% per year.

Given: $(0.02)^9 = 5.12 \times 10^{-16}$, $(1.02)^9 = 1.19$

$(0.02)^{10} = 1.024 \times 10^{-17}$, $(1.02)^{10} = 1.22$

Which one of the following is the best approximation of the population after 10 years?

A. 81.9 million C. 95.2 million

B. 86.8 million D. 97.6 million

Solution: Currently = 80 million

After one year = 80 million + $\frac{2}{100} \times$ 80 million

= 80 million (1 + 0.02)

= 80 million $(1.02)^1$

After two years = 80 million (1.02) + $\frac{2}{10}$(80 million (1.02))

= 80 million $(1.02)^2$

Using this technique, after 10 years implies 80 million $(1.02)^{10}$

= 80 million × 1.22

= 97.6 million

∴ ***Answer D***

25. In the set of complex numbers, which one of the following is the solution set of $z^3 - iz^2 + 2z = 0$?

A. {0} B. {0,−i} C. {0,−i, 2i} D. {0, i,− 2i}

Solution: $z^3 - iz^2 + 2z = 0$

$\Rightarrow z(z^2 - iz + 2) = 0$

$\Rightarrow z = 0$ or $z^2 - iz + 2 = 0$

$\Rightarrow z = \frac{i \pm \sqrt{(-i)^2 - 8}}{2} = \frac{i \pm \sqrt{-9}}{2} = \frac{i \pm 3i}{2}$

$\Rightarrow z = -i$, or $z = 2i$

Solution set = {0, −i, 2i}

∴ ***Answer C***

26. What is the solution set of $\frac{2}{x} - \frac{x-2}{x^2-2x} = 1 - \frac{2x-2}{3x-2}$?

A. {1,-2} B. {1,2} C. {-1} D. {1}

Solution: $\frac{2}{x} - \frac{x-2}{x^2-2x} = 1 - \frac{2x-2}{3x-2}$

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$$\Rightarrow \frac{2}{x} - \frac{x-2}{x(x-2)} = 1 - \frac{2x-2}{3x-2}$$

$$\Rightarrow \frac{2}{x} - \frac{1}{x} = \frac{3x-2-(2x-2)}{3x-2}, \; x \neq 2, \frac{2}{3}$$

$$\Rightarrow \frac{1}{x} = \frac{x}{3x-2}$$

$$\Rightarrow 3x - 2 = x^2$$

$$\Rightarrow x^2 - 3x + 2 = 0$$

$$\Rightarrow (x-2)(x-1) = 0$$

$$\Rightarrow x = 2, x = 1$$

T.S = {1}

∴ ***Answer D***

27. At which value(s) of x does, $f(x) = \frac{1}{4}x^4 - 2x^2$ have a local maximum?

A. x = 4 B. x = 0 C. x = -2 and x = 2 D. x = 0 and x = 2

Solution: $f(x) = \frac{1}{4}x^4 - 2x^2$

$$f'(x) = x^3 - 4x = x(x^2 - 4) = x(x-2)(x+2)$$

Therefore , critical numbers: x = −2, 0, 2

	−2		0		2	
f'(x)	−	0	+	0	−	0	+
	decrease		increase		decrease		increase

⇒ f has a local maximum at x = 0.

∴ ***Answer B***

28. What is an antiderivative of $f(x) = \frac{2}{4x^2+4x+1}$?

A. $\frac{1}{2x+1}$ B. $\frac{-2}{2x+1}$ C. $\frac{-1}{2x+1}$ D. $\ln(4x^2 + 4x + 1)$

Solution: $f(x) = \frac{2}{4x^2+4x+1}$

$$\Rightarrow \int f(x)dx = \int \frac{2}{4x^2+4x+1} dx = 2\int \frac{1}{(2x+1)^2} dx$$

Let u = 2x + 1

$$\frac{du}{dx} = 2$$

$$\Rightarrow \int f(x)\,dx = 2\int \frac{1}{u^2} \times \frac{du}{2} = \int \frac{1}{u^2} du = \int u^{-2} du$$

$$\Rightarrow \int f(x)\,dx = \frac{u^{-2+1}}{-2+1} + c$$

$$\Rightarrow \int f(x)\,dx = -\frac{1}{u} + c$$

$$\Rightarrow \int f(x)\,dx = -\frac{1}{2x+1} + c$$ .. Where c is arbitrary constant

∴ ***Answer C***

29. How many four digit even numbers can be formed from 1,2,3,4 and 5 if the numbers start with 3?

A. 40 B. 50 C. 100 D. 120

Solution:

3	5	5	2

(start with 3) Unit place. (ending with even №s: 2,4)

№ of four digit №s: 1×5×5×2 = 50. (If repetition is allowed)

∴ ***Answer B***

30. A satellite moves along a hyperbolic curve whose horizontal transverse axis is 24 km and an asymptote $y = \frac{5}{12}x + 2$. Then, what is the eccentricity of the hyperbola?

A. $\frac{5}{13}$ B. $\frac{12}{13}$ C. $\frac{13}{12}$ D. $\frac{15}{3}$

Solution: 2a = 24km, ⇒ a = 12km

Asymptote: $y = \frac{5}{12}x + 2$

$$y = \frac{b}{a}(x - h) + k \Rightarrow \frac{b}{a} = \frac{5}{12} \Rightarrow b = 5$$

$$c^2 = a^2 + b^2 = 12^2 + 5^2$$

$$c = \sqrt{169} = 13$$

$$\Rightarrow e = \frac{c}{a} = \frac{13}{12}$$

∴ ***Answer C***

31. The mark that students scored in an examination is grouped in class intervals as shown in the following table.

Class Interval (Mark)	Number of students
55 – 64	8
65 – 74	12
75 – 84	20
85 – 94	6
95 – 100	4

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What is the median of the mark?

A. 25.0 B. 75.5 C. 77.0 D. 79.5

Solution:

Cl	f	Cf
55 – 64	8	8
65 – 74	12	20
75 – 84	20	40
85 – 94	6	46
95 – 100	4	50

Median class: $\left(\frac{1}{2}\right)^{th}$ item, where $n = \sum f = 50 = 25^{th}$ item

∴ Median class: 75 – 84

$$\Rightarrow M_d = B_L + \frac{\left[\frac{n}{2} - cf_b\right]}{f_c} \times i$$

$$= 74.5 + \frac{[25 - 20]}{20} \times 10$$

$$= 74.5 + \frac{5}{20} \times 10$$

$$= 74.5 + 2.5 = 77$$

∴ ***Answer C***

32. Given $f(x) = \ln(x - 1)$ and $g(x) = \sqrt{1 - 2x}$, which one of the following is the domain of fog?

A. $\{x \in \mathbb{R}: x > 1\}$ C. $\{x \in \mathbb{R}: x < 0\}$

B. $\{x \in \mathbb{R}: x \leq \frac{1}{2}\}$ D. $\{x \in \mathbb{R}: x > \frac{1}{2}\}$

Solution: $f(x) = \ln(x - 1)$, $g(x) = \sqrt{1 - 2x}$

dom fog = $\{x: x \in \text{dom } g \text{ and } g(x) \in \text{dom } f\}$

$= \{x: x \in \left(-\infty, \frac{1}{2}\right] \text{ and } \sqrt{1 - 2x} \in (1, \infty)\}$

$= \{x: x \in \left(-\infty, \frac{1}{2}\right] \text{ and } \sqrt{1 - 2x} > 1\}$

$= \{x: x \in \left(-\infty, \frac{1}{2}\right] \cap x \in (-\infty, 0)\}$

$= \{x: x \in (-\infty, 0)\} = \{x \in \mathbb{R}: x < 0\}$

∴ ***Answer C***

33. A committee consisting of 3 students is to be selected from 10 candidates among which 4 are girls. What is the probability that at least one girl is selected?

A. $\frac{5}{6}$ B. $\frac{2}{3}$ C. $\frac{1}{3}$ D. $\frac{1}{6}$

Solution: Number of candidates = 10

Number of girls = 4

Then, number of boys = 6 and number of sample space = $10C_3$

$\Rightarrow n(S) = 120$

At least one girl = 1 girl and 2 boys or 2 girls and 1 boy or 3 girls and 0 boy

$$P(\text{at least one girl}) = \frac{4C_1 6C_2 + 4C_2 6C_1 + 4C_3 6C_0}{10C_3}$$

$$= \frac{\frac{4!}{1!3!}\frac{6!}{2!4!} + \frac{4!}{2!2!}\frac{6!}{1!5!} + \frac{4!}{3!1!}\frac{6!}{0!6!}}{\frac{10!}{3!7!}}$$

$$= \frac{100}{120} = \frac{5}{6}$$

∴ ***Answer A***

34. Which one of the following is equal to $\int (1 + x)3^x dx$?

A. $(1 + x)3^x - 3^x + c$

B. $(1 + x)3^x + (\log_3 e)3^x + c$

C. $(1 + x)3^x \log_3 e - (\log_3 e)^2 3^x + c$

D. $(1 + x)3^x \log_3 e - 3^x(\log_3 e) + c$

Solution: $\int (1 + x)3^x dx$

Let $u = 1 + x \Rightarrow dv = 3^x dx$

$du = dx \Rightarrow V = \int 3^x dx = \frac{3^x}{\ln 3}$

using integration by parts: $\int u dv = uv - \int v du$

$$\int (1 + x)3^x dx = (1 + x) \times \frac{3^x}{\ln 3} - \int \frac{3^x}{\ln 3} dx$$

$$\int (1 + x)3^x dx = (1 + x) \times \frac{3^x}{\log_e^3} - \frac{1}{\log_e^3} \int 3^x dx$$

$$\int (1 + x)3^x dx = (1 + x)3^x \log_3^e - (\log_3^e)^2 3^x + C$$

∴ ***Answer C***

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35. Let $f(x) = \begin{cases} 3^x + k, & \text{if } x \le 0 \\ 3\frac{\sin(2x)}{x}, & \text{if } x > 0 \end{cases}$

If f is continuous at x = 0, then what is the value of k?

A. 6 B. 5 C. 2 D. 0

Solution: $f(x) = \begin{cases} 3^x + k, & x \le 0 \\ \frac{3\sin 2x}{x}, & x > 0 \end{cases}$

f is continuous at 0 implies $\lim_{x\to 0} f(x) = f(0)$

$\Rightarrow \lim_{x\to 0^-} f(x) = \lim_{x\to 0^+} f(x) = f(0)$

$\Rightarrow \lim_{x\to 0^-} 3^x + k = \lim_{x\to 0^+} \frac{3\sin 2x}{x} = 3^0 + k$

$6 = 1 + k$

$\mathbf{k = 5}$

$\therefore$ ***Answer B***

36. What is the value of $\int_0^{\frac{\pi}{2}} 2x\cos x\, dx$?

A. $\pi - 2$ B. $\frac{\pi}{2} + 1$ C. $\pi + 2$ D. $\frac{\pi}{2} - 1$

Solution: $\int_0^{\frac{\pi}{2}} 2x\cos x\, dx$

To apply integration by parts

Let $u = 2x \Rightarrow dv = \cos x\, dx$

$\frac{du}{dx} = 2 \Rightarrow V = \int \cos x\, dx = \sin x$

$\int udv = uv - \int vdu$

$\int_0^{\frac{\pi}{2}} 2x\cos x\, dx = 2x \sin x - \int \sin x \times 2\, dx$

$\int_0^{\frac{\pi}{2}} 2x\cos x = 2x \sin x - 2\int \sin x\, dx$

$\int_0^{\frac{\pi}{2}} 2x\cos x = 2x \sin x + 2\cos x$

$\int_0^{\frac{\pi}{2}} 2x\cos x = 2x \sin x + 2\cos x \Big|_0^{\frac{\pi}{2}}$

$\int_0^{\frac{\pi}{2}} 2x\cos x\, dx = 2x\frac{\pi}{2}\sin\frac{\pi}{2} + 2\cos\frac{\pi}{2} - (2\times 0\times\sin 0 + 2\cos 0)$

$\int_0^{\frac{\pi}{2}} 2x\cos x\, dx = \pi - 2$

$\therefore$ ***Answer A***

37. Consider the following argument:

"If he does not love her, she will not marry him.

He loves her. Therefore, she will marry him."

If "p ≡ He loves her" and "q ≡ she will marry him", which one of the following is the correct representation of the argument and its validity?

A. $\neg p \Rightarrow \neg q, p \vdash q$; Valid argument

B. $\neg p \Rightarrow \neg q, p \vdash q$; Invalid argument

C. $p \Rightarrow q, p \vdash q$; Valid argument

D. $p \Rightarrow q, p \vdash q$; Invalid argument

Solution: P ≡ He loves her

q ≡ She will marry her

Argument: If he does not love her, she will not marry him.

He loves her

Therefore, she will marry him.

$\equiv \neg p \Rightarrow \neg q, P \vdash q$

Validity test: Let P ≡ T .....premise

$\neg p \Rightarrow \neg q \equiv T$ .....premise

$F \Rightarrow \neg q \equiv T$

$\therefore \neg q$ = either T or F.

$\therefore$ The argument is invalid

$\therefore$ ***Answer B***

38. A group of six students take their seats at random in a round table for a discussion. What is the probability that two specific students do **NOT** sit together?

A. $\frac{3}{5}$ B. $\frac{2}{3}$ C. $\frac{2}{5}$ D. $\frac{1}{3}$

Solution:

Number of seats of 6 student around a table = (6-1)! = 5! = 120

If two specific students sit together, then there are 5 students so (5-1)! = 4! = 24

$\Rightarrow$ P(two specific students do not sit together) $= 1 - 2\left(\frac{24}{120}\right)$

$= 1 - \frac{48}{120} = \frac{3}{5}$

$\therefore$ ***Answer A***

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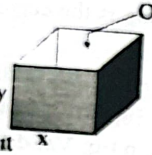
39. A box seen below is to have a square base, an open top and volume of 32 cubic unit. If x is the length of each side of its base and y is its height, how many units should x and y be in order to make the box with the smallest amount of material?

A. $x = 4, y = 2$

B. $x = 2, y = 8$

C. $x = \sqrt{8}, y = 4$

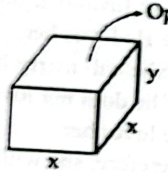
D. $x = \sqrt{2}, y = 16$



Solution: $V = x^2y = 32$ cubic unit

$S = x^2 + 4xy$

$S = x^2 + 4x\left(\frac{32}{x^2}\right)$



$S(x) = x^2 + \frac{128}{x}$

$\frac{dS}{dx} = 2x - \frac{128}{x^2}$

Critical number: $2x - \frac{128}{x^2} = 0$

$\Rightarrow 2x = \frac{128}{x^2} \Rightarrow x^3 = 64 \Rightarrow x = 4$

$\Rightarrow y = \frac{32}{x^2} = \frac{32}{16} = 2$

$\therefore$ ***Answer A***

40. Which one of the following is equal to $\int \frac{1}{x^2+x}\,dx$?

A. $\ln|x^2 + x| + c$

B. $2\ln|x + 1| + \ln|x| + c$

C. $\ln|x| - \ln|x + 1| + c$

D. $\ln|x| + \ln|x + 1| + c$

Solution: $\int \frac{1}{x^2+x}\,dx = \int \frac{1}{x(x+1)}\,dx$

Using partial fraction $\frac{1}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1}$

$\frac{1}{x(x+1)} = \frac{1}{x} - \frac{1}{x+1}$

$\Rightarrow \int \frac{1}{x^2+x}\,dx = \int \frac{1}{x}\,dx - \int \frac{1}{x+1}\,dx$

$\Rightarrow \int \frac{1}{x^2+x}\,dx = \ln|x| - \ln|x + 1| + c$

$\therefore$ ***Answer C***

41. Which one of the following is the equation of the line tangent to the graph of $f(x) = \frac{1}{x+1} + \cos x$ at $(0, f(0))$?

A. $x + y = 1$

B. $x - y = -2$

C. $x + y = 2$

D. $x + 4y = 2$

Solution: $f(x) = \frac{1}{x+1} + \cos x$, at $(0, f(0))$

$f'(x) = \left(\frac{1}{x+1} + \cos x\right)' = \frac{-1}{(x+1)^2} - \sin x$

$m = f'(0) = \frac{-1}{(0+1)^2} - \sin 0 = -1$

Equation of the tangent line at $(0, f(0)) = y - f(0) = f'(0)(x - 0)$

$y - 2 = -1(x)$

$y = -x + 2$

$\Rightarrow x + y = 2$

$\therefore$ ***Answer C***

42. Which one of the following is equal to $\int \frac{\ln(xe^x)}{x}\,dx$?

A. $\ln|x| + \frac{1}{2}e^x + c$

B. $\frac{1}{2}(\ln x)^2 + x + c$

C. $\ln|x| + e^{2x} + c$

D. $-\frac{1}{x^2} + (\ln x)^2 + c$

Solution: $\int \frac{\ln(xe^x)}{x}\,dx = \int \left(\frac{\ln x + \ln e^x}{x}\right)dx$

$= \int \left(\frac{\ln x}{x} + \frac{x \ln e}{x}\right)dx$

$= \int \left(\frac{\ln x}{x} + 1\right)dx$

$= \int \frac{\ln x}{x}\,dx + \int 1\,dx$

$= \int \frac{1}{x}\ln x\,dx + x + c$

Let $u = \ln x \Rightarrow dv = \frac{1}{x}\,dx$

$\frac{du}{dx} = \frac{1}{x} \Rightarrow v = \int \frac{1}{x}\,dx = \ln x$

$\Rightarrow \int \frac{1}{x} \ln x\,dx = \ln x \times \ln x - \int \ln x \times \frac{1}{x}\,dx$

$\Rightarrow \int \frac{1}{x} \ln x\,dx + \int \ln x \times \frac{1}{x}\,dx = (\ln x)^2$

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$$\Rightarrow 2\int \frac{1}{x}\ln x\,dx = (\ln x)^2$$

$$\Rightarrow \int \frac{1}{x}\ln x\,dx = \frac{1}{2}(\ln x)^2$$

$$\therefore \int \frac{\ln(xe^x)}{x}dx = \frac{1}{2}(\ln x)^2 + x + c$$

∴ ***Answer B***

43. Which one of the following is equal to $\lim_{x\to 1}\frac{1-x}{1-\frac{1}{x^2}}$?

A. 1 B. 0 C. $\frac{-1}{2}$ D. Doesn't exist

Solution: $\lim_{x\to 1}\frac{1-x}{1-\frac{1}{x^2}} = \lim_{x\to 1}\frac{1-x}{\frac{x^2-1}{x^2}}$

$$= \lim_{x\to 1}\frac{(1-x)x^2}{x^2-1}$$

$$= \lim_{x\to 1}\frac{(1-x)x^2}{(x-1)(x+1)}$$

$$= \lim_{x\to 1}\frac{-x^2}{x+1} = \frac{-1}{2}$$

∴ ***Answer C***

44. Consider the system $\begin{cases} \alpha x + y + z = 1 \\ x + 2y + 4z = 0 \\ 5x - y + z = 0 \end{cases}$

If the determinant of the coefficient matrix is 2, then what is the solution of the system of equations?

A. $\left(3\alpha, \frac{19\alpha}{2}, \frac{-11\alpha}{2}\right)$ C. $\left(3, \frac{-19}{2}, \frac{11}{2}\right)$

B. $\left(3, \frac{19}{2}, \frac{-11}{2}\right)$ D. $\left(\frac{3}{2}, \frac{19}{2}, \frac{-9}{2}\right)$

Solution: $\begin{cases} \alpha x + y + z = 1 \\ x + 2y + 4z = 0 \\ 5x - y + z = 0 \end{cases}$

Coefficient matrix $= \begin{pmatrix} \alpha & 1 & 1 \\ 1 & 2 & 4 \\ 5 & -1 & 1 \end{pmatrix}$ and its determinant is 2.

Using cramer's rule, $x = \frac{D_x}{D}$, $y = \frac{D_y}{D}$, and $z = \frac{D_z}{D}$

Where $D_x = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & -1 & 1 \end{vmatrix} = 6$

$D_y = \begin{vmatrix} \alpha & 1 & 1 \\ 1 & 0 & 4 \\ 5 & 0 & 1 \end{vmatrix} = -19$

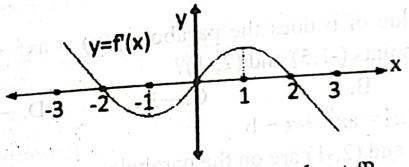
$D_z = \begin{vmatrix} \alpha & 1 & 1 \\ 1 & 2 & 0 \\ 5 & -1 & 0 \end{vmatrix} = -11$

$\Rightarrow x = \frac{6}{2} = 3,\ y = \frac{19}{2},\ z = \frac{-11}{2}$

Solution set $= \left\{\left(3, \frac{19}{2}, \frac{-11}{2}\right)\right\}$

∴ ***Answer B***

45. Suppose f is differentiable on $(-\infty, \infty)$ and the graph of its derivative is as shown below.



Which one of the following is true about f?

A. f is decreasing on $(-\infty,-1] \cup [1,\infty)$.

B. f has a local minimum at x = -2.

C. f is concave down on $[0,\infty)$.

D. f is concave up on (-1,1).

Solution: From the graph of f '(x), the zero's of f '(x) are -2,0,2 called **critical numbers.**

∴ f has a local maximum at x = -2 and x = 2 f has a local minimum at x = 0. On $(-\infty,-1) \cup (1,\infty)$, the slope of f '(x) is negative .Therefore, f "(x) < 0.

This implies, f is concave downward on $(-\infty,-1) \cup (1,\infty)$ similarly, on (-1, 1), the slope of f '(x) is positive. So, f "(x) > 0.

⇒ f is concave upward on (-1,1).

∴ ***Answer D***

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46. A box contains 5 white, 6 red and 4 black balls of identical size. If 3 balls are randomly taken out of the box one after the other, what is the probability that the first ball is white and both the second and third balls are red?

A. $\frac{2}{15}$ B. $\frac{3}{15}$ C. $\frac{4}{75}$ D. $\frac{5}{91}$

Solution: 3 balls are taken randomly are after the other implies there are 3 successive experiments is performed.

$n(s) = 5w + 6R + 4B = 15$

$\Rightarrow P(1^{st} \text{ ball is white}) = \frac{5}{15}$

$P(2^{nd} \text{ ball is red}) = \frac{6}{14}$

$P(3^{rd} \text{ ball is red}) = \frac{5}{13}$

$\therefore P(1^{st} \text{ ball is w and both } 2^{nd} \text{ and } 3^{rd} \text{ balls are red})$

$= \frac{5}{15} \times \frac{6}{14} \times \frac{5}{13} = \frac{5}{91}$

∴ Answer D

47. For what value of b does the parabola $p(x) = ax^2 + x + b$ pass through the points (-1,5) and (2,-1)?

A. 9 B. 3 C. −3 D. −15

Solution: $P(x) = ax^2 + x + b$

Let (-1,5) and (2,-1) are on the parabola

$\Rightarrow P(-1) = a(-1)^2 + (-1) + b$

$5 = a + b - 1$

$a + b = 6$.....**Equation1**

$\Rightarrow P(2) = a(2)^2 + 2 + b$

$-1 = 4a + 2 + b$

$4a + b = -3$.....**Equation2**

Solving Equation1and Equation2simultaneously,

$$\begin{cases} a+b=6 \\ 4a+b=-3 \end{cases}$$

$3a = -9$

$\Rightarrow a = -3$

$a+b = 6$

$-3+b = 6$

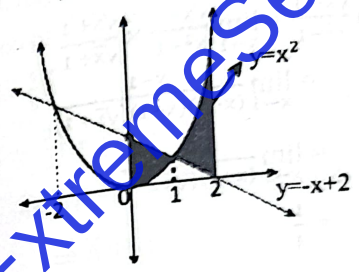
$\Rightarrow b = 9$

∴ Answer A

48. What is the area of the region between the graphs of $y = x^2$ and $y = -x + 2$, where $0 \le x \le 2$?

A. 3 B. 2 C. $\frac{3}{2}$ D. $\frac{2}{3}$

Solution:



Area of shaded region between two curves is

$A = \int_a^b [f(x) - g(x)]dx$, where $f(x) \ge g(x)$

$A = \int_0^1 [(-x+2) - x^2]dx + \int_1^2 (x^2 - (-x+2))dx$

$A = \frac{-x^3}{3} - \frac{x^2}{2} + 2x]_0^1 + \frac{x^3}{3} + \frac{x^2}{2} - 2x]_1^2$

$\therefore A = 3$

∴ Answer A

49. Which one of the following is true about the derivative of $f(x) = x|x|$?

A. f is not differentiable at $x = 0$.

B. $f'(x) = 2|x|$, for every $x \in (-\infty,\infty)$.

C. $f'(x) = 2x$, for every $x \in (-\infty,\infty)$.

D. $f'(x) = |x| + x$, for every $x \in (-\infty,\infty)$.

Solution: $f(x) = x|x|$

$$f(x) = \begin{cases} x^2, & \text{if } x \ge 0 \\ -x^2, & \text{if } x < 0 \end{cases}$$

$$f'(x) = \begin{cases} 2x, & \text{if } x \ge 0 \\ -2x, & \text{if } x < 0 \end{cases}$$

$f'(x) = 2|x|$ for every $x \in \mathbb{R}$

∴ Answer B

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50. Which one of the following is equal to $\lim_{x\to 1}\frac{\sqrt{x}-1}{x^2-1}$?

A. ∞ B. $\frac{1}{2}$ C. $\frac{1}{4}$ D. 0

Solution: $\lim_{x\to 1}\frac{\sqrt{x}-1}{x^2-1} = \lim_{x\to 1}\frac{\sqrt{x}-1}{x^2-1}\times\frac{\sqrt{x}+1}{\sqrt{x}+1}$

$$= \lim_{x\to 1}\frac{x-1}{(x-1)(x+1)(\sqrt{x}+1)}$$

$$= \lim_{x\to 1}\frac{1}{(x+1)(\sqrt{x}+1)}$$

$$= \frac{1}{4}$$

$\therefore$ ***Answer C***

51. Suppose P(1,2,1) and Q(1,0,2) are points in space and $\vec{A} = \overrightarrow{PQ}$. If $\vec{B}$ is parallel to $\overrightarrow{PQ}$ and $\vec{A}.\vec{B} = -10$, then which one of the following is true?

A. $\vec{A}$ and $\vec{B}$ has the same direction C. $\|\vec{B}\| = \frac{1}{10}\|\vec{A}\|$

B. $\|\vec{B}\| = 10\|\vec{A}\|$ D. $\|\vec{B}\| = 2\|\vec{A}\|$

Solution: P(1,2,1), Q(1,0,2) .... are points in space and

$\vec{A} = \overrightarrow{PQ} = (1,0,2) - (1,2,1) = (0,-2,1)$

$\vec{B} \parallel \overrightarrow{PQ} \Rightarrow \vec{B} = k\overrightarrow{PQ} = k(0,-2,1) = (0,-2k,k)$

$\vec{A}.\vec{B} = -10$

$(0,-2,1)\times(0,-2k,k) = -10$

$0 + 4k + k = -10$

$k = -2$. Thus, $\vec{B} = (0,4,-2)$

$\therefore \vec{A}$ and $\vec{B}$ are opposite in direction

$|\vec{A}| = \sqrt{0^2+(-2)^2+1^2} = \sqrt{5}$

$|\vec{B}| = \sqrt{0^2+4^2+(-2)^2} = \sqrt{20} = 2\sqrt{5}$

$\therefore |\vec{B}| = 2|\vec{A}|$

$\therefore$ ***Answer D***

52. Suppose $\vec{A} = 2\vec{j} - \vec{k}$ and $\vec{B} = 5\vec{i} + 15\vec{k}$, where $\vec{i}$, $\vec{j}$ and $\vec{k}$ are the standard unit vectors in the directions of positive x, y and z axis, respectively. Which one of the following is the unit vector in the direction of $\vec{A} + \frac{1}{5}\vec{B}$?

A. $\frac{3}{5}\vec{i} + \frac{4}{5}\vec{k}$ C. $\frac{4}{5}\vec{j} - \frac{3}{5}\vec{k}$

B. $\frac{1}{3}\vec{i} + \frac{2}{3}\vec{j} + \frac{2}{3}\vec{k}$ D. $\frac{2}{3}\vec{i} - \frac{1}{3}\vec{j} + \frac{2}{3}\vec{k}$

Solution: $\vec{A} = 2j - k$, $\vec{B} = 5i + 15k$

$\Rightarrow \vec{A} + \frac{1}{5}\vec{B} = (2j - k) + \frac{1}{5}(5i + 15k)$

$\Rightarrow \vec{A} + \frac{1}{5}\vec{B} = 2j - k + i + 3k$

$\Rightarrow \vec{A} + \frac{1}{5}\vec{B} = i + 2j + 2k$

$\Rightarrow$ The unit vector in the direction of $\vec{A} + \frac{1}{5}\vec{B}$ is

$$\frac{i+2j+2k}{|i+2j+2k|} = \frac{i+2j+2k}{\sqrt{1^2+2^2+2^2}}$$

$$= \frac{i+2j+2k}{3}$$

$$= \frac{1}{3}i + \frac{2}{3}j + \frac{2}{3}k$$

$\therefore$ ***Answer B***

53. A line given by a vector equation $r(t) = (0,3) + t(1,1)$ is tangent to a circle at point (0,3). If the radius of the circle is $\sqrt{2}$, which one of the following is the center of the circle?

A. (1,4) B. (1,−4) C. (−1,2) D. (1,2)

Solution:

$r(t) = (0,3) + t(1,1)$ ..... Vector equation of the tangent line.

$P_1(0,3)$ .....Point of tangency.

$r = \sqrt{2}$

$r(t) = (t, 3+t)$

$\Rightarrow (x,y) = (t, 3+t)$

$\Rightarrow \begin{cases} x = t \\ y = 3 + t \end{cases}$ ..... parametric equation of ℓ

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$\Rightarrow$ Rectangular form of ℓ becomes $y = x + 3$

Suppose $P_1(0,3)$ be a point of tangency, center of the circle $C(h, k)$ and $P(x, y)$ any arbitrary point on the tangent line.

$\Rightarrow \overrightarrow{P_1P} \cdot \overrightarrow{P_1C} = 0$

$\Rightarrow [(x, x+3) - (0, 3)] \times [(h, k-3)] = 0$

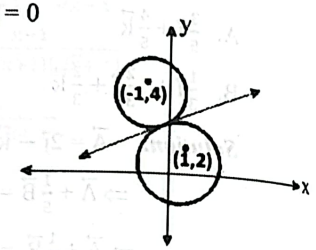
$\Rightarrow (x, x) \times (h, k-3) = 0$

$\Rightarrow hx + kx - 3x = 0$

$\Rightarrow x(h + k - 3) = 0$

$\Rightarrow h + k - 3 = 0$

$\Rightarrow h + k = 3 \ldots (*)$



Distance from the center to the tangent line is

$h^2 + (k - 3)^2 = 2$

$\Rightarrow h^2 + (3 - h - 3)^2 = 2$

$\Rightarrow h^2 + h^2 = 2$

$\Rightarrow 2h^2 = 2 \Rightarrow h = \pm 1$

If $h = 1$, then $k = 2$, if $h = -1$, then $k = 4$.

$\therefore$ Center of the circle are $(1, 2)$ or $(-1, 4)$

$\therefore$ ***Answer D***

54. What is cot(arcsin x) if $0 < x < 1$?

A. $\frac{x}{\sqrt{1-x^2}}$ B. $\frac{1}{x}\sqrt{1-x^2}$ C. $\sqrt{1-x^2}$ D. $\frac{1}{\sqrt{1-x^2}}$

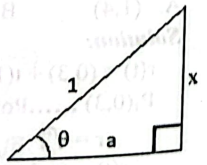
Solution: cot(arcsinx), if $0 < x < 1$

Let θ = arc sinx

$\Rightarrow \sin\theta = x = \frac{opp}{adj}$

By Pythagoras relation, $1^2 = x^2 + a^2$

$\Rightarrow a = \sqrt{1-x^2}$



$\Rightarrow \cot\theta = \frac{\cos\theta}{\sin\theta} = \frac{\frac{\sqrt{1-x^2}}{1}}{\frac{x}{1}} = \frac{\sqrt{1-x^2}}{x}$

$\therefore$ ***Answer B***

55. Which one of the following describes the principle of Mathematical Induction on a set of natural numbers?

A. If an assertion is true for a natural number n, then it is true for $n + 1$.

B. If an assertion is true for 1 and it is true for $n + 1$, then it is true for some n.

C. If an assertion holds for $n = 20$ and for any $n \geq 20$, then it is true for n implies true for $n + 1$.

D. If an assertion is true for $n = 1$, and is true for $n = k$, whenever is true for $n = k + 1$.

Solution: Principle of mathematical Induction on a set of natural numbers states that "If the given assertion is true for $n = 1$, and also true for $n = k + 1$, whenever it is true for $n = k$, then the assertion is true for every natural numbers n."

$\therefore$ **No answer**

56. If $\vec{u} = (-3, x)$ and $\vec{v} = (x, y - 2)$ are vectors, what is the value of y so that $\vec{u} + \vec{v} = 3\vec{u} - \frac{1}{2}\vec{v}$?

A. $\frac{2}{3}$ B. $-\frac{10}{3}$ C. -4 D. $-\frac{22}{3}$

Solution: If $\vec{u} = (-3, x)$ and $\vec{v} = (x, y - 2)$

$\Rightarrow \vec{u} + \vec{v} = 3\vec{u} - \frac{1}{2}\vec{v}$

$\Rightarrow \vec{v} + \frac{1}{2}\vec{v} = 3\vec{u} - \vec{u}$

$\Rightarrow \frac{3}{2}\vec{v} = 2\vec{u}$

$\Rightarrow \frac{3}{2}(x, y - 2) = 2(-3, x)$

$\Rightarrow \left(\frac{3}{2}x, \frac{3}{2}(y - 2)\right) = (-6, 2x)$

$\Rightarrow \frac{3}{2}x = -6$ and $\frac{3}{2}(y-2) = 2x$

$\Rightarrow x = -4$ and $\frac{3}{2}y - 3 = -8$

$\Rightarrow y = -\frac{10}{3}$

$\therefore$ ***Answer B***

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57. What is the solution set of $\cos^2 x + \frac{1}{2}\sin 2x = 1$ in the interval $[0,2\pi)$?

A. $\left\{0, \frac{\pi}{4}, \pi, \frac{5\pi}{4}\right\}$ C. $\{0,\pi\}$

B. $\left\{0, \frac{\pi}{4}, \frac{3\pi}{4}, \pi\right\}$ D. $\left\{0, \frac{\pi}{4}, \pi\right\}$

Solution: $\cos^2 x + \frac{1}{2}\sin 2x = 1, x \in [0,2\pi)$

$\Rightarrow \cos^2 x + \frac{1}{2} \times 2\sin x \cos x = 1$

$\Rightarrow \cos^2 x + \sin x \cos x = \cos^2 x + \sin^2 x$

$\Rightarrow \sin x \cos x = \sin^2 x$

$\Rightarrow \sin x \cos x - \sin^2 x = 0$

$\Rightarrow \sin x[\cos x - \sin x] = 0$

$\Rightarrow \sin x = 0$ or $\cos x - \sin x = 0$

$x = 0, \pi$ or $\cos x = \sin x \Rightarrow \frac{\cos x}{\cos x} = \frac{\sin x}{\cos x}$

$\Rightarrow \tan x = 1$

$\Rightarrow x = \frac{\pi}{4}, \frac{5\pi}{4}$

Solution set $= \left\{0, \frac{\pi}{4}, \pi, \frac{5\pi}{4}\right\}$

∴ Answer A

58. Which one of the following points is closer to the sphere $x^2 + y^2 + z^2 - 2x + 6z + 9 = 0$?

A. (1,0,0) B. (0,0,0) C. (0,-1,0) D. (0,0,-1)

Solution:

$x^2 + y^2 + z^2 - 2x + 6z + 9 = 0$

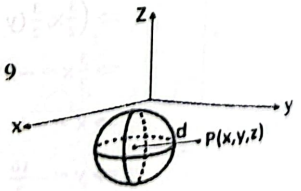
$x^2 - 2x + y^2 + z^2 + 6z = -9$

$(x-1)^2 + y^2 + (z+3)^2 = -9 + 1 + 9$

$(x-1)^2 + y^2 + (z+3)^2 = 1$

Center (1, 0, -3)

r = 1



Let P(x, y, z) be any arbitrary point in space, then

$d = \sqrt{(x-1)^2 + y^2 + (z+3)^2}$

if $(x-1)^2 + y^2 + (z+3)^2$ is minimum then P is very close to the sphere.

$(x-1)^2 + y^2 + (z+3)^2$ is minimum if $0 < x < 2$, $y = 0$ and $-3 \le z < 0$.

∴ Answer D

59. Which one of the following is equal to $\sec\left(\frac{\pi}{2} - x\right)\sin^3 x + \cos 2x$

A. $2\cos x$ B. $2\sin x$ C. $\cos^2 x$ D. $\sin^2 x$

Solution: $\sec\left(\frac{\pi}{2} - x\right)\sin^3 x + \cos 2x$

$= \frac{\sin^3 x}{\cos\left(\frac{\pi}{2} - x\right)} + \cos^2 x - \sin^2 x$

$= \frac{\sin^3 x}{\cos\frac{\pi}{2}\cos x + \sin\frac{\pi}{2}\sin x} + \cos^2 x - \sin^2 x$

$= \frac{\sin^3 x}{\sin x} + \cos^2 x - \sin^2 x$

$= \sin^2 x + \cos^2 x - \sin^2 x$

$= \cos^2 x$

∴ Answer C

60. If $f(x) = 2 - \frac{1}{2}\sin\left(\frac{\pi}{2}x\right)$, then which one of the following is the amplitude and period of f, respectively?

A. $\frac{1}{2}$ and 4 B. $\frac{-1}{2}$ and 4 C. 2 and π D. $\frac{1}{2}$ and π

Solution: $f(x) = 2 - \frac{1}{2}\sin\left(\frac{\pi}{2}x\right)$

$\Rightarrow f(x) = -\frac{1}{2}\sin\left(\frac{\pi}{2}x\right) + 2$

This has a form $f(x) = a \sin bx + c$

Amplitude $= |a| = \left|\frac{-1}{2}\right| = \frac{1}{2}$

Period $= \frac{2\pi}{|b|} = \frac{2\pi}{\pi/2} = 4$

∴ Answer A

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61. Which one of the following is necessarily true?

A. If $\|\vec{A}\| = \|\vec{B}\|$, then $\vec{A} = \vec{B}$.

B. $\|k\vec{A}\| = k\|\vec{A}\|$, for any real number k.

C. If $\vec{A}$ is parallel to $\vec{B}$, then $\vec{A}.\vec{B} = 0$.

D. If $\vec{u}$ is a unit vector in the direction of $\vec{A}$, then $\vec{A}.\vec{u} = \|\vec{A}\|$.

Solution:

✓ $|\vec{A}| = |\vec{B}|$, then $\vec{A}$ is not necessarily equal to $|\vec{B}|$

Example : $\vec{A} = (1,-1)$, $\vec{B} = (1,1)$

$|\vec{A}| = |\vec{B}| = \sqrt{2}$ but $\vec{A} \neq \vec{B}$.

✓ $|k\vec{A}| = k|\vec{A}|$ if $k \geq 0$ and $|kA| = |k||\vec{A}|$ if $k < 0$

✓ If $\vec{A} \parallel \vec{B}$, then $\vec{A} = k\vec{B}$, $k \in \mathbb{R}$ and $\vec{A} \perp \vec{B}$, then $\vec{A} \cdot \vec{B} = 0$.

✓ If $\vec{u}$ is a unit vector, then $\vec{A} \cdot \vec{u} = |\vec{A}|\,|\vec{u}|\cos\theta$

$= |\vec{A}|\cos 0°$ (since $\vec{u}$ and $\vec{A}$ are in the same direction)

$= |\vec{A}|$

∴ ***Answer D***

62. Let $\vec{a} = 2\vec{i} + (x-1)\vec{j} + \vec{k}$ and $\vec{c} = \vec{i} - \vec{j} + y\vec{k}$ be vectors. If $\vec{a}.\vec{c} = 0$ and $\|\vec{a}\| = 3$, which one of the following is a possible value of y?

A. -4 B. -1 C. 3 D. 4

Solution: $\vec{a} = 2i + (x-1)j + k$

$\vec{c} = i - j + yk \Rightarrow \vec{a}.\vec{c} = 0$ and $|\vec{a}| = 3$

$\Rightarrow \vec{a} = (2, x-1, 1)$; $\vec{c} = (1,-1,y)$ …In component form

$\vec{a}.\vec{c} = 2 + 1 - x + y = 0$

$\Rightarrow y = x - 3$

$|\vec{a}| = \sqrt{2^2 + (x-1)^2 + 1^2}$

$\Rightarrow 9 = 5 + (x-1)^2$

$\Rightarrow (x-1)^2 = 4$

$\Rightarrow x - 1 = \pm 2$

$\Rightarrow x = 3$ or $x = -1$

∴ The possible values of y are 0 and -4.

∴ ***Answer A***

63. If $\vec{A}$ is perpendicular to $\vec{B}$, what is the cosine of the angle between $\vec{A}$ and $\vec{A} - \vec{B}$?

A. $\frac{\|\vec{A}-\vec{B}\|}{\|\vec{A}\|}$ B. $\frac{\|\vec{A}\|}{\|\vec{A}-\vec{B}\|}$ C. $\frac{\|\vec{A}-\vec{B}\|}{\|\vec{B}\|}$ D. $\frac{\|\vec{B}\|}{\|\vec{A}-\vec{B}\|}$

Solution: $\vec{A} \perp \vec{B}$ and Let θ be the angle between $\vec{A}$ and $\vec{A} - \vec{B}$

$\Rightarrow \cos\theta = \frac{\vec{A}.(\vec{A}-\vec{B})}{|\vec{A}||\vec{A}-\vec{B}|} = \frac{\vec{A}.\vec{A} - \vec{A}.\vec{B}}{|\vec{A}||\vec{A}-\vec{B}|}$

$= \frac{|\vec{A}|^2}{|\vec{A}||\vec{A}-\vec{B}|}$ (since $\vec{A} \perp \vec{B}$; implies $\vec{A}.\vec{B} = 0$ and $\vec{A}.\vec{A} = \vec{A}^2 = |\vec{A}|^2$)

$= \frac{|\vec{A}|}{|\vec{A}-\vec{B}|}$

∴ ***Answer B***

64. Suppose the following statements are the premises of an argument.

"He was lazy or he did not like the classroom.

If he was lazy, he could not pass the exam.

He passed the exam."

Which one of the following can be a conclusion that makes the argument valid?

A. He did like the classroom.

B. He did not like the classroom.

C. If he was not lazy, he did like the classroom.

D. He was not lazy and he did like the classroom.

Solution: Let p = He was lazy

q = He like the class room

r = He pass the exam.

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Argument form: $p \vee \neg q$

$p \Rightarrow \neg r$

r

valid argument

$r = T$ .....premise

$p \Rightarrow \neg r = T$..... premise

$p \Rightarrow F = T$

$\therefore p = F$ .....law of $\Rightarrow$

$p \vee \neg q = T$ .....premise

$F \vee \neg q = T$

$\neg q = T$ .....valid

$\therefore$ Conclusion = He did not like the class room

$\therefore$ ***Answer B***

65. What is the image of the ellipse whose equation is $2(x+2)^2 + (y-1)^2 = 2$ under a translation that takes (2, 1) to (4,0) followed by a rotation of 90°?

A. $x^2 + 2y^2 = 2$ C. $2(x-4)^2 + y^2 = 2$

B. $2x^2 + y^2 = 2$ D. $(x-4)^2 + 2y^2 = 2$

Solution: $2(x+2)^2 + (y-1)^2 = 2$

$\Rightarrow \frac{(x+2)^2}{1} + \frac{(y-1)^2}{2} = 1$..... Vertical ellipse

$(T(2,1) = (4,0)$

$\Rightarrow$ Translation vector, $\vec{v} = (4-2, 0-1) = (2,-1)$

$\Rightarrow T(\text{center}) = T(-2,1) = (0,0)$

Rotate the ellipse 90° implies the ellipse becomes horizontal whose center is (0, 0) since $a = \sqrt{2}$, $b = 1$

$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\Rightarrow \frac{x^2}{2} + \frac{y^2}{1} = 1$

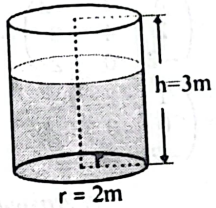
$\Rightarrow x^2 + 2y^2 = 2$

$\therefore$ ***Answer A***

1. A water tank is a circular cylinder with base radius 2m and height 3m. If the tank is empty and water is pumped into it at rate of $2m^3$/min, how long does it take for the tank to be full?

A. 1.5 min B. $\frac{3}{2}\pi$ min C. 6π min D. 12 min

Solution: $\frac{dv}{dt} = 2m^3/\text{min}$



$v = \pi r^2 h$

$\frac{dv}{dt} = \pi r^2 \frac{dh}{dt}$

$2 = \pi(2)^2 \frac{dh}{dt}$

$\frac{dh}{dt} = \frac{1}{2\pi}$

$\Rightarrow dh = \frac{1}{2\pi} dt$

$\int dh = \frac{1}{2\pi} \int dt$

$h = \frac{1}{2\pi} t$

$\Rightarrow 3 = \frac{1}{2\pi} t$

$\Rightarrow t = 6\pi$ min

$\therefore$ ***Answer C***

2. If $A = \begin{bmatrix} 2 & 0 & -1 \\ 1 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ and $(2A + B)^T = A^T A$, then which one of the following is equal to B?

A. $\begin{bmatrix} 1 & 0 & -2 \\ 2 & 0 & 0 \\ 0 & 0 & 4 \end{bmatrix}$ C. $\begin{bmatrix} 8 & 0 & -4 \\ 4 & 8 & 0 \\ 0 & 0 & -4 \end{bmatrix}$

B. $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ D. $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ -2 & 0 & 4 \end{bmatrix}$

Solution: $A = \begin{bmatrix} 2 & 0 & -1 \\ 1 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

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$\Rightarrow (2A+B)^T = A^TA \Rightarrow (2A)^T + B^T = A^TA \Rightarrow B^T = A^TA - 2A^T$

$\Rightarrow B = (A^TA - 2A^T)^T = (A^TA)^T - (2A^T)^T = A^T(A^T)^T - 2(A^T)^T$

$\Rightarrow B = A^TA - 2A$

$$= \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 0 & -1 \end{pmatrix}\begin{pmatrix} 2 & 0 & -1 \\ 1 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix} - 2\begin{pmatrix} 2 & 0 & -1 \\ 1 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 5 & 2 & -2 \\ 2 & 4 & 0 \\ -2 & 0 & 2 \end{pmatrix} - \begin{pmatrix} 4 & 0 & -2 \\ 2 & 4 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ -2 & 0 & 4 \end{pmatrix}$$

∴ *Answer D*

3. Which one of the following represents a geometric sequence?

A. $3,1,\frac{1}{3},\frac{1}{9},\frac{1}{27},\ldots$ C. 1,3,6,10,15,...

B. $\frac{1}{2},-\frac{1}{3},\frac{1}{4},-\frac{1}{5},\frac{1}{6},\ldots$ D. -3,6,-9,12,-15,...

Solution: The sequence G is called a geometric sequence, if every term $G_{n+1} = r\,G_n$, for all n = 1,2,3, ... and $r \in \mathbb{R}|\{0\}$.

This implies, $3,1,\frac{1}{3},\frac{1}{9},\frac{1}{27},\ldots$ is a geometric sequence.

∴ *Answer A*

4. The sequence $\left\{\frac{(n-1)(2n+1)}{1-n^2}\right\}_{n=1}^{\infty}$ converges to:

A. $-\infty$ B. -2 C. 0 D. 1

Solution: $\left\{\frac{(n-1)(2n+1)}{1-n^2}\right\}_{n=1}^{\infty}$

Let $a_n = \frac{(n-1)(2n+1)}{1-n^2}$

$\Rightarrow a_n = \frac{2n^2-n-1}{1-n^2}$

$\Rightarrow a_n = \frac{2-\frac{1}{n}-\frac{1}{n^2}}{\frac{1}{n^2}-1}$

$\Rightarrow \lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{2-\frac{1}{n}-\frac{1}{n^2}}{\frac{1}{n^2}-1} = -2$

∴ *Answer B*

5. If F(x) is an antiderivative of $f(x) = 1 - \frac{2}{x^2}$, and F(1) = 0, then F(2) is equal to:

A. 0 B. $\frac{1}{2}$ C. $\frac{-1}{2}$ D. 3

Solution: $f(x) = 1 - \frac{2}{x^2}$, F(1) = 0

$F(x) = \int f(x)dx = \int \left(1 - \frac{2}{x^2}\right)dx = \int 1dx - \int \frac{2}{x^2}dx$

$F(x) = x - 2\int x^{-2}dx = x - 2\left(\frac{x^{-2+1}}{-2+1}\right) + c$

$F(x) = x + \frac{2}{x} + c$

$F(1) = 1 + \frac{2}{1} + c$

$0 = 3 + c$

$c = -3$

∴ $F(x) = x + \frac{2}{x} - 3$

$F(2) = 2 + \frac{2}{2} - 3 = 0$

∴ *Answer A*

6. Which of the following expression is a polynomial expression?

A. $x^2 - 3x + \sin x$ C. $\frac{2+\pi}{1+\pi^2}$

B. $\frac{4x^3+12x^2-x}{\pi x^2}$ D. $2 - 3x^{2/3} + 7x^{5/2} + 3x^{-1}$

Solution: A polynomial expression is an expression that can be written in the form $a_nx^n + a_{n-1}x^{n-1} + \ldots + a_1x^1 + a_0$ where $n \in \mathbb{N}$, and $a_n, a_{n-1}, \ldots, a_1, a_0$ are constants.

∴ *Answer C*

7. What is the distance from the origin to the line that passes through (1,0) and (0,1)?

A. $\frac{1}{2}\sqrt{2}$ B. 1 C. $\frac{1}{2}$ D. $\sqrt{2}$

Solution: Equation of a line that passes through (1,0) and (0,1) is

$y - 0 = \frac{1-0}{0-1}(x-1)$ (i.e $y - y_1 = \frac{y_2-y_1}{x_2-x_1}(x - x_1)$

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$\Rightarrow y = -x + 1$

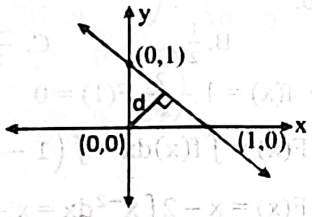
$\Rightarrow y + x - 1 = 0$

$d = \left|\frac{Ax_0 + By_0 + c}{\sqrt{A^2 + B^2}}\right|$.. where A = 1, B = 1, c = -1 and $x_0 = 0, y_0 = 0$

$= \left|\frac{1\times 0 + 1\times 0 - 1}{\sqrt{1^2 + 1^2}}\right|$

$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$

$= \frac{\sqrt{2}}{2}$



∴ ***Answer A***

8. If the list of a measurement is 10, α, 5, α,5, 10,20,15,20,5 with mean $\bar{x}$, then what is the value of α in terms of $\bar{x}$?

A. $10\bar{x} - 90$ B. $9\bar{x} - 90$ C. $5\bar{x} - 90$ D. $5\bar{x} - 45$

Solution: $\bar{x} = \frac{10 + \alpha + 5 + \alpha + 5 + 10 + 20 + 15 + 10 + 5}{10}$

$\bar{x} = \frac{90 + 2\alpha}{10} = \frac{45 + \alpha}{5}$

$5\bar{x} - 45 = \alpha$

∴ ***Answer D***

9. The total cost (in Birr) of producing x iron sheets per day is $C(x) = 1{,}000 + 100x - 0.5x^2$, $0 \le x \le 100$. What is the marginal (rate of change of) cost at a production level of 80 iron sheets?

A. 8.5 B. 20 C. 1,800 D. 5,800

Solution: $C(x) = 1000 + 100x - 0.5x^2$, $0 \le x \le 100$

$\Rightarrow M(x) = C'(x) = 100 - x$

$\Rightarrow M(80) = 100 - 80 = 20$

∴ ***Answer B***

10. If z = x + yi is a complex number, then $|z|^2 + \frac{1}{2}(z - \bar{z})^2 = 1$ is equivalent to which one of the following equations?

A. $x^2 - y^2 = 1$ C. $x^2 - y^2 = 2$

B. $x^2 - 3iy^2 = 1$ D. $2x^2 - y^2 = 2$

Solution: z = x + yi

$\Rightarrow |z|^2 = x^2 + y^2, \bar{z} = x - yi$

$\Rightarrow |z|^2 + \frac{1}{2}(z - \bar{z})^2 = 1$

$\Rightarrow x^2 + y^2 + \frac{1}{2}(x + yi - (x - yi))^2 = 1$

$\Rightarrow x^2 + y^2 + \frac{1}{2}((2yi))^2 = 1$

$\Rightarrow x^2 + y^2 + \frac{1}{2} \times 4y^2 i^2 = 1$..... But $i^2 = -1$

$\Rightarrow x^2 - y^2 = 1$

∴ ***Answer A***

11. The following is the frequency distribution of a grouped data.

Class Intervals	Frequency (f)
3 – 7	2
8 – 12	2
13 –17	10
18 – 22	6

What is the mean and standard deviation of the distribution, respectively?

A. 15, $2\sqrt{5}$ B. 15, $\sqrt{7.5}$ C. 12.5, $5\sqrt{2}$ D. 12.5, $\sqrt{15}$

Solution:

CI	f	XC	$f_c x_c$	$(x_c - \bar{x})^2$	$f_c(x_c - \bar{x})^2$
3–7	2	5	10	100	200
8 – 12	2	10	20	25	50
13 – 17	10	15	150	0	0
18 – 22	6	20	120	25	150
	$\sum f = 20$		$\sum f_c x_c = 300$		$\sum f_c(x_c - \bar{x})^2 = 400$

Where f_c = frequency of the class

x_c = class mark of the class

$\Rightarrow \bar{x} = \frac{\sum f_c x_c}{\sum f} = \frac{300}{20} = 15$

$\Rightarrow \sigma = \sqrt{\frac{\sum f_i(x_c - \bar{x})^2}{\sum f}}$

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$\Rightarrow \sigma = \sqrt{\frac{400}{20}} = \sqrt{20} = 2\sqrt{5}$

∴ ***Answer A***

12. If $(p \vee q) \Rightarrow (\neg r \wedge r)$ is true, then which one of the following is necessarily true?

A. $(p \vee r) \Rightarrow q$ B. $\neg q \wedge r$ C. $\neg p \Leftrightarrow r$ D. $\neg p \vee r$

Solution: $(p \vee q) \Rightarrow (\neg r \wedge r) = T$

Since is $\neg r \wedge r$ always false, then $(p \vee q) \Rightarrow F = T$

$\Rightarrow p \vee q$ is F

$\Rightarrow p = F$ and $q = F$

$\neg p \vee r = T \vee r = T$

∴ ***Answer D***

13. What is the actual value of the sum $\sum_{n=1}^{\infty}\left(\frac{2^n+5^n}{10^n}\right)$?

A. 0.325 B. 1 C. $\frac{5}{4}$ D. $\frac{37}{9}$

Solution: $\sum_{n=1}^{\infty}\left(\frac{2^n+5^n}{10^n}\right) = \sum_{n=1}^{\infty}\left(\left(\frac{2}{10}\right)^n + \left(\frac{5}{10}\right)^n\right)$

$= \sum_{n=1}^{\infty}\left(\frac{1}{5}\right)^n + \sum_{n=1}^{\infty}\left(\frac{1}{2}\right)^n$

$\Rightarrow \frac{\left(\frac{1}{5}\right)^1}{1-\frac{1}{5}} + \frac{\left(\frac{1}{2}\right)^1}{1-\frac{1}{2}}, \ldots\ldots\ldots\ldots \left(S_\infty = \frac{a}{1-r}\right)$

$= \frac{\frac{1}{5}}{\frac{4}{5}} + \frac{\frac{1}{2}}{\frac{1}{2}}$

$= \frac{1}{4} + 1 = \frac{5}{4}$

∴ ***Answer C***

14. If $f(x) = \frac{x+1}{x-1}$ and $f(a) = 5$ then $f(2a)$ is equal to:

A. 2 B. 4 C. 6 D. 8

Solution: $f(x) = \frac{x+1}{x-1}$, $f(a) = 5$

$\Rightarrow f(a) = \frac{a+1}{a-1} \Rightarrow 5 = \frac{a+1}{a-1} \Rightarrow 5a - 5 = a + 1$

$\Rightarrow 4a = 6 \Rightarrow a = \frac{3}{2}$

$\Rightarrow f(2a) = f\left(2 \times \frac{3}{2}\right) = f(3) = \frac{3+1}{3-1} = \frac{4}{2} = 2$

∴ ***Answer A***

15. If $f(x) = \sqrt[3]{1+e^{-x}}$, which of the following is equal to $f^{-1}(x)$?

A. $\ln\left(\frac{1}{x^3-1}\right)$ B. $\frac{1}{\ln(x^3-1)}$ C. $\ln(1-x^3)$ D. $(1+e^{-x})^3$

Solution: (Le $y = f(x)$)

$y = \sqrt[3]{1+e^{-x}}$

$x = \sqrt[3]{1+e^{-y}}$ .....Interchange x and y

$x^3 = 1 + e^{-y}$ .....Solve for y

$x^3 - 1 = e^{-y}$

$\ln(x^3-1) = \ln e^{-y}$.....Multiply both sides by ln

$\Rightarrow -y = \ln(x^3-1)$

$\Rightarrow y = -\ln(x^3-1)$

$\Rightarrow y = \ln(x^3-1)^{-1}$

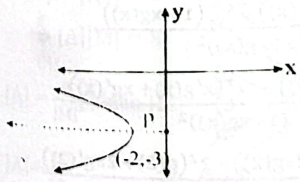
$\Rightarrow f^{-1}(x) = \ln\left(\frac{1}{x^3-1}\right)$

∴ ***Answer A***

16. What is the equation of the directrix for the parabola whose equation is $y^2 + 8x + 6y + 25 = 0$?

A. $y = 3$ B. $x = 2$ C. $x = 0$ d. $x = 4$

Solution: Parabola, $y^2 + 8x + 6y + 25 = 0$



$y^2 + 6y = -8x - 25$

$(y+3)^2 = -8x - 25 + 9$

$(y+3)^2 = -8(x+2)$(opening to the left.)

$-4p = -8 \Rightarrow p = 2$

Vertex (-2,-3)

Directrix $= x = h + p$

$= -2 + 2$

$= 0$

∴ ***Answer C***

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17. Given that $\lim_{x\to3} f(x) = 5$ and $\lim_{x\to3} g(x) = 11$, what is the value of

$$\lim_{x\to3}\left[\frac{(f(x)-g(x))(g(x)-2f(x))}{g(x)^2-f(x)^2}\right]?$$

A. $-\frac{66}{96}$ B. $-\frac{1}{16}$ C. 0 D. does not exist

Solution: $\lim_{x\to3} f(x) = 5, \lim_{x\to3} g(x) = 11$

$$\Rightarrow \lim_{x\to3}\left[\frac{(f(x)-g(x))(g(x)-2f(x))}{g(x)^2-f(x)^2}\right] = \lim_{x\to3}\left[\frac{(f(x)-g(x))(g(x)-2f(x))}{(g(x)-f(x))(g(x)+f(x))}\right]$$

$$= \lim_{x\to3}\frac{2f(x)-g(x)}{g(x)+f(x)}$$

$$= \frac{\lim_{x\to3}[2f(x)-g(x)]}{\lim_{x\to3}(g(x)+f(x))}$$

$$= \frac{2\lim_{x\to3} f(x)-\lim_{x\to3} g(x)}{\lim_{x\to3} g(x)+\lim_{x\to3} f(x)}$$

$$= \frac{2\times5-11}{11+5} = \frac{-1}{16}$$

∴ Answer B

18. If $f(x) = \frac{x^2}{1+xg(x)}$, $g(2) = 1$ and $g'(2) = 10$, then which one of the following is equal to $f'(2)$?

A. -8 B. $\frac{-8}{9}$ C. $\frac{4}{3}$ D. $\frac{8}{9}$

Solution: $f(x) = \frac{x^2}{1+xg(x)}$, $g(2) = 1$, $g'(2) = 10$

$$\Rightarrow f'(x) = \frac{(x^2)'(1+xg(x)) - x^2(1+xg(x))'}{(1+xg(x))^2}$$

$$\Rightarrow f'(x) = \frac{2x(1+xg(x)) - x^2(x'g(x)+xg'(x))}{(1+xg(x))^2}$$

$$\Rightarrow f'(2) = \frac{2\times2(1+2\times g(2)) - 2^2(g(2)+2\times g'(2))}{(1+2\times g(2))^2}$$

$$\Rightarrow f'(2) = \frac{4(1+2\times1) - 4(1+2\times10)}{(1+2\times1)^2}$$

$$\Rightarrow f'(2) = -8$$

∴ Answer A

19. If $w = \frac{16i}{1+i} + (1-3i)^2$ and $z = |w| + \bar{w}$, which one of the following is the simplest form of z?

A. $\sqrt{2}+2i$ B. $2+2i$ C. $4-2i$ D. $2-2i$

Solution: $w = \frac{16i}{1+i} + (1-3i)^2 = \frac{16i}{1+i}\times\frac{1-i}{1-i} + (1-6i+9i^2)$

$$w = \frac{16i-16i^2}{2} + (-8-6i)$$

$$w = \frac{16i}{1+i} + (1-3i)^2 = 8+8i-8-6i$$

$$w = \frac{16i}{1+i} + (1-3i)^2 = 2i$$

$$\Rightarrow Z = |w| + \bar{w}$$

$$\Rightarrow Z = |2i| + \bar{2i}$$

$$\Rightarrow Z = 2-2i$$

∴ Answer D

20. If $M = \begin{bmatrix}0 & 1 & 2\\ 3 & -1 & 0\\ 5 & 2 & 4\end{bmatrix}$, and $A^TM = 2I$, where A is a 3×3 matrix and I is the identity matrix of order 3, then what is det (A)?

A. 0.2 B. $\frac{4}{17}$ C. 0.8 D. $\frac{1}{17}$

Solution: $M = \begin{bmatrix}0 & 1 & 2\\ 3 & -1 & 0\\ 5 & 2 & 4\end{bmatrix}$ ..... $A^TM = 2I$

$$|A^TM| = |2I|$$

$$\Rightarrow |A^T||M| = 2^3\,|I|$$

$$\Rightarrow |A||M| = 8$$

$$\Rightarrow |A| = \frac{8}{|M|}, \text{ since } |M| = \begin{vmatrix}0 & 1 & 2\\ 3 & -1 & 0\\ 5 & 2 & 4\end{vmatrix}$$

$$\Rightarrow |A| = \frac{8}{10}$$

$$\Rightarrow |A| = 0.8$$

$$= 12 - 2$$

$$= 10$$

∴ Answer C

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21. If distinct codes (words) of eight letters are formed by rearranging the letters in the word "ABBEBAYE", how many of the codes begin with B or Y?

A. 840 B. 630 C. 1680 D. 420

Solution: "ABBEBAYE"

$n(\text{codes begin with B}) = 1 \times \frac{7!}{2!2!2!} = \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2!}{2 \times 2 \times 2!}$

(Since A,B,E occurs each 2 times) = 630

$n(\text{codes begin with y}) = 1 \times \frac{7!}{2!3!2!} = \frac{7 \times 6 \times 5 \times 4 \times 3!}{2 \times 3!}$

$= 210$

(since A,E occurs each 2 times and B occurs 3 times)

$\Rightarrow n(\text{codes begin with B or y}) = 630 + 210 = \mathbf{840}$

$\therefore$ ***Answer A***

22. What should be the value of k so that the system of equation

$$\begin{cases} x - y + z = 1 \\ -x + 5y - 4z = 1 \\ 2x + 2y - z = k \end{cases}$$ has a solution?

A. 0 B. 1 C. -4 D. 4

Solution: $\begin{cases} x - y + z = 1 \\ -x + 5y - 4z = 1 \\ 2x + 2y - z = k \end{cases}$

Using Gauss-Jordan elimination method

$$AM = \left(\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ -1 & 5 & -4 & 1 \\ 2 & 2 & -1 & k \end{array}\right)$$

$$\begin{matrix} R_2 \to R_1 + R_2 \\ R_3 \to R_3(-2R_1) \end{matrix} \left(\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 4 & -3 & 2 \\ 0 & 4 & -3 & k-2 \end{array}\right)$$

$$R_2 \to \tfrac{1}{4}R_2 \left(\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 1 & \frac{-3}{4} & \frac{1}{2} \\ 0 & 4 & -3 & k-2 \end{array}\right)$$

$$R_3 \to R_3 + (-4R_2) \left(\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 1 & \frac{-3}{4} & \frac{1}{2} \\ 0 & 0 & 0 & k-4 \end{array}\right)$$ - Echelon form.

The system has a solution of $k - 4 = 0$

$\Rightarrow k = 4$

$\therefore$ ***Answer D***

23. Let $f(x) = \begin{cases} \frac{a \sin x}{x - |x|}, & \text{if } x < 0 \\ e^{-x} + \cos x, & \text{if } x \geq 0 \end{cases}$

If f is continuous at $x = 0$, then what is the value of a?

A. 4 B. 2 C. $\frac{1}{2}$ D. -4

Solution: $f(x) = \begin{cases} \frac{a \sin x}{x-|x|}, & \text{if } x < 0 \\ e^{-x} + \cos x, & \text{if } x \geq 0 \end{cases}$

f is continuous at $x = 0$ iff $\lim_{x \to 0^-} f(x) = \lim_{x \to 0^+} f(x) = f(0)$

$\Rightarrow \lim_{x \to 0^-} \frac{a \sin x}{x-|x|} = e^0 + \cos 0$

$\Rightarrow \lim_{x \to 0^-} \frac{a \sin x}{x+x} = 2$

$\Rightarrow a \lim_{x \to 0^-} \frac{\sin x}{2x} = 2$

$\Rightarrow a \frac{1}{2} = 2$

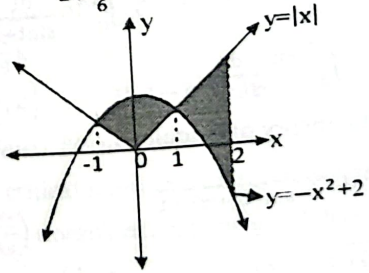
$\Rightarrow a = 4$

$\therefore$ ***Answer A***

24. What is the area of the region between the graphs of $y = -x^2 + 2$, and $y = |x|$, where $-1 \leq x \leq 2$?

A. $\frac{11}{6}$ B. $\frac{25}{6}$ C. $\frac{7}{3}$ D. $\frac{11}{3}$

Solution:



Intersection points of $\begin{cases} y = |x| \\ y = -x^2 + 2 \end{cases}$

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$-x = -x^2 + 2$, for $x < 0$, $\quad x = -x^2 + 2$, for $x > 0$

$x^2 - x - 2 = 0$, $\quad x^2 + x - 2 = 0$

$(x-2)(x+1) = 0$, $\quad (x+2)(x-1) = 0$

$x = -1$, $\quad x = 1$

Area of the shaded region

$$= \int_{-1}^{0}(-x^2 + 2 - (-x))dx + \int_{0}^{1}(-x^2 + 2 - x)dx +$$

$$\int_{1}^{2}\left((x - (-x^2 + 2)\right)dx$$

$$= 2\int_{-1}^{0}(x - x^2 + 2)dx + \int_{1}^{2}(x + x^2 - 2)dx$$

$$= 2\left[\frac{x^2}{2} - \frac{x^3}{3} + 2x\right]_{-1}^{0} + \left[\frac{x^2}{2} + \frac{x^3}{3} - 2x\right]_{1}^{2}$$

$$= -2\left(\frac{1}{2} + \frac{1}{3} - 2\right) + \left[\frac{4}{2} + \frac{8}{3} - 4 - \left(\frac{1}{2} + \frac{1}{3} - 2\right)\right]$$

$$= -2\left(\frac{-7}{6}\right) + \frac{11}{6}$$

$$= \frac{25}{6}$$

$\therefore$ ***Answer B***

25. What is the derivative of $f(x) = \int_{0}^{x^2+\pi} \frac{dt}{\sin t + 1}$?

A. $\frac{\cos x}{\sin(x^2+\pi) + 1}$ $\quad$ C. $\frac{2x\cos x}{\sin(x^2+\pi)+1}$

B. $\frac{2x}{\sin(x^2+\pi) + 1}$ $\quad$ D. $\int_{0}^{2x} \frac{dt}{\sin t + 1}$

Solution: $f(x) = \int_{0}^{x^2+\pi} \frac{dt}{\sin t + 1}$

Using fundamental theory of calculus I,

$$f'(x) = (x^2 + \pi)' \times \frac{1}{\sin(x^2+\pi)+1}$$

$$f'(x) = \frac{2x}{\sin(x^2+\pi)+1}$$

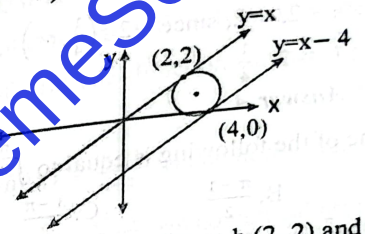
$\therefore$ ***Answer B***

26. If two lines $y = x$ and $y = x - 4$ are tangent to a circle at (2, 2) and (4,0), respectively, then what is the equation of the circle?

A. $(x-2)^2 + y^2 = 4$ $\quad$ C. $(x-3)^2 + (y-1)^2 = 2$

B. $(x-4)^2 + (y-2)^2 = 4$ $\quad$ D. $(x-1)^2 + (y+1)^2 = 10$

Solution:



Equation of a line which passes through (2, 2) and (4,0) is

$y - 2 = \frac{0-2}{4-2}(x-2)$

$y - 2 = -(x-2)$

$\Rightarrow y = -x + 4$ which is $\perp$ to both lines. (since $y = x \,\|y = x - 4$)

$\Rightarrow y = -x + 4$ contain the radius.

$\Rightarrow$ Center $= \left(\frac{2+4}{2}, \frac{2+0}{2}\right) = (3, 1)$

$$\Rightarrow r = \frac{1}{2}\sqrt{(4-2)^2 + (0-2)^2}$$

$$\Rightarrow r = \sqrt{2}$$

$\therefore$ Equation of the circle $= (x-3)^2 + (y-1)^2 = 2$

$\therefore$ ***Answer C***

27. Which one of the following is the set of all critical numbers of $f(x) = \frac{1}{3}x^3 - |4x - 1|$?

A. $\left\{\frac{1}{4}, 2\right\}$ $\quad$ B. $\left\{-2, \frac{1}{4}, 2\right\}$ $\quad$ C. $\{-2,2\}$ $\quad$ D. $\left\{\frac{1}{4}\right\}$

Solution: $f(x) = \frac{1}{3}x^3 - |4x - 1|$

C is a critical number iff $f'(c)$ doesn't exist or $f'(c) = 0$

$\Rightarrow f'\left(\frac{1}{4}\right)$ doesn't exist. (The graph has a sharp corner at $x = \frac{1}{4}$)

and $f(x) = \begin{cases} \frac{1}{3}x^3 - (4x-1), & \text{if } x \geq \frac{1}{4} \\ \frac{1}{3}x^3 + (4x-1), & \text{if } x < \frac{1}{4} \end{cases}$

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$$\Rightarrow f'(x) = \begin{cases} x^2 - 4, & \text{if } x > \frac{1}{4} \\ x^2 + 4, & \text{if } x < \frac{1}{4} \end{cases}$$

$\Rightarrow x^2 - 4 = 0$

$\Rightarrow x = 2, -2$, since $-2 \in \left(\frac{1}{4}, \infty\right)$

$\therefore C = 2, \frac{1}{4}$

$\therefore$ ***Answer A***

28. Which one of the following is equal to $\int_0^{\frac{\pi}{2}} \frac{x - \sin x}{\sec x} dx$?

A. $\frac{\pi - 3}{2}$ B. $\frac{\pi - 1}{2}$ C. $\frac{3 - \pi}{2}$ D. $\frac{\pi + 3}{2}$

Solution: $\int_0^{\frac{\pi}{2}} \frac{x - \sin x}{\sec x} dx = \int_0^{\frac{\pi}{2}} (x - \sin x)\cos x \, dx$

$= \int_0^{\frac{\pi}{2}} x\cos x \, dx - \int_0^{\frac{\pi}{2}} \sin x \cos x \, dx$

Let $u = x \Rightarrow dv = \cos x \, dx$

$\Rightarrow du = dx \Rightarrow v = \int \cos x \, dx$

$\Rightarrow v = \sin x$

$\Rightarrow \int x \cos x \, dx = x \sin x - \int \sin x \, dx$

$\Rightarrow \int x \cos x \, dx = x \sin x + \cos x + c$

$\therefore \Rightarrow \int_0^{\frac{\pi}{2}} x\cos x \, dx = x\sin x + \cos x \Big|_0^{\pi/2}$

$\Rightarrow \int_0^{\frac{\pi}{2}} x\cos x \, dx = \frac{\pi}{2}\sin\frac{\pi}{2} + \cos\frac{\pi}{2} - (0 \times \sin 0 + \cos 0)$

$\Rightarrow \int_0^{\frac{\pi}{2}} x\cos x \, dx = \frac{\pi}{2} - 1$

and $\int_0^{\frac{\pi}{2}} \sin x \cos x \, dx$,

Let $u = \sin x \Rightarrow du = \cos x \, dx$

$\Rightarrow \int \sin x \cos x \, dx = \int u \times du = \frac{u^2}{2} = \frac{(\sin x)^2}{2} \Big|_0^{\pi/2}$

$\Rightarrow \int \sin x \cos x \, dx = \frac{\left(\sin\frac{\pi}{2}\right)^2}{2} - \frac{(\sin 0)^2}{2} = \frac{1}{2}$

$\therefore \int_0^{\frac{\pi}{2}} \frac{x - \sin x}{\sec x} dx = \frac{\pi}{2} - 1 - \frac{1}{2} = \frac{\pi}{2} - \frac{3}{2} = \frac{\pi - 3}{2}$

$\therefore$ ***Answer A***

29. Which one of the following is true?

A. A polynomial can have infinitely many vertical asymptotes.

B. The graph of a rational function can never cross its horizontal asymptote.

C. The graph of $f(x) = \frac{3x - 1}{x - 1}$ has no horizontal asymptote.

D. The graph of $f(x) = \frac{x^3 - x}{x^2 - x}$ has no vertical asymptote.

Solution: **Facts:** The graph of a polynomial functions have no vertical asymptotes, because its domain is $\mathbb{R}$.

✓ The graph of a rational function can cross its horizontal asymptote.

✓ The graph of $f(x) = \frac{ax + b}{cx + d}$ has a horizontal asymptote of $y = \frac{a}{c}$.

$\therefore$ ***Answer D***

30. The simplified form of the derivative of $f(x) = \frac{1 + \sin x}{\cos x}$ is

A. $\sec x + \tan x$ B. $\frac{1 + \sin x}{\cos^2 x}$ C. $\frac{1}{1 + \tan x}$ D. $\frac{\cos x}{\sin^2 x}$

Solution: $f(x) = \frac{1 + \sin x}{\cos x}$

$f'(x) = \frac{(1 + \sin x)' \cos x - (1 + \sin x)(\cos x)'}{(\cos x)^2}$

$= \frac{\cos x \times \cos x - (1 + \sin x)(-\sin x)}{(\cos x)^2}$

$= \frac{\cos^2 x + \sin x + \sin^2 x}{(\cos x)^2}$

$= \frac{1 + \sin x}{(\cos x)^2}$

$\therefore$ ***Answer B***

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31. If Q_i, D_i and P_i are respectively the i^{th} – quartile, decile and percentile of a data arranged in an increasing order, then which one of the following is necessarily true?

A. $Q_2 = \frac{Q_1 + Q_3}{2}$ C. $P_{25} > Q_1$

B. $D_3 > P_{25}$ D. Q_2 = mean of the data

Solution: The data is arranged in increasing order .

$\Rightarrow P_{25} = Q_1$, $P_{50} = Q_2 = D_5$ = median

$\Rightarrow P_{75} = Q_3$ and $Q_2 = \frac{Q_1 + Q_3}{2}$

$\therefore$ ***Answer A***

32. If $f(x) = x^2\sqrt{2x+12}$, what is the slope of the tangent line to the graph of f at x = 2?

A. −4 B. 2 C. 18 D. 17

Solution: $f(x) = x^2\sqrt{2x+12}$

$f'(x) = (x^2)'\sqrt{2x+12} + x^2(\sqrt{2x+12})'$

$f'(x) = 2x\sqrt{2x+12} + \frac{x^2}{2\sqrt{2x+12}} \times (2x+12)'$

$f'(x) = 2x\sqrt{2x+12} + \frac{x^2}{\sqrt{2x+12}}$

$\Rightarrow m = f'(2) = 2\times2\times\sqrt{2\times2+12} + \frac{2^2}{\sqrt{2\times2+12}}$

$\Rightarrow m = 17$

$\therefore$ ***Answer D***

33. If $f(x) = e^{2x}\sin x$, then f "(x) is equal to

A. $3e^{2x}\sin x - 4e^{2x}\cos x$ C. $e^{2x}(3\sin x + 4\cos x)$

B. $4e^{2x}\sin x + 2e^{2x}\cos x$ D. $e^{2x}(4\sin x - 3\sin x)$

Solution: $f(x) = e^{2x}\sin x$

$f'(x) = (e^{2x})'\sin x + e^{2x}(\sin x)'$

$f'(x) = 2e^{2x}\sin x + e^{2x}\cos x$

$f''(x) = (2e^{2x}\sin x)' + (e^{2x}\cos x)'$

$f''(x = 2(e^{2x})'\sin x + 2e^{2x}(\sin x)' + (e^{2x})'\cos x + e^{2x}(\cos x)'$

$f''(x = 4e^{2x}\sin x + 2e^{2x}\cos x + 2e^{2x}\cos x - e^{2x}\sin x$

$f''(x = 3e^{2x}\sin x + 4e^{2x}\cos x$

$f''(x = e^{2x}(3\sin x + 4\cos x)$

$\therefore$ ***Answer C***

34. If the equation $(x-2)^2 - (y-2)^2 = 1$ represents a hyperbola which one of the following represents equation of an asymptote to the hyperbola?

A. $y = 4 - x$ B. $x + y = 1$ C. $x = 2 - y$ D. $x + 2y = 3$

Solution: $(x-2)^2 - (y-2)^2 = 1$ ..... is an x – hyperbola

$\Rightarrow$ Asymptotes: $y - k = \pm\frac{b}{a}(x-h)$

$\Rightarrow y - 2 = \pm\,1(x-2)$

$\Rightarrow y = x - 2 + 2$ and $y = -x + 4$

$\Rightarrow y = x$

$\therefore$ ***Answer A***

35. What is the sum of the series $\sum_{n=1}^{\infty}(-1)^n 3^{-2n}$?

A. $\frac{-1}{8}$ B. −0.13 C. −0.1 D. $\frac{1}{8}$

Solution: $\sum_{n=1}^{\infty}(-1)^n 3^{-2n} = \sum_{n=1}^{\infty}(-1)^n\left(\frac{1}{9}\right)^n$

$= \sum_{n=1}^{\infty}\left(-\frac{1}{9}\right)^n$

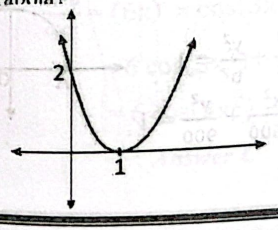
$\Rightarrow S_\infty = \frac{G_1}{1-r}$

$= \frac{-\frac{1}{9}}{1-\left(-\frac{1}{9}\right)}$

$= \frac{-1}{10}$

$\therefore$ ***Answer C***

36. Which of the equations below is represented by the following parabola?



A. $y = x^2 + 2$

B. $y = (2x-1)^2$

C. $y = 2(x-1)^2$

D. $y = (2x+1)^2$

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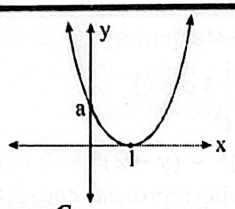
Solution:



$\Rightarrow y = a(x-1)^2$

$\Rightarrow y = 2(x-1)^2$

$\therefore$ ***Answer C***

37. A company produced 25,000 bulbs and randomly tested 2% of the product. Among the tested bulbs, if 40 have defect of type D_1, 60 have defect of type D_2 and 25 have both types of defects, what is the probability that a bulb produced by the company has **none** of the defects?

A. 0.95 B. 0.80 C. 0.85 D. 0.20

Solution: Number of bulbs = 25,000

Number of tested bulbs $= \frac{2}{100} \times 25{,}000 = 500$

$n(D_1) = 40,\ n(D_2) = 60,\ n(D_1 \cap D_2) = 25$

$\Rightarrow$ P(defect bulb) $= \frac{40}{500} + \frac{60}{500} - \frac{25}{500} = \frac{75}{500}$

$\Rightarrow$ P(none of the defects) $= 1 - \frac{75}{500} = \frac{500-75}{500} = \frac{425}{500} = 0.85$

$\therefore$ ***Answer C***

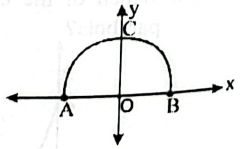
38. A semi-elliptical arc over a tunnel for a road through a mountain has a major axis of length 80 meters and a height of 30 meters at the center. What is the equation of the semi-elliptical arc over the tunnel, if the center is considered as the origin?

A. $\frac{x^2}{6400} + \frac{y^2}{900} = 1$ C. $\frac{x^2}{900} + \frac{y^2}{6400} = 1$

B. $\frac{x^2}{1600} + \frac{y^2}{900} = 1$ D. $\frac{x^2}{8100} + \frac{y^2}{6400} = 1$

Solution: AB = 2a = 80 $\Rightarrow$ a = 40

OC = b = 30



Equation of a semi-ellipse $= \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$= \frac{x^2}{1600} + \frac{y^2}{900} = 1$

$\therefore$ ***Answer B***

39. Suppose $AX = b$, where A is a 3×3 matrix, $b = (b_1, b_2, b_3)^T$ and $X = (x, y, z)^T$. Which one of the following is necessarily true about this system of linear equations?

A. The system has a solution only when $\det(A) \neq 0$.

B. The Cramer's rule is suitable to solve the system if two rows of A are identical

C. If $\det A \neq 0$ and the second column of A is a multiple of b, then $x = 0$.

D. If $b = 0$, then $X = (0,0,0)^T$ is the only solution of the system.

Solution: $Ax = b$

$A\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ .... Equation of a system of linear equations.

This system has a solution if $\det(A) \neq 0$

i.e $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1}\begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$

$\therefore$ ***Answer A***

40. If $y = \sin(3x^2)$, then the simplified form of $\frac{d^2y}{dx^2}$ is:

A. $-6\sin(3x^2)$ C. $6\cos(3x^2) - 36x^2\sin(3x^2)$

B. $\cos(6x) - 6\sin(3x^2)$ D. $x^2\cos(3x^2) + 6\sin(3x^2)$

Solution: $y = \sin(3x^2)$

$\frac{dy}{dx} = \cos(3x^2) \times (3x^2)'$

$\frac{dy}{dx} = \cos(3x^2) \times 6x$

$\frac{d^2y}{dx^2} = (6x)' \times \cos(3x^2) + 6x\,(\cos(3x^2))'$

$= 6\cos(3x^2) + 6x(-\sin(3x^2) \times (3x^2)')$

$= 6\cos(3x^2) - 36x^2\sin(3x^2)$

$\therefore$ ***Answer C***

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41. If S is a set with 10 elements and A⊆ S, what is the probability that A has 3 or more elements?

A. $\frac{7}{10}$ B. $\frac{8}{11}$ C. $\frac{121}{128}$ D. $\frac{7}{128}$

Solution: $A \subseteq S$, $n(S) = 2^{10}$

$$P(\geq 3) = 1 - P(< 3)$$

$$= 1 - [P(0) + P(1) + P(2)]$$

$$= 1 - \left[\frac{C(10,\ 0) + C(10,\ 1) + C(10,\ 2)}{2^{10}}\right]$$

$$= 1 - \left[\frac{1 + 10 + 45}{2^{10}}\right]$$

$$= 1 - \frac{56}{2^{10}} = 1 - \frac{2^3 \times 7}{2^{10}}$$

$$= \frac{121}{128}$$

∴ ***Answer C***

42. If $a_n = \left(\frac{n+3}{n+1}\right)^n$, then the limit of the sequence $\{a_n\}_{n=1}^{\infty}$ is equal to:

A. 1 B. $\frac{1}{2}e$ C. e^2 D. $+\infty$

Solution: $a_n = \left(\frac{n+3}{n+1}\right)^n = \left(\frac{1+\frac{3}{n}}{1+\frac{1}{n}}\right)^n$

$$\lim_{x\to\infty} a_n = \frac{\lim_{x\to\infty}\left(1+\frac{3}{n}\right)^n}{\lim_{x\to\infty}\left(1+\frac{1}{n}\right)^n}$$

$$\lim_{x\to\infty} a_n = \frac{e^3}{e} = e^2$$

∴ ***Answer C***

43. Which one of the following is equal to $\lim_{x\to\infty}\left(\frac{x}{x+2}\right)^{-3x}$?

A. e^6 B. e^{-3} C. $e^{-3/2}$ D. e^{-6}

Solution: $\lim_{x\to\infty}\left(\frac{x}{x+2}\right)^{-3x} = \lim_{x\to\infty}\left(\frac{x+2}{x}\right)^{3x}$

$$= \lim_{x\to\infty}\left(1 + \frac{2}{x}\right)^{3x}$$

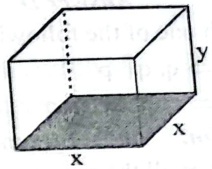
$$= (e^2)^3 = e^6$$

∴ ***Answer A***

44. If a box with square base and open top is made from 1,200cm² material, what is the largest volume of the box in cm³?

A. 4,000 B. 8,000 C. 15,000 D. 3,000

Solution: $V = x^2y$



Surface area $= x^2 + 4xy = 1200$

$$4xy = 1200 - x^2$$

$$y = \frac{1200 - x^2}{4x}$$

$$V = x^2\left(\frac{1200 - x^2}{4x}\right)$$

$$V = x\left(\frac{1200 - x^2}{4}\right)$$

$$V = \frac{-x^3}{4} + \frac{1200x}{4}$$

$$V(x) = -\frac{1}{4}x^3 + 300x$$

$$V'(x) = -\frac{3}{4}x^2 + 300$$

$$V'(x) = -\frac{3}{4}x^2 + 300 = 0$$

$$\Rightarrow -\frac{3}{4}x^2 = -300$$

$$x^2 = \frac{4\times300}{3} \Rightarrow x = 20$$

$$y = \frac{1200 - (20)^2}{4(200)} \Rightarrow y = 10$$

$$V = x^2y = (20)^2 \times 10 = 4000\text{cm}^2$$

∴ ***Answer A***

45. Which one of the following is true about the graph of $f(x) = \frac{x^2 - 1}{x - x^2}$?

A. $x = 0$ and $x = 1$ are its vertical asymptotes.
B. $y = 1$ is its horizontal asymptote.
C. $y = x - 1$ is its oblique asymptote.
D. It is almost the same as the horizontal line $y = -1$ as $x \to \pm\infty$

Solution: $f(x) = \frac{x^2-1}{x-x^2} = \frac{(x-1)(x+1)}{x(1-x)} = \frac{-(x+1)}{x}$, $x \neq 1$

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- ✓ $x = 0$ is its vertical asymptote
- ✓ there is a hole at $x = 1$
- ✓ $y = -1$ is its horizontal asymptote. i.e when $x \to \pm\infty$, $y \to -1$.

∴ ***Answer D***

46. Which one of the following is a valid logical argument?

A. $p \Rightarrow q, q \mid -p$ C. $\neg p \wedge q, q \Rightarrow r \mid -r$

B. $p \Leftrightarrow q, p \Rightarrow q \mid -q$ D. $\neg p, p \vee q, r \Rightarrow q \mid -r$

Solution: Valid argument: The conclusion is always true whenever all the premises are true.

Example : $\neg p \wedge q, q \Rightarrow r \vdash r$

$\neg p \wedge q = T$.....premise

$\neg p = T, q = T$

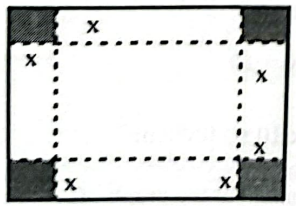
$q \Rightarrow r = T$

$T \Rightarrow r = T$

$\therefore r = T$.....Valid

∴ ***Answer C***

47. Suppose that equal squares are cut from each of the four corners of a square cardboard whose sides are 72 centimeters long. (See the figure below.) The resulting flaps are then folded up to form a box without a top. How long should be each of the four squares that has to be cut off to maximize the volume of the box?



A. 6 cm

B. 12 cm

C. 15 cm

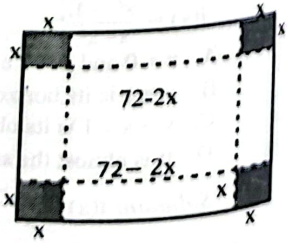
D. 24 cm

Solution:

The box is formed by turning up the sides has height x cm, and base $72-2x$ by $72-2x$. Therefore, its volume is given by $V = x(72-2x)(72-2x)$

$\Rightarrow V(x) = x(72-2x)^2$

$\Rightarrow V(x) = x(5184 - 288x + 4x^2)$



$\Rightarrow V(x) = 5184x - 288x^2 + 4x^3$

$\Rightarrow V'(x) = 5184 - 576x + 12x^2 = 0$

$\Rightarrow x^2 - 48x + 432 = 0$

$\Rightarrow x = 12cm$

∴ ***Answer B***

48. What is the area of the region between the graph of $f(x) = -x^2 + 4x - 3$ and the x-axis from $x = 0$ to $x = 3$?

A. $-\frac{2}{3}$ B. $\frac{2}{3}$ C. $\frac{4}{3}$ D. $\frac{8}{3}$

Solution: $f(x) = -x^2 + 4x - 3$

The zero's of $f(x)$: $-x^2 + 4x - 3 = 0$

$\Rightarrow x^2 - 4x + 3 = 0$

$\Rightarrow (x-3)(x-1) = 0$

$\therefore x = 1$ and $x = 3$

$$\Rightarrow A = \left|\int_0^1 f(x)dx\right| + \left|\int_1^3 f(x)dx\right|$$

$$= \left|\int_0^1 (-x^2 + 4x - 3)dx\right| + \left|\int_1^3 (-x^2 + 4x - 3)dx\right|$$

$$= \left|\frac{-x^3}{3} + \frac{4x^2}{2} - 3x\right]_0^1\right| + \left|\frac{-x^3}{3} + \frac{4x^2}{2} - 3x\right]_1^3\right|$$

$$= \frac{8}{3}$$

∴ ***Answer D***

49. Which one of the following is equal to $\int \frac{x + \ln(x+1)}{(x+1)^2} dx$?

A. $\ln(x+1) + \frac{x}{x+1} + c$ C. $(x+1)^2 - \frac{1}{x+1} + c$

B. $(x+1)^2 + \frac{1}{x+1} + c$ D. $\frac{x\ln(x+1)}{x+1} + c$

Solution: $\int \frac{x + \ln(x+1)}{(x+1)^2} dx = \int \frac{x}{(x+1)^2} dx + \int \frac{\ln(x+1)}{(x+1)^2} dx$

Let $u = x + 1 \Rightarrow du = dx$

$$\Rightarrow \int \frac{x}{(x+1)^2} dx = \int \frac{u-1}{u^2} du$$

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$$\Rightarrow \int \frac{x}{(x+1)^2}dx = \int \frac{1}{u}du - \int \frac{1}{u^2}du = \ln|u| + \frac{1}{u} + c$$

$$\Rightarrow \int \frac{\ln(x+1)}{(x+1)^2}dx = \int \frac{\ln u}{u^2}du = \int \frac{1}{u^2} \times \ln u \, du$$

Let $w = \ln u \Rightarrow dw = \frac{1}{u}du \Rightarrow dv = \frac{1}{u^2}du$ and $V = \int \frac{1}{u^2}du = \frac{-1}{u}$

$$\Rightarrow \int \frac{\ln u}{u^2}du = \frac{-1}{u}\ln u - \int \frac{-1}{u} \times \frac{1}{u}du$$

$$\Rightarrow \int \frac{\ln u}{u^2}du = \frac{-1}{u}\ln u + \int \frac{1}{u^2}du = \frac{-1}{u}\ln u - \frac{1}{u} + c$$

$$\Rightarrow \int \frac{x + \ln(x+1)}{(x+1)^2}dx = \ln|u| + \frac{1}{u} - \frac{1}{u}\ln u - \frac{1}{u} + c$$

$$= \ln(x+1) - \frac{1}{(x+1)}\ln(x+1) + c$$

$$= \ln(x+1)(1 - \frac{1}{x+1}) + c$$

$$= \frac{x\ln(x+1)}{x+1} + c$$

∴ ***Answer D***

50. If $F(x) = f(2x+2).g(1-x^2)$, with $f(2) = -3$, $f'(2) = 4$, $g(1) = -5$, and $g'(1) = 1$, then what is the actual value of $F'(0)$?

A. -40 B. -20 C. 0 D. 19

Solution: $F(x) = f(2x+2).g(1-x^2)$, $f(2) = -3$, $f'(2) = 4$

$g(1) = -5$, $g'(1) = 1$

$F'(x) = f'(2x+2).2.g(1-x^2) + f(2x+2).g'(1-x^2).(-2x)$

$F'(0) = 2f'(2)g(1) + f(2)g'(1).0$

$= 2f'(2)g(1)$

$= 2.4.(-5)$

$= -40$

∴ ***Answer A***

51. Suppose that an airplane is descending at a speed of 50 miles per hour at an angle of 30° below the horizontal line. What is the

x- and y- components, respectively, of the velocity of the

A. $50\sqrt{3}$, 25 C. 25, $-25\sqrt{3}$

B. -25, $50\sqrt{3}$ D. $-25\sqrt{3}$, -25

Solution: $x = V\cos 30°$

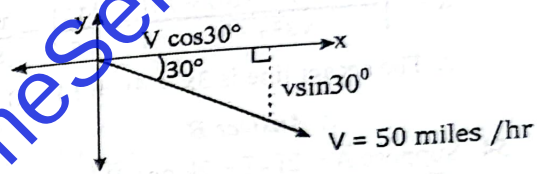
$x = -50 \times \frac{\sqrt{3}}{2}$

$x = -25\sqrt{3}$ and

$y = V\sin 30°$

$y = -50 \times \frac{1}{2}$

$y = -25$



∴ ***Answer D***

52. If a point (2, 5) is reflected under a line to the point (-3,1), what is the line of reflection?

A. $2x + 3y = 7$ C. $8y + 10x = 19$

B. $x + 3y = 7$ D. $2x + 3y + 5 = 0$

Solution: M(2,5) = (− 3, 1)

Mid-point $= \left(\frac{2-3}{2}, \frac{5+1}{2}\right) = \left(-\frac{1}{2}, 3\right)$

Slope of a line that passes through (2, 5) and (−3, 1) is given by

$m = \frac{5-1}{2-(-3)} = \frac{4}{5}$.

⇒ Slope of the line of reflection is $\frac{-5}{4}$

⇒ Its equation is $y - 3 = \frac{-5}{4}\left(x + \frac{1}{2}\right)$

$\Rightarrow 4y - 12 = -5x - \frac{5}{2}$

$\Rightarrow 4y + 5x = 12 - \frac{5}{2}$

$\Rightarrow 4y + 5x = \frac{19}{2}$

$\Rightarrow 8y + 10x = 19$

∴ ***Answer C***

53. Let the equation $x^2 + 2x + y^2 = 8$ represents a circ one of the following lines cut the circle at exactly two

A. $4x + 3y + 19 = 0$ C. $2y = 5x + 43$

B. $3x + 4y + 14 = 0$ D. $2x = y - 50$

Solution: $x^2 + 2x + y^2 = 8$

$(x+1)^2 + y^2 = 9$

C (−1, 0) and r = 3

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If $d = \left|\frac{Ax_0 + By_0 + C}{\sqrt{A^2+B^2}}\right| < r$, then ℓ: $Ax + By + C = 0$

Intersects the circle at exactly two points

$$d = \left|\frac{3\times(-1) + 4(0) + 14}{\sqrt{3^2+4^2}}\right| = \left|\frac{11}{5}\right| = \frac{11}{5} < 3$$

$\therefore$ The secant line is $3x + 4y + 14 = 0$

$\therefore$ ***Answer B***

54. Suppose $\vec{A} = 2\vec{i} - \vec{j} + 2k$ and $\vec{B}$ is a vector in space such that $|\vec{B}| = \vec{A} \cdot \vec{B}$. If $\vec{u}$ is the unit vector in the direction of $\vec{B}$, then $|\vec{A} + \vec{u}|^2$ is equal to:

A. 16 B. 12 C. 10 D. 14

Solution: $\vec{A} = 2i - j + 2k$, $|\vec{B}| = \vec{A} \cdot \vec{B}$

$$|\vec{A} + \vec{u}|^2 = (\vec{A} + \vec{u}).(\vec{A} + \vec{u})$$
$$= \vec{A}.\vec{A} + \vec{A}.\vec{u} + \vec{U}.\vec{A} + \vec{U}.\vec{U}$$
$$= |\vec{A}|^2 + 2\vec{A}.\vec{u} + |\vec{u}|^2$$
$$= (2^2 + (-1)^2 + 2^2) + \vec{A}.2\vec{u} + 1$$
$$= 10 + 2\vec{A}.\vec{u}, \quad \vec{u} = \frac{\vec{B}}{|\vec{B}|}$$
$$= 10 + 2\vec{A}.\frac{\vec{B}}{|\vec{B}|}$$
$$= 10 + \frac{2|\vec{B}|}{|\vec{B}|} = 12$$

$\therefore$ ***Answer B***

55. If A = (−2, 3), B = (3, 1) and C is any other point on the plane, then which one of the following is the coordinate form of $\overrightarrow{AC} - \overrightarrow{BC}$?

A. (−5, 2) B. (5,−2) C. (1, 4) D. (−1,−4)

Solution: A = (-2, 3), B = (3,1), C(x, y)

$$\overrightarrow{AC} - \overrightarrow{BC} = \overrightarrow{AC} + \overrightarrow{CB} = \overrightarrow{AB} = (3,1) - (-2,3)$$
$$= (3+2, 1-3) = (5, -2)$$

$\therefore$ ***Answer B***

56. What is the equation of a line that passes through the point (−1, 2) and parallel to the vector (1, −1)?

A. $2x - y = 1$ C. $x - 2y = 3$

B. $x + y - 1 = 0$ D. $y - 2x + 1 = 0$

Solution: Vector equation of the line becomes

$$(x, y) = (-1, 2) + t(1, -1)$$
$$\Rightarrow \begin{cases} x = -1 + t \\ y = 2 - t \end{cases} \Rightarrow \begin{cases} x + 1 = t \\ y = 2 - (x + 1) \end{cases}$$
$$\Rightarrow y = 2 - x - 1$$
$$\Rightarrow y + x - 1 = 0$$

$\therefore$ ***Answer B***

57. Suppose P and Q are points in space such that the midpoint of $\overline{PQ}$ is on the negative z-axis and the distance between P and Q is 6. If P = (2,-1, 0), then what is the coordinate of Q?

A. (-2, 1, 4) B. (2,-1,6) C. (2,-1,-6) D. (-2,1,-4)

Solution: P(2,-1,0), Q(a, b, c)

Midpoint of PQ = $\left(\frac{2+a}{2}, \frac{-1+b}{2}, \frac{0+c}{2}\right) = (0, 0, t)$

Where t is on the negative z-axis

$$\Rightarrow \frac{2+a}{2} = 0 \Rightarrow a = -2$$
$$\Rightarrow \frac{-1+b}{2} = 0 \Rightarrow b = 1$$
$$\Rightarrow \frac{c}{2} = t \Rightarrow t = \frac{c}{2}$$
$$PQ = d = \sqrt{(a-2)^2 + (b+1)^2 + c^2}$$
$$6^2 = (-2-2)^2 + (1+1)^2 + c^2$$
$$36 = 16 + 4 + c^2$$
$$c^2 = 16$$
$$c = \pm 4$$

Since t is on the negative z-axis, then $c = -4$

$\therefore$ ***Answer D***

58. Which one of the following is equivalent to

$\neg[(\forall x)(P(x) \Rightarrow Q(x))]$?

A. $(\forall x)(\neg P(x) \Rightarrow \neg Q(x))$ C. $(\exists x)(\neg P(x) \wedge Q(x))$

B. $(\exists x)(\neg P(x) \Rightarrow \neg Q(x))$ D. $(\exists x)(P(x) \wedge \neg Q(x))$

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Solution: $\neg[(\forall x)(P(x) \Rightarrow Q(x)] \equiv \neg[(\forall x)(\neg P(x) \vee Q(x))]$
$\equiv (\exists x)(\neg(\neg p(x)) \wedge \neg Q(x))$
$\equiv (\exists x)(\ p(x) \wedge \neg Q(x))$

$\therefore$ ***Answer D***

59. An observer on level ground is at a distance $10\sqrt{3}$ m from a building. The angles of elevation to the bottom of the windows on the second and third floors are 30° and 60°, respectively. What is the distance h between the bottoms of the windows?

$\left[\because \sin 30^\circ = \cos 60^\circ = \frac{1}{2} \text{ and } \sin 60^\circ = \cos 30^\circ = \frac{\sqrt{3}}{2}\right]$

A. 15m B. 20m C. $15\sqrt{3}$m D. 32m

Solution:

$\tan 30^\circ = \frac{y}{10\sqrt{3}}$

$y = 10\sqrt{3} \times \tan 30^\circ$

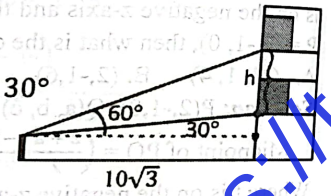
$y = 10m$

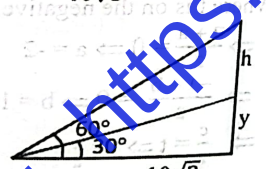
$\tan 60^\circ = \frac{y+h}{10\sqrt{3}}$

$\Rightarrow 10 + h = 10\sqrt{3} \tan 60^\circ$

$10 + h = 30$

$h = \mathbf{20m}$





$\therefore$ ***Answer B***

60. The following is an assertion of a person and his proof. "For any natural numbers n, $n! < 10^n$.

Proof: **Step 1.** Let n = 1. Since $1! = 1$ and $10^1 = 10$, it is true that $1! < 10^1$.

Step 2. Let n = 2. Since $2! = 2$ and $10^2 = 100$, it is true that $2! < 10^2$.

Step 3. Let n = 3. Since $3! = 6$ and $10^3 = 1000$, it is true that $3! < 10^3$.

Step 4. Continuing in this manner, we can see that whenever $k! < 10^k$ is true, then $(k + 1)! < 10^{k+1}$ is also true.

Therefore, by induction, $n! < 10^n$ for all natural numbers"

Which one of the following is true about the proof?

A. The proof is correct by the principle of mathematical induction, though Step 2 and Step 3 can be omitted.

B. The proof is correct by the principle of mathematical induction; and step 2 and Step 3 are necessary since they provide additional information.

C. The proof is invalid because Step 4 did not justify the desired induction step.

D. The proof follows the technique of a proof by exhaustion.

Solution:

Basis step: Show that the statement is true for n = 1.

Inductive step: Assume that the statement is true for n = k and show that it is true for n = k + 1

Conclusion: the given statement is true for all natural numbers.

$\therefore$ ***Answer C***

61. What is the image of the line given by (x, y) = (−1, 0) + t(3,6), $t \in \mathbb{R}$, under the translation that takes (1,0) to (0,1) followed by the reflection about the line y = 2x?

A. y = 2x + 3 B. y = 2x − 3 C. y = 2x + 6 D. y = 2x − 5

Solution:

$\ell = (x, y) = (-1, 0) + t(3, 6)$

$\Rightarrow x = -1 + 3t, y = 6t$

$\Rightarrow$ The rectangular form of ℓ becomes $y = 6\frac{(x+1)}{3} = 2x + 2.$

T(1,0) = (0,1), then the translation vector becomes $\vec{v} = (-1,1)$

Now, take any point on ℓ.

Let (1, 4)

$\Rightarrow T(1,4) = (-1+1, 1+4) = (0,5)$

Then, reflect (0, 5) by the line y = 2x.

$\Rightarrow M(0, 5) = (4,3)$

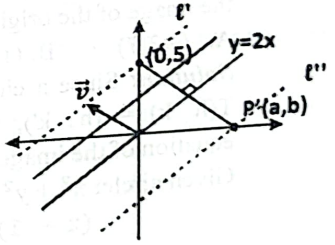
Since image of ℓ' under the mirror line y = 2x

$y - 3 = 2(x - 4)$

$y - 3 = 2x - 8$

$y = 2x - 5$

$\therefore$ ***Answer D***



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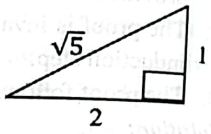
62. If $\theta = 2\arctan\left(\frac{1}{2}\right)$, then which one of the following is equal to sec(θ)?

A. $\frac{25}{3}$ B. $\frac{4}{5}$ C. $\frac{5}{3}$ D. $\frac{5}{4}$

Solution: $\theta = 2\arctan\left(\frac{1}{2}\right)$

Let $\alpha = \arctan\left(\frac{1}{2}\right)$



$\tan\alpha = \frac{1}{2}$

$\text{Sec}\,\theta = \sec 2\alpha$

$= \frac{1}{\cos 2\alpha}$

$= \frac{1}{\cos^2\alpha - \sin^2\alpha}$

$= \frac{1}{\left(\frac{2}{\sqrt{5}}\right)^2 - \left(\frac{1}{\sqrt{5}}\right)^2}$

$= \frac{5}{3}$

∴ ***Answer C***

63. If a translation T takes the circle $x^2 + y^2 - 2x + 6y + 3 = 0$ into the circle whose equation is $(x + 2)^2 + (y - 4)^2 = 7$, then what is the image of the origin under T?

A. (−3,7) B. (1,2) C. (1,−3) D. (−2, 4)

Solution: Since a circle has center (h ,k) and radius r such that T(h, k) = (h', k'), then $(x - h')^2 + (y - k')^2 = r^2$ is the equation of the image circle.

Given circle: $x^2 + y^2 - 2x + 6y + 3 = 0$

$(x - 1)^2 + (y + 3)^2 = -3 + 1 + 9$

$(x - 1)^2 + (y + 3)^2 = 7$

Center (1,− 3)

Image circle $(x + 2)^2 + (y - 4)^2 = 7$

Center (−2, 4)

∴ T(1, −3) = (−2, 4)

$\Rightarrow \vec{V} = (-2 - 1, 4 - (-3)) = (-3, 7)$

$\Rightarrow$ T (0, 0) = (0 +(−3), 0+7) = (−3,7)

∴ ***Answer A***

64. If ℓ is the line passing through (0, 2) and parallel to $V = \vec{i} + 3\vec{j}$, which one of the following is true about ℓ and the circle $(x - 2)^2 + (y - 1)^2 = 5$?

A. ℓ is tangent to the circle at (0, 2).

B. ℓ is tangent to the circle at some point P, where P ≠ (0,2).

C. ℓ intersects the circle at two distinct points.

D. The distance between ℓ and the center of the circle is greater than $\sqrt{5}$

Solution: Vector equation of ℓ becomes (x, y) = (0, 2) + t(1,3) and its rectangular form is y = 3x + 2 equation of the circle: $(x - 2)^2 + (y - 1)^2 = 5$ and its center is (2,1)

$\Rightarrow$ The point (0, 2) is also on the circle.

(i.e $(0 - 2)^2 + (2 - 1)^2 = 5$)

Distance between ℓ and center of the circle is given by

$$d = \left|\frac{Ax_0 + By_0 + c}{\sqrt{A^2 + B^2}}\right| = \left|\frac{-3 \times 2 + 1 \times 1 - 2}{\sqrt{(-3)^2 + (1)^2}}\right| = \left|\frac{-7}{\sqrt{10}}\right| = \frac{7}{\sqrt{10}} < \sqrt{5}$$

∴ ℓ intersects the circle at two distinct points

∴ ***Answer C***

65. Suppose $\vec{A} = 3i - 4j$ and $\vec{B}$ is a vector in the xy-plane such that the angle between $\vec{A}$ and $\vec{B}$ is $\frac{\pi}{3}$. If $\vec{u}$ is the unit vector in the direction of $\vec{B}$, then $\vec{A}.(\vec{A} - 2\vec{u})$ is equal to:

A. 20 B. 5 C. 15 D. 30

Solution: $\vec{A} = 3i - 4j$

$\vec{A}.(\vec{A} - 2\vec{u}) = \vec{A}.\vec{A} - 2\vec{A}.\vec{u}$

$= |\vec{A}|^2 - \frac{2\vec{A}.\vec{B}}{|\vec{B}|}$

$= |\vec{A}|^2 - \frac{2|\vec{A}||\vec{B}|}{|\vec{B}|}\cos\frac{\pi}{3}$

$= |\vec{A}|^2 - 2|\vec{A}|\cos\frac{\pi}{3}$

$= 5^2 - 2\times 5\times\frac{1}{2} = 20$

∴ ***Answer A***

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Directions: This exam contains a total of 50 questions. Each of the questions is followed by four possible alternatives. Read each question and carefully **blacken** only the letter of your choice on the separate answer sheet provided.

Constant	Symbol	Value
Acceleration due to gravity	g	$10 m/s^2$
Permittivity of vacuum	ε_o	$8.85 \times 10^{-12} F/m$
Faraday's constant	F	$96500 Cmol^{-1}$
Charge of an electron	e	1.6×10^{-19} Coulomb
Universal gravitational constant	G	$6.67 \times 10^{-11} \frac{Nm^2}{Kg^2}$
$\mu_o = 4\pi \times 10^{-7}$ Tm/A		
$\pi = 3.14$		
Sin 30^0= cos60^0 = 0.5		
Sin 37^0=cos53^0=0.6		
Sin 53^0=cos37^0=0.8		
Sin 60^0=cos30^0=0.866		

1. Two parallel copper wires are 1 cm apart. Lightning sends a 10,000 A pulse of current along each conductor. What is the force per unit length on one conductor?

A .1000 N/m B .200N/m C.20N/m D.2000N/m

Given : r = 1 cm $=10^{-2}$ m

$I_1 = I_2 = 10{,}000$ A

Required : Force per unit length ($\frac{F}{\ell} = ?$)

Solution: Consider the second current carrying wire in a magnetic field B_1 produced by the 1st current carrying wire.

$F = I_2 \vec{\ell} \times \vec{B_1}$

$F = I_2 \ell B_1 \sin 90^0$ ..... $\sin 90^0 = 1$

$F = I_2 \ell \left(\frac{\mu_o I_1}{2\pi r}\right)$ ..... Since $B_1 = \frac{\mu_o I_1}{2\pi r}$

$\frac{F}{\ell} = I_2\left(\frac{\mu_o I_1}{2\pi r}\right)$

$\frac{F}{\ell} = \frac{10000A \times 10000A \times 4\pi \times 10^{-7} Tm/A}{2\pi \times 10^{-2} m}$

$\frac{F}{\ell} = \frac{1 \times 10^8 A \times 4 \times 10^{-7} T}{2 \times 10^{-2}}$

$\frac{F}{\ell} = \frac{10 \times 4 \times 100 N/m}{2}$

$\therefore \frac{F}{\ell} = 2000$ N/m

$\therefore$ ***Answer B***

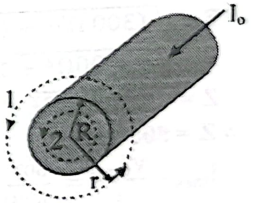
2. A long ,straight wire of radius R carries a steady current I that is uniformly distributed through the cross-section of the wire. What is the magnetic field a distance r from the center of the wire in the region r > R ?

A $\frac{\mu_o I}{2\pi R^2} r$ B $\frac{\mu_o I}{2\pi r^2} R$ C $\frac{\mu_o I}{\pi r}$ D. $\frac{\mu_o I}{2\pi r}$

Solution: The magnetic field a distance r from the center of the wire in the regions $r \geq R$.

i) For $r \geq R$ let us choose path 1 in the given figure a circle of radius r centered at the wire .The total current passing through the plane of the circle I_0 Amperes Law applied to the circular path gives:

$\therefore B = \frac{\mu_o I}{2\pi r}$ ..... **For r > R**



ii) Consider the interior of the wire ,where r < R. We choose the circular path 2 shown in the given figure. Here the current I passing through the plane of the circle is less than the total current I_o. Because the current uniform over the cross- section of the wire ,the fraction of the current enclosed by the circle of radius r < R must equal the ratio of the area πr^2 enclosed by circular path 2 and the cross- sectional area πR^2 of the wire.

$\frac{I}{I_o} = \frac{\pi r^2}{\pi R^2}$

$I = \frac{r^2 I_o}{R^2}$

$(2\pi r)B = \mu_o I$

$(2\pi r)B = \mu_o \frac{r^2 I_o}{R^2}$

$\therefore B = \frac{\mu_o r I}{2\pi R^2}$ ..... r < R

$\therefore$ ***Answer D***

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3. In a RLC series circuit, suppose R = 300Ω , L = 60 mH , C = 0.5 μF, V = 50 V , and ω = 10,000 rad/s. What is the average power delivered to the entire circuit?

A .1.5 W B. 2.3 W C .3.5 W D .1.0 W

Given: ***Required*** **: Average power ($P_{av} = ?$)**

$R = 300\ \Omega$

$L = 60mH = 60 \times 10^{-3}\ H$

$C = 0.5\mu F = \frac{1}{2} \times 10^{-6}\ F$

$V_{max} = 50V$

$\omega = 10{,}000$ rad/s

Solution: $X_L = \omega L = 10000\ \text{rad/sec} \times 60 \times 10^{-3} H$

$X_L = 600\ \Omega$

$$X_C = \frac{1}{w_O C} = \frac{1}{10000\ \text{rad/sec} \times \frac{1}{2} \times 10^{-6} F}$$

$\therefore X_C = 200\Omega$

$$Z = \sqrt{R^2 + (x_L - X_C)^2}$$

$$Z = \sqrt{(300\ \Omega)^2 + (600\ \Omega - 200\Omega)^2}$$

$$Z = \sqrt{90000\Omega^2 + 160000\Omega^2}$$

$$Z = \sqrt{250000\Omega^2}$$

$\therefore Z = 500\ \Omega$

$$I_{max} = \frac{V_{max}}{z} = \frac{50V}{500\Omega} = 0.1\ A$$

$$P_{av} = (I_{rms})^2 \times R$$

$$P_{av} = \left(\frac{0.1A}{\sqrt{2}}\right)^2 \times 300\ \Omega$$

$\therefore P_{av} = 1.5\ W$

$\therefore$ ***Answer A***

4. The vector product of two non-collinear vectors $\vec{A}$ and $\vec{B}$.

A . is perpendicular to the plane formed by $\vec{A}$ and $\vec{B}$.

B . is on the plane formed by $\vec{A}$ and $\vec{B}$.

C . must be zero.

D . is a scalar quantity.

Solution:

- $\vec{A} \times \vec{B} = \vec{C}$. Implies $\vec{C}$ perpendicular to both $\vec{A}$ and $\vec{B}$.
- It is also perpendicular to the plane formed by $\vec{A}$ and $\vec{B}$.
- $\vec{A} \times \vec{B}$ will be zero if $\vec{A} \parallel \vec{B}$.

$\therefore \vec{C} = \vec{A} \times \vec{B}$ is a vector.

$\therefore$ ***Answer A***

5. Temperature of a body is a measure of the _____.

A . heat content of a body.

B . average molecular kinetic energy.

C .average pressure in a gas.

D . heat of vaporization of a liquid.

Solution: **Temperature** of a body is a measure of the Average molecular kinetic energy.

$\therefore$ ***Answer B***

6. Two negative point charges are 2 meters apart and repel each other with a force of 2 Newton. When the distance between the charges is doubled, the force between them is _____.

A . one fourth as great. C . twice as great.

B . one half as great. D .four times as great

Given: Let $q_1 = -Q_1$ and $q_2 = -Q_2$ ***Required:*** $F_2 = ?$

$r_1 = r = 2m$

$F_1 = F = 2N$..... Repulsion

$r_2 = 2r_1$

Solution: $F_2 = \frac{Kq_1q_2}{r_2^2}$.....Equation1

$F_1 = \frac{Kq_1q_2}{r_1^2}$..... Equation 2

Dividing Equation 1 by Equation 2 we get:

$$\frac{F_2}{F_1} = \frac{r_1^2}{r_2^2}$$

$$F_2 = \left(\frac{r_1}{r_2}\right)^2 . F_1$$

$$F_2 = \left(\frac{r_1}{2r_1}\right)^2 \times 2\ N$$

$$F_2 = \frac{1}{4} \times 2N$$

$\therefore F_2 = 0.5\ N$

Therefore, the force between them is one fourth as great.

$\therefore$ ***Answer A***

7. The terminal voltage of a cell supplying energy to a circuit is usually less than its emf because of the cell's

A .size C .mass

B .internal resistance D . energy.

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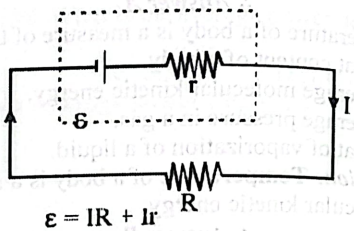
Required:

$P_T = \frac{V_T^2}{R}$

$P_\varepsilon = \frac{\varepsilon^2}{R}$

$P_T < P_E$. why ?

Solution:



$\varepsilon = IR + Ir$

$IR = \varepsilon - Ir$

$V_T = \varepsilon - I.r$

$\Rightarrow V_T < \varepsilon$ because of internal resistance

$\therefore P_T < P_\varepsilon$

∴ Answer B

8. The peak voltage of an alternating emf is 141 volts. The effective value of the voltages in volt is:

A .70.5 B .100 C.141 D. 200

Given: $V_{max(Peak)} = 141V$

Required : $V_{effective} = ?$

Solution : In alternating current circuit the peak voltage (maximum) voltage and effective or root mean square values (rms) voltage is given by :

$V_{rms} = V_{effective} = \frac{V_{max}}{\sqrt{2}} = \frac{141\ V}{1.414}$

$V_{effective} = 99.68\ V$

$\therefore V_{effective} = 100V$

∴ Answer B

9. The voltage output of a generator is given by V= (300 V) sin(ωt). What is the rms current in the circuit when this generator is connected to a 100 Ω resistor?

A .1.50 A B .2.12 A C .1.12 A D .1.41 A

Given : V = (300 Volt) sin ωt

R =100Ω

Required: $I_{rms} = ?$

Solution: $I_{max} = \frac{V_{max}}{R}$

$I_{max} = \frac{300}{100\Omega}$ volt sin ωt.

$I_{max} = (3A)\sin \omega t$

$\therefore I_{max} = 3A$

$I_{rms} = \frac{I_{max}}{\sqrt{2}}$

$I_{rms} = \frac{3\ A}{\sqrt{2}}$

$\therefore I_{rms} = 2.12\ A$

∴ Answer B

10. Which of the following is NOT true of vector product?

A. $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$ C. $\vec{A} \perp \vec{B} \Rightarrow |\vec{A} \times \vec{B}| = AB$

B. $\vec{A} \times (\vec{B} + \vec{C}) = \vec{B} \times \vec{A} + \vec{A} \times \vec{C}$ D. $\vec{A} \parallel \vec{B} \Rightarrow \vec{A} \times \vec{B} = 0$

Solution: **Properties of Vector product:**

i) Unlike the scalar product , the order in which the two vectors are multiplied in a cross product is important i.e

$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$

ii) Vector product obeys the distributive law i.e

$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$

iii) If $\vec{A}$ is parallel to $\vec{B}$ (i.e $\theta = 0^0$ or 180^0) ,then $\vec{A} \times \vec{B} = 0$.

iv) If $\vec{A} \perp \vec{B} \Rightarrow |\vec{A} \times \vec{B}| = AB$

∴ Answer B

11. If a single constant force F acts on a particle of mass m and the particle starts from rest, what is the instantaneous power delivered by the force at time t ?

A. $2\frac{F^2}{m}t$ B. $\frac{F^2}{2m}t$ C. $\frac{F^2}{m}t$ D. $3\frac{F^2}{m}t$

Given : Constant force = F

u = 0

Mass = m

time = t

Required: Instantaneous power delivered = ?

Solution: $P = \frac{w}{t} = \frac{FS}{t} = F(\frac{S}{t}) = FV$

$P_{Instantaneous} = F.V_{Instantaneous}$

$V_{inst} = at = \frac{Ft}{m}$ ..... But $a = \frac{F}{m}$

$P_{inst} = F \times V_{inst}$

$P_{inst} = F \times \frac{Ft}{m}$

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$$\therefore P_{inst} = \frac{F^2}{m}.t$$

∴ *Answer C*

12. Which of the following statement is **NOT** true?

A .If a heat Q is transferred from a hot reservoir at temperature T_h to a cold reservoir at temperature T_c, the entropy of the cold reservoir increases.

B .The increase in entropy of the cold reservoir is greater than the decrease in entropy of the hot reservoir.

C .The change in entropy of the system between any two equilibrium states does not depend only on the initial and final states.

D .Metals are good heat conductors because of their large number of electrons.

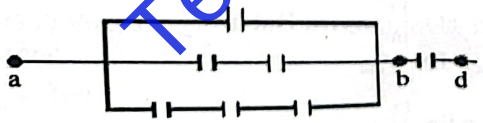
Solution:

- The second law of thermodynamics is the **law of entropy,** and every process that takes place in nature , whether it be mechanical, electrical ,chemical or biological.
- The function is called **entropy** of the system ,and is denoted by S. We then have $S_2 - S_1 = \Delta S = \frac{\Delta Q}{T}$..... For any reversible path.
- **Entropy** is the measure of disorder of molecules of a body. When heat is added to a system its entropy increases.
- The second law of thermodynamics, that when all systems taking part in a process are included , the entropy either remains constant or increases. In other words ,no process is possible in which the entropy decreases ,when all systems taking part in the process are included.
- The change in entropy of the system between any two equilibrium states depend only on the initial and final states.

∴ *Answer C*

13.Consider the network of identical capacitors, each having capacitance C between terminal a and b and a capacitor of capacitance $\frac{11}{5}C$ between terminal b and d shown in figure below. What is the effective capacitance between terminal a and d?

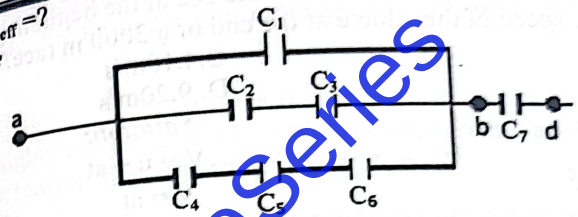
A . $\frac{121}{30}C$ B. 1C C . $\frac{11}{6}C$ D . 3 C



Required: Effective capacitance between terminal a and d or

$C_{total} = C_{eff} = ?$

Solution:



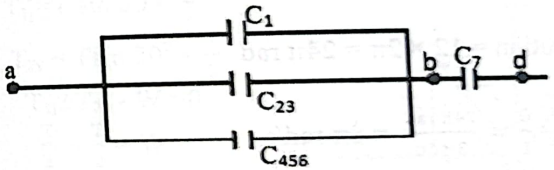
$C_1 = C_2 = C_3 = C_4 = C_5 = C_6 = C$

$C_7 = \frac{11}{5}C$

$$C_{23} = \frac{C_2 \times C_3}{C_2 + C_3} = \frac{C \times C}{C + C} = \frac{C}{2}$$

$$\frac{1}{C_{456}} = \frac{1}{C_4} + \frac{1}{C_5} + \frac{1}{C_6} = \frac{1}{C} + \frac{1}{C} + \frac{1}{C}$$

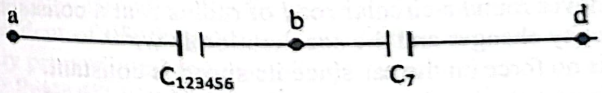
$$\therefore C_{456} = \frac{C}{3}$$



$$C_{123456} = C_1 + C_{23} + C_{456}$$

$$C_{123456} = C + \frac{C}{2} + \frac{C}{3}$$

$$C_{123456} = \frac{11}{6}C$$



$$C_{eff} = \frac{C_{123456} \times C_7}{C_{123456} + C_7}$$

$$C_{eff} = \frac{\frac{11}{6}C \times \frac{11}{5}C}{\frac{11}{6}C + \frac{11}{5}C}$$

$$\therefore C_{eff} = 1C$$

∴ *Answer B*

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14.An athlete accelerates at 1.9 m/s^2 for 2.2 sec at the beginning of a race. What is the speed of the athlete at the end of a 3000 m race?

A .0.86m/s
B .1.16m/s
C.4.18m/s
D. 9.20m/s

Given :
U = 0 m/s
a = 1.9 m/s^2
t =2.2 sec
S =3000m

Solution:
$V = u + at$
$V = at$
$V = 1.9 m/s^2 \times 2.2s$
$\therefore V = 4.18 m/s$

Required :Speed (V =?)

∴ Answer C

15.A body is rotated with a uniform speed in a circle of radius 1m.What is the centripetal acceleration in m/s^2 if it completes 12 rotations in 3 seconds?

A .$36\pi^2$ B .$64\pi^2$ C.$72\pi^2$ D. $144\pi^2$

Given :
r =1m
θ =12 revolution = $12 \times 2\pi = 24\pi$ rad
t = 3s

Required : a_c = ?

Solution: $\omega = \frac{\theta}{t} = \frac{24\pi \text{ rad}}{3 \text{ sec}} = 8\pi$ rad/s

$a_c = \frac{V_t^2}{r} = \omega^2 r$

$a_c = (8\pi \text{ rad/s})^2 \times 1 \text{ m}$

$\therefore a_c = 64\pi^2 m/s^2$

∴ Answer B

16.If a car moves round a circular road of radius r at a constant speed

A its velocity changes and the acceleration is v/r^2
B .there is no force on the car since its speed is constant.
C .there is no velocity change since the speed is constant.
D .the force per unit mass on the car is towards the center and is v^2/r

Solution: **Centripetal force.** F_C is the force that produces centripetal acceleration.

Centripetal force F_C is the product of mass of the body and the centripetal acceleration.

$\Rightarrow F_C = ma_c$

$\Rightarrow F_C = \frac{mV_t^2}{r}$

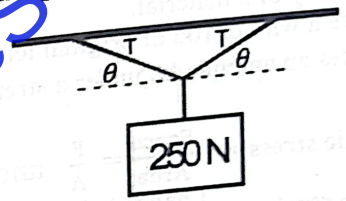
$\Rightarrow \frac{F_C}{m} = \frac{V_t^2}{r}$

Therefore, the force per unit mass on the car is towards the center is $\frac{V_t^2}{r}$.

∴ Answer D

17.Traffic lights weighing 250 N are suspended from a cable as shown. Find the tension T in the string if $\theta_1 = \theta_2 = 30^0$

A .120 N
B .165 N
C .196 N
D .250 N



Given : W =250 N
$\theta_1 = \theta_2 = 30^0$

Required : $T_1 = T_2 = T$ =?

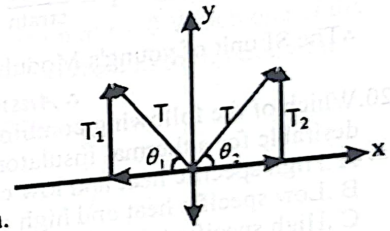
Solution: $T_{1y} = T \sin 30^0 = \frac{T}{2}$

$T_{2y} = T \sin 30^0 = \frac{T}{2}$

$T_{1y} + T_{2y} - W = 0$

$\frac{T}{2} + \frac{T}{2} = W$

$\therefore$ **T =W = 250N in each.**



∴ Answer D

18. The period of oscillation of a particle undergoing simple harmonic motion is:

A .independent of the amplitude of the motion.
B .directly proportional to the frequency of oscillation.
C .independent of the frequency of oscillation.
D .directly proportional to mass of oscillating object.

Solution: **Period(T)** is a body of mass m attached to the end of a vibrating spring of stiffness K is given by $T = 2\pi\sqrt{\frac{m}{K}}$.

$f = \frac{1}{T} \Rightarrow f = \frac{1}{2\pi}\sqrt{\frac{K}{m}}$.

The period or frequency of oscillation of a particle undergoing simple harmonic motion is independent of the amplitude of the motion .It is inversely proportional to the frequency of oscillation.

∴ Answer A

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19. Young's Modulus is the
A .ratio of stress to strain.
B .force per unit area
C .ratio of strain to stress.
D .force per unit elongation
Solution: **Young's modulus (tensile modulus ,Y)** describes the length of elasticity of a material.

❖ Suppose a wire or rod of original length L and cross-sectional area A elongates an amount ΔL under a stretching force F applied to its end. Then

❖ **Tensile stress** $= \frac{\text{Force}}{\text{Area}} = \frac{F}{A}$ and

❖ **Tensile strain** $= \frac{\text{Change in Length}}{\text{Original Length}} = \frac{\Delta L}{L}$

❖ Young's Modulus $= Y = \frac{\text{stress}}{\text{strain}} = \frac{\frac{F}{A}}{\frac{\Delta L}{L}} = \frac{FL}{A\Delta L}$

∴The SI unit of young's Modulus is **Pa** .

∴ Answer A

20.Which of the following combinations of properties would be most desirable for a thermal insulator?
A .High specific heat and low conductivity.
B .Low specific heat and high conductivity.
C .High specific heat and high conductivity.
D .Low specific heat and low conductivity.
Solution: **Thermal insulator** is high specific heat and low conductivity.
Example: devices (house hold)are most of the time are good thermal insulator.

∴ Answer A

21. An air – filled parallel – plate capacitor has a capacitance of 1PF.The plate separation is doubled and the space between them is completely filled with wax. If the capacitance increases to 2pF, what is the dielectric constant of wax?
A .2.0 B. 4.0 C. 6.0 D. 8.0
Given : $C_1 = 1pF$
$C_2 = 2pF$
$d_1 = d_o$
$d_2 = 2d_1 = 2d_o$
Required : dielectric constant of wax $= K = ?$

Solution: For parallel- plate capacitor , $C_o = \frac{\varepsilon_o A}{d}$.

We can express the capacitor when the capacitor is filled with a dielectric constant of wax is given by : $C = K\frac{\varepsilon_o A}{d}$.

$$C_1 = \frac{\varepsilon_o A}{d_1}$$

$$C_2 = K\frac{\varepsilon_o A}{d_1}$$

$$C_2 = K\frac{\varepsilon_o A}{2d_1}$$

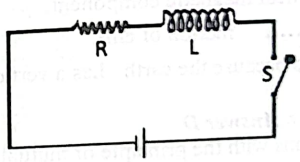
$$C_2 = \frac{K}{2}\left(\frac{\varepsilon_o A}{d_1}\right)$$

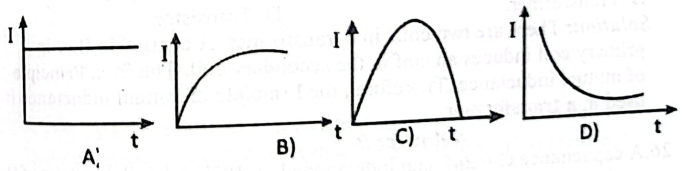
$$C_2 = \frac{K}{2}(C_1)$$

$$\therefore K = \frac{2C_2}{C_1} = \frac{2\times 2PF}{1PF} = 4$$

∴ Answer B

22. In the circuit shown below the switch is on at t = 0.Which one of the graphs shown represent the variation of current in the circuit with time?





Solution: When the switch S is closed the current increases exponentially .

∴ Answer B

23. A charged particle is moving in a UNIFORM magnetic field. If the direction of motion of the charged particle is parallel to the magnetic field, the shape of the charged particle's path will be:
A .circular C. spiral
B .straight D. elliptical.

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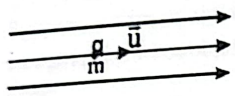
Solution: A charged particle moving in a magnetic field experiences a deflecting force F is given by : $\vec{F} = q(\vec{V} \times \vec{B})$ where

q = positive change

θ = angle between $\vec{V}$ and $\vec{B}$

$F = qVB \sin\theta \ldots\ldots \theta = 0^0$ and $\vec{v} \parallel \vec{B}$



$\Rightarrow F = qVB \sin 0°$

$\therefore F = 0$

$\Rightarrow$ no deflecting force acts on it. i.e it travels in a straight line.

∴ Answer B

24. An induced e.m.f is obtained between the ends of a horizontal steel axle X of a train moving due east. This is because
A .X points due east.
B .the earth's magnetic field has a horizontal component.
C .X moves parallel to the earth's field.
D .the earth has a vertical magnetic component.

Solution: $\mathcal{E} = B\ell V$..... motion of emf.

$\mathcal{E} = B\ell V$ is induced because the earth has a vertical magnetic field component.

∴ Answer D

25. Which device functions with the principle of mutual induction?
A .Electromagnet C .Electric motor.
B .Transformer. D .Transistor

Solution: There are two coils in a transformer. A charge in flux in primary coil induces an emf in the secondary coil. This is a Principle of mutual inductance .Therefore , the Principle of mutual inductance is used in **a transformer.**

∴ Answer B

26. A capacitance C = 2μF and inductance L = 10H and coil resistance 5Ω are in series in a circuit. When an alternating current of 2A rms flows, the power in watts in the circuit is
A .100 B. 50 C. 20 D. 2

Given : C = 2μF
L =10H
R =5Ω
I_{rms} = 2A

Solution:
$P_{av} = (I_{rms})^2 \times R$
$P_{av} = (2A)^2 \times 5\Omega$
$\therefore P_{av} = 20W$

Required: Average power = P_{av} = ?

∴ Answer C

27. A car with initial speed of 4m/s starts accelerating in a straight line at $3.2m/s^2$ for 10 seconds. If the brakes are then applied and it travels a further 20m before stopping, what is the acceleration in m/s^2?
A .-48.6 B. -16.2 C .-8.1 D .-32.4

Given: u =4 m/s
a_1 =$3.2m/s^2$
t = 10 sec

Required : a = ?

Solution: For the first part of the motion:

$V = u + a_1 t$

$V = 4\ m/s + 3.2m/s^2 \times 10sec$

$\therefore V = 36\ m/sec$

For the second part of the motion:

U = 36 m/sec
S =20 m
V = 0 m/sec

$S = \frac{V^2 - U^2}{2a}$

$S = \frac{-U^2}{2a}$

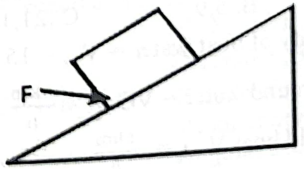
$a = \frac{-U^2}{2S}$

$a = \frac{-(36\ m/sec)^2}{2 \times 20\ m}$

$\therefore a = -32.4m/s^2$

∴ Answer D

28. A 100 kg box is pushed at a constant speed up the smooth 30^0 inclined plane shown below. What horizontal force F is required?
A .577.35 N
B .1000.35N
C. 1732.35N
D. 1154.70N



Given : m = 100 kg
θ =30^0

Required : Horizontal force = F= ?

Solution: F = Fcos θ

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$mg \sin\theta = F\cos\theta$

$F = mg \tan 30^0$

$F = 100\ kg \times 10 m/s^2 \tan 30^0$

$\therefore F = 577.36\ N$

∴ *Answer A*

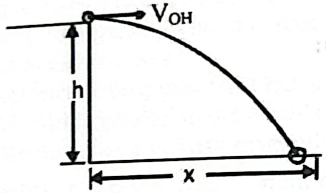
29. A ball rolls off the edge of a horizontal table top 4m high. It strikes the floor at a point 5m horizontally away from the edge of the table. How long was the ball in the air in seconds?

A. $\frac{3}{\sqrt{5}}$ B. $\frac{1}{\sqrt{5}}$ C. $\frac{2}{\sqrt{3}}$ D. $\frac{2}{\sqrt{5}}$

Given: h= 4m
x = 5m

Required : t =?

Solution:



$\frac{1}{2}gt^2 = h$

$t = \sqrt{\frac{2h}{g}}$

$t = \sqrt{\frac{2\times 4m}{10 m/s^2}}$

$t = \sqrt{0.8\ s^2}$

$t = \sqrt{\frac{4}{5}}$

$\therefore t = \frac{2}{\sqrt{5}} S$

∴ *Answer D*

30. A boat is travelling at 15km/hr with respect to the water of a river in the upstream direction. The water itself is flowing at 6km/hr with respect to the ground. A child on the boat walks from rear seat to front seat at 5km/hr with respect to the boat. What are the velocities of the boat and the child in km/hr with respect to the ground respectively?

A .6 ,12 B. 5,9 C .21,12 D .9,14

Given : Velocity of boat water $= V_{b.w} = 15\frac{km}{h}$ ....up the stream

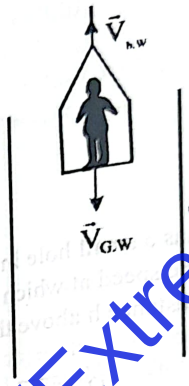
Velocity of ground water $= V_{G.W} = \frac{6km}{h}$ ..... down the stream.

Velocity of child boat $V_{c.b} = \frac{5km}{h}$ .....up the stream.

Required: Velocity of boat to ground $= V_{b.G} = ?$

Velocity of child to ground $= V_{c.G} = ?$

Solution:



$V_{b.w} = V_{b.G} - V_{G.W}$

$V_{b.G} = V_{b.w} + V_{G.W}$

$V_{b.G} = 15\frac{km}{h}$ up $+ \frac{6km}{h}$ down

$V_{b.G} = \frac{15km}{h}$ up $- \frac{6km}{h}$ up

$\therefore V_{b.G} = 9$ km/h up

$V_{c.b} = V_{c.G} - V_{b.G}$

$V_{c.G} = V_{c.b} + V_{b.G}$

$V_{c.G} = \frac{5km}{h}$ up $+ \frac{9km}{h}$ up

$\therefore V_{c.G} = 14$km/h up

$\therefore$ (9,14) km/h respectively

∴ *Answer D*

31. A body has moved from initial position having a coordinate of (0,1,1.5) meters to a final position with coordinate given by (1.2 ,2 ,1.5) meters under the influence of a constant force given by $\vec{F} = (-\hat{\imath} + 2\hat{\jmath} + 3\hat{k})$N. What is the work done by the force in moving the body from its initial position to its final position?

A .8J B .0.8 J C .4 J D .0.4 J

Given :

$\vec{S_1} = (0,1,1.5)$ m

$\vec{S_2} = (1.2,2,1.5)$ m

$\vec{F} = (-\hat{\imath} + 2\hat{\jmath} + 3\hat{k})$ N

$\Delta\vec{S} = \vec{S_2} - \vec{S_1}$

$\Delta\vec{s} = (1.2\hat{\imath} + \hat{\jmath})$ m $= \vec{s}$

Required : W =?

Solution:

$W = \vec{F}.\Delta\vec{S}$

$W = \vec{F}.\vec{S}$

$W = (-\hat{\imath} + 2\hat{\jmath} + 3\hat{k})$N $\times (1.2\hat{\imath} + \hat{\jmath})$m

$W = (-1\times 1.2 + 2\times 1 + 3\times 0)$ J

$\therefore W = 0.8$ J

∴ *Answer B*

32. The angular position of a swinging door is described by $\theta = 5+10t+2t^2$ rad. What is the angular speed and acceleration of the door at t =3 s?

A .22 rad/s, 4 rad/s^2 C. 22 rad/s, 12rad/s^2

B .-17 rad/s , 4 rad/s^2 D.12rad/s , 3 rad/s^2

Given : $\theta = 5 + 10t + 2t^2$ rad
t = 3sec

Required :
Angular speed = $\omega = ?$
Angular acceleration = $\alpha = ?$

Solution: $\omega = \frac{d\theta}{dt}$

$\omega = \frac{d}{dt}(5 + 10t + 2t^2)$ rad

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$\omega = 10 + 4t$

$\omega = 10 + 4 \times 3$ sec

$\therefore \omega = 22$ rad/s

$\propto = \frac{d\omega}{dt}$

$\propto = \frac{d(10+4t)}{dt}$

$\therefore \propto = 4 m/s^2$

∴ Answer A

33. An open tank containing a liquid of density ρ has a small hole in its side at a distance y_1 from the bottom. What is the speed at which the fluid leaves the hole when the liquid level is a distance h above the hole?

A. $\sqrt{2gh}$ B. $\sqrt{2gy_1}$ C. $\sqrt{gh}$ D. $\sqrt{gy_1}$

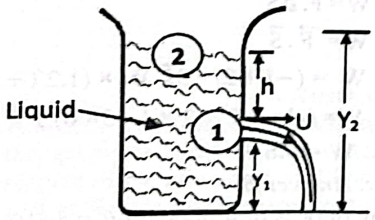
Given : A liquid of density ρ in a tank
d side at a distance y_1

Required : Speed = U =?

Solution :

$P + \frac{1}{2}\rho u^2 + \rho gh = \text{constant}.$

$P_1 + \frac{1}{2}\rho u_1^2 + \rho g y_1 = p_2 + \frac{1}{2}\rho u_2^2 + \rho g y_2$



$A_2 u_2 = A_1 u_1$

$u_2 = \frac{A_1}{A_2} u_1$

$A_2 >> A_1 \Longrightarrow u_2 = 0$

$P_1 = P_2 =$ atmospheric.

$\frac{1}{2}\rho u_1^2 + \rho g y_1 = \rho g y_2$

$U = \sqrt{2g(y_2 - y_1)}$

$\therefore u = \sqrt{2gh}$ ..... $y_2 - y_1 = h$

∴ Answer A

34. A gas is compressed at a constant pressure of $8 \times 10^{+4} N/m^2$ from a volume of $9 \times 10^{-3} m^3$ to a volume of $3 \times 10^{-3} m^3$. In the process 400J of heat energy flows out of the gas. What are the work done by the gas and the change in internal energy of the gas?

A. -567 J, -167 J
B. -560 J, -160 J
C. -589 J, -189 J
D. -540 J, -140J

Given : $P = 8 \times 10^4 N/m^2$
$V_o = 9 \times 10^{-3} m^3$
$V = 3 \times 10^{-3} m^3$
$\Delta Q = -400J$
$\Delta V = V - V_o$
$\Delta V = (3-9) \times 10^{-3} m^3$
$\Delta V = -6 \times 10^{-3} m^3$

Required : W =? ΔU = ?

Solution:

$W = P\Delta V$

$W = 8 \times 10^4 N/m^2 (-6 \times 10^{-3} m^3)$.

$\therefore W = -480$ J

$\Delta Q = \Delta U + W$

$\Delta U = \Delta Q - W$

$\Delta U = -400J - (-480\ J)$

$\therefore \Delta U = 80$ J

∴ No Answer

Answer -480J , 80 J

35. An object having a net charge of 24μC is placed in a uniform electric field of 610N/C directed vertically. What is the mass of the object if it floats in this electric field?

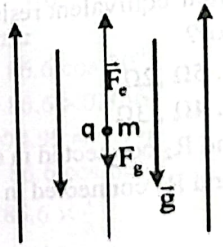
A. 14.640 kg
B. 1.464 gm
C. 14.460 gm
D. 14.460kg

Given : $q = 24\ \mu F$

$\vec{E} = 610$ N/c

Required : mass = m = ?

Solution : When the object floats in the electric field electric force on it must be equal to its weight.



$F_g = F_e$

$mg = qE$

$m = \frac{qE}{g}$

$m = \frac{24 \times 10^{-6}\ F \times 610\ N/C}{10\ m/s^2}$

$m = 1.464 \times 10^{-3} kg$

$\therefore m = 1.464$ gm

∴ Answer B

36. Which of the following is **NOT** property of a conductor in electrostatic equilibrium?

A. The electric field is zero everywhere inside the conductor.
B. Any net charge on the conductor resides entirely on its surface.
C. On an irregularly shaped conductor, the surface charge density is greatest where the radius of curvature of the surface is the smallest.
D. The electric field just outside the conductor is tangent to its surface.

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***Solution :* Property of a conductor in electrostatic equilibrium:**

- The electric field is zero everywhere inside the conductor.
- Any net charge on the conductor resides entirely on its surface.
- On an irregularly shaped conductor, the surface charge density is greatest where the radius of curvature of the surface is the smallest.
- The electric field just outside the conductor is perpendicular to its surface.

∴ ***Answer D***

37. Which of the following statement is **NOT** true?

A .Any device such as battery or generator can be considered as source of emf.

B .The terminal voltage can be equal to emf in real system.

C .Emf forces electrons move in a direction opposite to the electrostatic force.

D .Emf can be defined as work done per unit charge.

Solution: In any circuit (real system) the terminal voltage (V_T) is always less than the EMF. Because the source has internal resistance.

Or EMF = Internal resistance (Ir) + External resistance (IR)

$\Longrightarrow$ EMF = Ir +IR

$\Longrightarrow$ EMF = Ir +V_T

$\Longrightarrow$ V_T = EMF − Ir

$\Longrightarrow$ V_T < EMF..... Is always true.

∴ ***Answer B***

38. Two resistors connected in series have an equivalent resistance of 7Ω. When they are connected in parallel, their equivalent resistance is $\frac{10}{7}$ Ω. What is the resistance of each resistor?

A .5.5Ω, 1.5 Ω

B .6Ω , 1Ω

C. 5Ω ,2Ω

D. 4Ω , 3Ω

***Given* :** $R_T = 7\Omega$ when two resistors R_1 and R_2 connected in series.

$R_T = \frac{10}{7}\Omega$ when two resistors R_1 and R_2 connected in parallel.

Required : R_1 = ? and R_2 =?

Solution: $R_1 + R_2 = R_T$ ..... Series connection

$R_1 + R_2 = 7\Omega$..... Equation 1

$\frac{R_1 \times R_2}{R_1 + R_2} = R_T$ ..... Parallel connection

$\frac{R_1 \times R_2}{7\Omega} = \frac{10}{7}\Omega$ ..... Equation 2

$\Longrightarrow R_1 = \frac{10}{R_2}$ .....From equation 2

$\frac{10}{R_2} + R_2 = 7$..... From equation 1 and 2

$R_2^2 - 7R_2 + 10 = 0$

$R_2 = \frac{-(-7) \pm \sqrt{49 - 4\times10\times1}}{2\times1}$

$R_2 = 3.5\,\Omega \pm 1.5\Omega$

∴ $R_2 = 5\Omega$ or 2Ω

$R_1 + R_2 = 7$

∴ $R_1 = 2\Omega$ or 5Ω

If $R_2 = 5\Omega$ then $R_1 = 2\Omega$ and if $R_2 = 2\Omega$ then $R_1 = 5\Omega$

∴ ***Answer C***

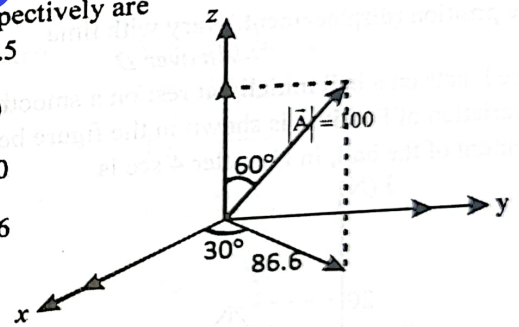
39. The x ,y ,and z components of the vector $|\vec{A}| = 100$m shown in the figure below, respectively are

A .86.6, 21.6, 37.5

B .50.0,43.3,75.0

C .75.0,43.3, 50.0

D .37.5,21.6, 86.6



Given: $|\vec{A}| = 100$

Required : A_x = ? Ay = ? Az = ?

Solution:

$A_x = 86.6 \cos 30^0$

$A_x = 86.6 \times 0.866$

∴ $A_x = 74.99 \approx 75$ m

$A_y = 86.6 \cos 60^0$

$A_y = 86.6 \times \frac{1}{2}$

∴ $A_y = 43.3$m

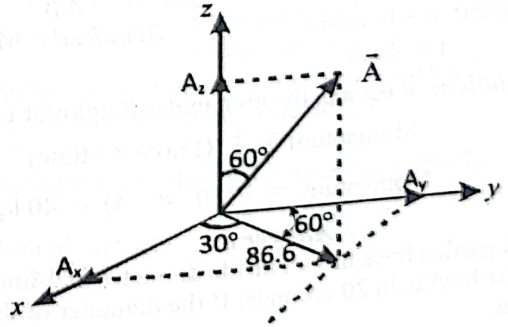
$A_z = A \cos 60^0$

$A_z = 100 \times \frac{1}{2}$

∴ $A_z = 50$ m

∴ $\vec{A} = 75\hat{i} + 43.3\hat{j} + 50\hat{k}$

∴ ***Answer C***



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40. The net force acting on a system of particles is zero. Which one of the following quantities may vary with time.?

A . The momentum of the system.

B . The kinetic energy of the system.

C . The velocity of the center of mass of the system.

D . The position of the center of mass of the system.

Solution: The net force acting on a system of particles is zero ,the acceleration is zero .

- ✓ The momentum (p = mv) is constant.
- ✓ The kinetic energy ($KE = \frac{1}{2}m\,v^2$) is constant..
- ✓ The velocity is constant..
- ✓ The position (displacement) vary with time

∴ Answer D

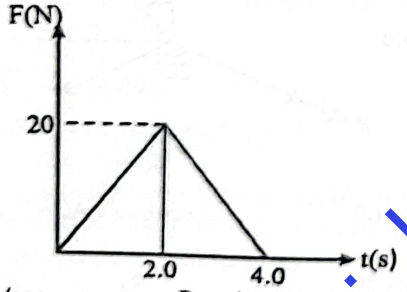
41.A force F acts on a ball initially at rest on a smooth surface for a time t .The variation of F with t is shown in the figure below. The momentum of the ball, in Ns ,after 4 sec is

A .10

B .20

C .30

D .40



Given: u = 0 m/sec
t = 4 sec

Required : Momentum = P = ?

Solution: $\vec{P}$ equals the area under F against t graph

$$\text{Momentum} = \frac{1}{2}(\text{Force} \times \text{time})$$

$$\text{Momentum} = \frac{1}{2}(20 \times 4) = 40\ \text{kg.m/sec}$$

∴ Answer D

42. A garden hose has an inside diameter of 16mm.The hose can fill a 10 liter bucket in 20 seconds. If the diameter of the nozzle is reduced to 8 cm.

A .Water exits with a speed twice the speed inside the hose.

B .a 10 liter bucket can be filled in 10 seconds.

C .a 20 liter bucket can be filled in 40 seconds.

D .water exits with a speed half the speed inside the hose.

Solution:

Given:

$d_1 = 16mm = 1.6cm = 5d_1$

$V = 10L$

$AU = \frac{v}{t} = \frac{10\ liter}{20S} = 0.5\frac{\ell}{s}$

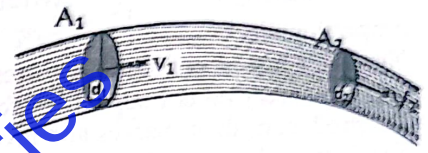
$\Rightarrow V = 0.5\frac{\ell}{s} \times t$

$V = 10L$

$t = 20\ Sec$

$d_2 = 8cm$

$d_2 > d_1$ ....not reduced



$A_1U_1 = \text{constant}$

$A_1U_1 = A_2U_2$

$A_1 = \pi\left(\frac{d_1}{2}\right)^2$

$A_2 = \pi\left(\frac{d_2}{2}\right)^2$

$A_2 = \pi\left(5.\frac{d_1}{2}\right)^2$

$A_2 = 25A_1$

$U_2 = \frac{A_1U_1}{A_2}$

$U_2 = \frac{A_1U_1}{25A_1}$

$U_2 = \frac{1U_1}{25}$

$U = 0.5\frac{\ell}{s} \times 40\ S = 20\ L$

$\Rightarrow$ It fills 20 L in 40 Sec

∴ Answer C

43.When stretched beyond its elastic limit, a metal rod such as steel

A .becomes elastic

B .has no energy.

C .obeys hook's law

D .becomes plastic.

Solution: Ductile material can be stretched beyond its elastic limit. Therefore ,a metal rod such as steel becomes plastic.

∴ Answer D

44.Coulomb's law for the force between electric charges most closely resembles

A .the law of conservation of energy.

B .Newton's second law of motion.

C .Newton's law of gravitation.

D .the law of conservation of mass.

Solution: **Coulomb's law:** state that the force between two point charges placed in vacuum is directly proportional to the product of the charges and inversely proportional to the square of their separation.

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